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AIUB. IEC Notes.

Final Term

03

Ch 15:- Impedance And The Phasor Diagram / Resistive Elements:-

Average value And RMS Value:-

- ① Average value (for full cycle) = 0
- ② Average value (for half cycle) = $0.636 \times V_m = 0.636 \times V_{peak}$.
- ③ RMS Value, $V_{rms} = 0.707 \times V_m$ or $0.707 \times V_{peak} = \frac{V_m}{\sqrt{2}} = \frac{V_{peak}}{\sqrt{2}}$

Instantaneous Value to phasor or polar form:-

Instantaneous Value	Phasor or polar Form
① $V(t) = V_m \sin(\omega t + \theta)$	$V = V_{rms} \angle \theta$ or $V = \frac{V_m}{\sqrt{2}} \angle \theta$
② $V(t) = V_m \sin(\omega t - \theta)$	$V = V_{rms} \angle -\theta$ or $V = \frac{V_m}{\sqrt{2}} \angle -\theta$
③ $i(t) = I_m \sin(\omega t + \theta)$	$I = I_{rms} \angle \theta$ or $I = \frac{I_m}{\sqrt{2}} \angle \theta$
④ $i(t) = I_m \sin(\omega t - \theta)$	$I = I_{rms} \angle -\theta$ or $I = \frac{I_m}{\sqrt{2}} \angle -\theta$
⑤ $V(t) = V_m \sin(\omega t)$	$V = V_{rms} \angle 0^\circ$ or $\frac{V_m}{\sqrt{2}} \angle 0^\circ$

#problem:-1:-

⇒ The following instantaneous voltage convert to

(i) The phasor or polar form.

(ii) The V_{rms} value?

(iii) The Average value for half cycle:

$$V(t) = 100 \sin(\omega t + 45^\circ)$$

⇒ In given:-

$$V(t) = 100 \sin(\omega t + 45^\circ)$$

$$\therefore V_m = 100$$

(i)

We know:-

$$V = \frac{V_m}{\sqrt{2}} \angle \theta$$

$$= \frac{100}{\sqrt{2}} \angle 45^\circ V$$

$$= 70.71 \angle 45^\circ$$

(ii)

We know:-

$$V_{rms} = \frac{V_m}{\sqrt{2}} = \frac{100}{\sqrt{2}} = 70.71 V$$

(iii)

We know:-

$$\text{The Average value for half cycle} = 0.636 V_m$$

$$= 0.636 \times 100$$

$$= 63.6$$

Polar Form Multiplication:—

$$A \angle \theta_1 \times B \angle \theta_2 = (A \times B) \angle (\theta_1 + \theta_2)$$

Polar Form Division:—

$$A \angle \theta_1 / B \angle \theta_2 = A/B \angle (\theta_1 - \theta_2)$$

Euler's Identity:—

$$e^{j\theta} = \cos \theta + j \sin \theta$$
$$e^{-j\theta} = \cos \theta - j \sin \theta$$

here, $j = \sqrt{-1}$

Exponential Form:—

Instantaneous Value	Exponential Form
1. $V(t) = V_m \sin(\omega t + 0^\circ)$	1. $V = V_{rms} \times e^{j0^\circ} = \frac{V_m}{\sqrt{2}} \times 1$
2. $V(t) = V_m \sin(\omega t + \theta)$	2. $V = V_{rms} \times e^{j\theta} = \frac{V_m}{\sqrt{2}} e^{j\theta}$
3. $V(t) = V_m \sin(\omega t - \theta)$	3. $V = V_{rms} \times e^{-j\theta} = \frac{V_m}{\sqrt{2}} e^{-j\theta}$

Rectangular Form:-

Instantaneous Value	Rectangular Form
1. $v(t) = V_m \sin(\omega t + \theta)$	1. $V = V_{rms} (\cos \theta + j \sin \theta) = \frac{V_m}{\sqrt{2}} (\cos \theta + j \sin \theta)$ $= \frac{V_m}{\sqrt{2}} \angle \theta$
2. $v(t) = V_m \sin(\omega t + \theta)$	2. $V = V_{rms} (\cos \theta + j \sin \theta) = \frac{V_m}{\sqrt{2}} (\cos \theta + j \sin \theta)$
3. $v(t) = V_m \sin(\omega t - \theta)$	3. $V = V_{rms} (\cos \theta - j \sin \theta) = \frac{V_m}{\sqrt{2}} (\cos \theta - j \sin \theta)$

Rectangular Form Addition:-

$$(a + jb) + (c + jd) = (a + c) + j(b + d)$$

Rectangular Form Subtraction:-

$$(a + jb) - (c + jd) = (a - c) + j(b - d)$$

Problems:-

⇒ A Voltage and current for a circuit is given as $220 \angle 20^\circ$ and $14 \angle -36^\circ$.

- Find V_m and I_m .
- Convert it into exponential and Rectangular Form
- Write the instantaneous expression for $v(t)$ and $i(t)$

⇒ In given:-

$$V = 220 \angle 0^\circ \quad \text{--- (i)}$$

$$I = 14 \angle -36^\circ \quad \text{--- (ii)}$$

(i)

we know:-

$$V = V_{rms} \angle \theta \quad \text{--- (iii)}$$

∴ Comparing the (i) and (iii) eq:-

$$V_{rms} = 220$$

$$\frac{V_m}{\sqrt{2}} = 220$$

$$\therefore V_m = 220 \sqrt{2} = 311.12 \text{ V}$$

Again:-

$$I = I_{rms} \angle -\theta \quad \text{--- (iv)}$$

∴ Comparing the (ii) and (iv) eq:-

$$I_{rms} = 14$$

$$\therefore \frac{I_m}{\sqrt{2}} = 14$$

$$I_m = 14 \sqrt{2} = 19.798 \text{ A}$$

$$\therefore I_m = 19.798 \text{ A}$$

(ii)

we know:-

$$V = V_{rms} e^{j\theta}$$

$$= 220 e^{j0}$$

$$= 220 \text{ V}$$

$$I = I_{rms} e^{-j\theta}$$

$$= 14 e^{-j36^\circ} \text{ A}$$

we know:-

$$V = V_{rms} (\cos \theta + j \sin \theta)$$

$$= 220 (\cos 0 + j \sin 0)$$

$$= 220 \text{ V}$$

$$I = I_{rms} (\cos \theta - j \sin \theta)$$

$$= 14 (\cos 36^\circ - j \sin 36^\circ)$$

(iii)

we know

$$V(t) = V_m \sin(\omega t + \theta')$$

$$\therefore V(t) = 311.12 \sin(\omega t + 0')$$

$$i(t) = I_m \sin(\omega t - \theta')$$

$$i(t) = 10.708 \sin(\omega t - 36')$$

Ans:-

Problems 2:-

⇒ The following instantaneous voltage convert to

(i) The phasor or polar form.

(ii) The exponential form.

(iii) The rectangular form.

$$V(t) = 100 \sin(\omega t + 45')$$

⇒ In given

$$V(t) = 100 \sin(\omega t + 45')$$

(i)

we know

polar form

$$V = V_{rms} \angle \theta$$

$$= \frac{V_m}{\sqrt{2}} \angle \theta$$

$$= \frac{100}{\sqrt{2}} \angle 45^\circ$$

$$= 70.71 \angle 45^\circ$$

ii)

we know:-

The exponential form:-

$$\begin{aligned} V &= V_{rms} e^{j\theta} \\ &= \frac{V_m}{\sqrt{2}} e^{j\theta} \\ &= \frac{100}{\sqrt{2}} e^{j45^\circ} \\ &= 70.71 e^{j45^\circ} \end{aligned}$$

iii)

we know:-

The rectangular form:-

$$\begin{aligned} V &= V_{rms} (\cos\theta + j\sin\theta) \\ &= \frac{V_m}{\sqrt{2}} (\cos\theta + j\sin\theta) \\ &= \frac{100}{\sqrt{2}} (\cos 45^\circ + j\sin 45^\circ) \\ &= 70.71 (0.707 + j0.707) \\ &= 50 + j50 \\ &= 50 + j50 \end{aligned}$$

Impedance:

⇒ According to ohm's Law, Impedance is the ratio of Voltage to current. The Unit of Impedance is ohm (Ω)

$$Z = \frac{V}{i} = \frac{V_m \sin(\omega t + \theta_v)}{I_m \sin(\omega t + \theta_i)} \quad \text{ohm } (\Omega)$$

$$Z = \frac{V}{i} = \frac{V_{rms} \angle \theta_v}{I_{rms} \angle \theta_i} = \frac{V_{rms}}{I_{rms}} \angle (\theta_v - \theta_i) = \frac{V}{I} \angle (\theta_v - \theta_i) \quad \Omega$$

$$Z = \frac{V_m}{I_m} \angle (\theta_v - \theta_i) = R + jX = Z \angle \theta_z$$

Resistance Reactance

$$\text{Magnitude of impedance } Z = \frac{V_m}{I_m} = \frac{V_{rms}}{I_{rms}} = \frac{V}{I} \quad \Omega$$

$$\text{Angle of impedance } \theta_z = \theta_v - \theta_i$$

$$\text{Resistance } R = Z \cos \theta_z$$

$$\text{Reactance } X = Z \sin \theta_z$$

problem 10 —

⇒ A Voltage and Current is given as $220 \angle 0^\circ$ and $14 \angle -36^\circ$ respectively. Determine the impedance in polar and rectangular form.

⇒ Is given: —

$$V_{rms} = 220 \text{ V}$$

$$\theta_v = 0^\circ$$

$$I_{rms} = 14 \text{ A}$$

$$\theta_i = -36^\circ$$

we know: —

$$\begin{aligned} \text{magnitude of impedance } |Z| &= \frac{V_{rms}}{I_{rms}} \\ &= \frac{220}{14} \\ &= 15.714 \Omega \end{aligned}$$

$$\begin{aligned} \text{Angle of impedance } \theta_z &= \theta_v - \theta_i \\ &= 0 - (-36) \\ &= 36^\circ \end{aligned}$$

In polar: —

$$\therefore \text{Impedance } Z = 15.714 \angle 36^\circ \Omega$$

In rectangular: —

$$\text{Impedance } Z = 12.712 + j0.236 \Omega$$

Compare it to the eq $Z = R + jX$

$$\therefore \text{Resistance } R = 12.712 \Omega$$

$$\text{Reactance } X = 0.236 \Omega$$

problems 2

⇒ A circuit has a resistance of 15Ω and a reactance of 22Ω . if the applied voltage is $300V$ peak value, with an initial angle of 15° and the frequency is 50 Hz . then,

① determine the impedance

② determine the current.

⇒ In given:

$$R = 15\Omega$$

$$X = 22\Omega$$

$$V_m = 300V$$

$$\theta = 15^\circ$$

$$f = 50\text{ Hz}$$

①

We know:

$$\text{Impedance } Z = R + jX$$

$$= 15 + j22\Omega$$

∴ in polar:

$$Z = 26.627 \angle 55.71^\circ \Omega$$

②

We know:

$$\omega = 2\pi f = 2\pi \times 50 = 100\pi = 314.159 \text{ rad/s}$$

∴ Instantaneous Voltage, $V = V_m \sin(\omega t + \theta)$

$$= 300 \sin(314.159t + 15^\circ) \text{ Volt}$$

∴ polar form of voltage:

$$V = V_{rms} \angle \theta$$

$$= \frac{V_m}{\sqrt{2}} \angle 15^\circ = \frac{300}{\sqrt{2}} \angle 15^\circ = 212.13 \angle 15^\circ \text{ Volt}$$

$$\therefore \text{Current } I = \frac{V}{Z}$$

$$= \frac{212.13 \angle 15^\circ}{26.626 \angle 55.71^\circ}$$

$$= 7.967 \angle (15 - 55.71^\circ)$$

$$= 7.967 \angle -40.71^\circ \text{ A}$$

power or

(Average or Real or True or Wattfull power)

Instantaneous power:

$$P(t) = P - P \cos 2\omega t + P \sin 2\omega t \text{ [W]}$$

$\Rightarrow$ The average values of $\cos$ and $\sin$ to the equation = 0

So the average value of total instantaneous power equal to the Constant term which is called power.

$$\therefore \boxed{P = P_{\text{re}} = \frac{V_m I_m}{2} \cos \theta = V_{\text{rms}} I_{\text{rms}} \cos \theta = VI \cos \theta \text{ Watt}}$$

Note:-

- ① This real power represents the power which is consumed by load (or resistor)
- ② The real power is the power which converts from electrical energy to other form of energy.
- ③ The real power is measured by wattmeter.

Reactive or imaginary or Quadrature or Wattless Power (or Reactive Volt-Ampere)

⇒ The maximum value of instantaneous Reactive power is called the reactive power. denote by Q and P_R Unit: var

$$\therefore Q = P_R = \frac{V_m I_m}{2} \sin \theta = V_{rms} I_{rms} \sin \theta = VI \sin \theta$$

Note:-

- ① The reactive power depends of the the angle of $\sin$.
 → It may be positive (for capacitor)
 → It may be negative (for inductor)

- ② The reactive power is measured by Varmeter.

Volt-Ampere or Apparent power:-

⇒ The volt-ampere or apparent power represents the maximum possible supply power by a source. denote S Unit in VA

$$\therefore S = \sqrt{P_R^2 + P_A^2} = \sqrt{P^2 + Q^2} = \frac{V_m I_m}{2} = V_{rms} I_{rms} = VI \quad \text{VA}$$

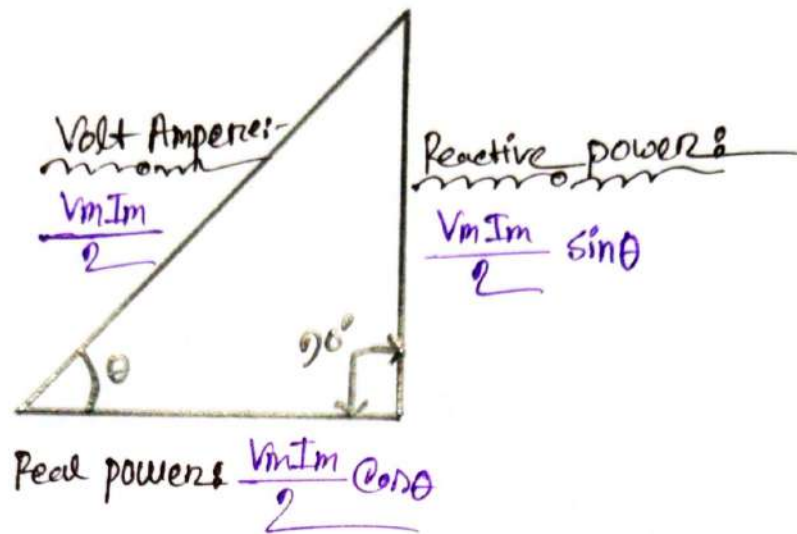
power factor:-

⇒ Cos θ which is a factor, by which volt-ampere are multiplied to given power is called power factor denote by pf

$$\therefore pf = \cos \theta = \frac{P}{\frac{V_m I_m}{2}} = \frac{P}{V_{rms} I_{rms}} = \frac{P}{S}$$

Note:- power factor is always positive.

Power Triangle:-



We know:-

$$V_{rms} = \frac{V_m}{\sqrt{2}}$$

$$I_{rms} = \frac{I_m}{\sqrt{2}}$$

$$\therefore V_{rms} \times I_{rms} = \frac{V_m}{\sqrt{2}} \times \frac{I_m}{\sqrt{2}} = \frac{V_m I_m}{2}$$

Reactive factor:-

$\Rightarrow \sin \theta$ which is a factor by which volt-amperes are multiplied to given reactive power is called reactive factor:-
denote by rf

$$\therefore rf = \sin \theta = \frac{P}{\frac{V_m I_m}{2}} = \frac{P}{V_{rms} I_{rms}} = \frac{P}{S}$$

power Calculation:-

⇒ power for AC circuits can be defined as a complex entity as well as impedance.

The Apparent power:-

$$S = P + jQ = \sqrt{P^2 + Q^2} \angle \theta = \tan^{-1}\left(\frac{Q}{P}\right)$$

in rectangular in polar

where $S = VI = \frac{V_m I_m}{2}$ VA

S is called apparent power in VA (Volt-ampere)

Real or active or wattful power:-

⇒ P in W (watts) is considered as the real or active or wattful power.

$$P = VI \cos \theta = \frac{V_m I_m}{2} \cos \theta \quad \text{in W}$$

Reactive or wattless power:-

$$Q = VI \sin \theta = \frac{V_m I_m}{2} \sin \theta \quad \text{in VAR}$$

Note:- Real power (P) is measured by wattmeter and Q is measured by Varimeter.

power factor:-

Here, $\cos\theta$ is called the power factor (Cosine of angle between Voltage and Current)

$$\therefore \text{power factor pf} = \cos\theta = \frac{P}{S} \text{ watt/VA}$$

Reactive factor:-

$\Rightarrow \sin\theta$ is called the reactive factor (Sine of Angle between Voltage and current).

$$\therefore \text{reactive factor} \sin\theta = \frac{Q}{S} \text{ VAR/VA}$$

pure Resistive load:-

We know:-

Euler's Identity:-

$$e^{j\theta} = \cos\theta + j\sin\theta$$

if $\theta=0$

$\rightarrow$ I and V are in phase

$$\text{So, Impedance } Z = R \quad \Omega$$

$$\text{Current } I = \frac{V}{Z} \quad A$$

$$A. \text{ power } S = P \text{ in Watt}$$

$$\text{power factor, } \cos\theta = 1$$

$$\text{reactive factor } \sin\theta = 0$$

(No Reactance,
No reactive
power)

$$\begin{aligned} S &= P + jQ \\ &= P + j0 \\ S &= P \end{aligned}$$

Note:- Hence, Resistive load are also called unity pf loads.

$$\text{Real power } P = V_{rms} I_{rms} = \frac{V_{rms}^2}{R} = \frac{I_{rms}^2 R}{1} \text{ in watt}$$

pure capacitive load:-

We know:-

Euler's identity:-

$$e^{j\theta} = \cos\theta + j\sin\theta$$

if $(\theta = 90^\circ)$

$\Rightarrow$ I leads V by 90° ($\theta = 90^\circ$) (pf is zero, reactive factor is +)

So, Impedance; $Z = -jX_C$ Ω

Capacitive reactance $X_C = \frac{1}{2\pi f C}$ Ω

Current $I = \frac{V}{Z}$ A

Power $S = -j\phi$ VAR

$$\begin{aligned} S &= P - j\phi \\ &= 0 - j\phi \\ &= -j\phi \end{aligned}$$

pure inductive load:-

if $(\theta = 90^\circ)$

$\Rightarrow$ I lags V by 90° ($\theta = 90^\circ$) (pf is zero, reactive factor is +)

So, Impedance; $Z = jX_L$ Ω

Inductive reactance $X_L = 2\pi f L$ Ω

Current $I = \frac{V}{Z}$ A

Power $S = P + j\phi$
 $= 0 + j\phi = j\phi$ VAR

problem:-

⇒ the voltage and current of a circuit are given as follows $v(t) = 100 \sin(314t + 60^\circ) \text{ V}$ and $i(t) = 10 \sin(314t + 30^\circ) \text{ A}$

- (i) Calculate the apparent power, the power or Active power, the reactive power, the power factor, the reactive factor.
- (ii) Write the expression of instantaneous real power, the expression of instantaneous reactive power, the instantaneous expression of total power.
- (iii) Draw the power triangle.

⇒ In given:-

$$V(t) = 100 \sin(314t + 60^\circ) \text{ V}$$

$$i(t) = 10 \sin(314t + 30^\circ) \text{ A}$$

here, $V_m = 100 \text{ V}$

$$I_m = 10 \text{ V}$$

$$\theta_v = 60^\circ$$

$$\theta_i = 30^\circ$$

$$\therefore \theta = \theta_2 = \theta_v - \theta_i = 60^\circ - 30^\circ = 30^\circ$$

①

We know:-

$$\text{Apparent power } S = \frac{V_m I_m}{2} = \frac{100 \times 10}{2} = 500 \text{ VA}$$

$$\text{Power factor pf} = \cos \theta = \cos 30^\circ = \frac{\sqrt{3}}{2} = 0.866 \text{ watt/VA}$$

$$\text{Reactive factor rf} = \sin \theta = \sin 30^\circ = \frac{1}{2} = 0.5 \text{ VAR/VA}$$

$$\text{The power / Active power } P = \frac{V_m I_m}{2} \times \cos \theta = 500 \cos 30^\circ = 433.01 \text{ W}$$

$$\text{Reactive power } Q = \frac{V_m I_m}{2} \sin \theta = 500 \sin 30^\circ = 250 \text{ VAR}$$

(ii)

we know:-

Instantaneous expression for real power:-

$$P = S \cos \theta$$

$$P = 500 \cos 30^\circ \text{ W}$$

Instantaneous expression for reactive power:-

$$Q = S \sin \theta$$

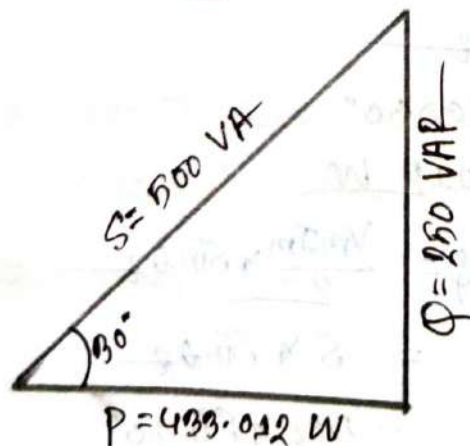
$$Q = 500 \sin 30^\circ \text{ VAR}$$

Instantaneous expression for total power:-

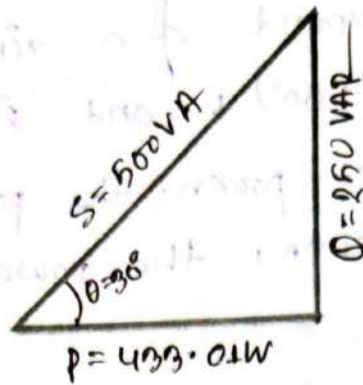
$$P(\theta) = P - P \cos 2\omega t + Q \sin 2\omega t \text{ [W]}$$

$$= 433.012 - 433.012 \cos 2\omega t + 250 \sin 2\omega t \text{ [W]}$$

(ii)



iii



power Triangle.

Problem:-

$\Rightarrow$ The voltage of $V(t) = 100 \sin(2\pi 4t - 60^\circ)$ V applied to an impedance of $Z = (8.66 + j5) \Omega$.

i) Calculate the apparent power, the power, the reactive power, the power factor, the reactive factor

ii) Draw the power triangle.

In given:-

$$V(t) = 100 \sin(2\pi 4t - 60^\circ) \text{ V}$$

$$Z = (8.66 + j5) \Omega$$

$$\therefore V = V_{rms} \angle -60^\circ$$

$$= \frac{V_m}{\sqrt{2}} \angle -60^\circ$$

$$= \frac{100}{\sqrt{2}} \angle -60^\circ$$

$$V = 70.71 \angle -60^\circ$$

$$Z = |Z| \angle \tan^{-1}\left(\frac{5}{8.66}\right)$$

$$Z = 10 \angle 30^\circ$$

$$|Z| = \sqrt{(8.66)^2 + (5)^2}$$

$$= 10$$

$$\theta = \tan^{-1}\left(\frac{5}{8.66}\right)$$

$$= 30^\circ$$

we know:-

$$\begin{aligned}\therefore I &= \frac{V}{Z} = \frac{70.71 \angle -60}{10 \angle 30} \\ &= \frac{70.71}{10} \angle (-60 - 30) \\ &= 7.071 \angle -90^\circ \text{ A}\end{aligned}$$

$$\begin{aligned}\therefore V_m &= 100 \text{ V}, V_{rms} = 70.71 \text{ V}, \theta_v = -60^\circ \\ I_{rms} &= 7.071 \text{ A} \quad \theta_i = -90^\circ, \theta_z = \theta_v - \theta_i = 30^\circ\end{aligned}$$

(i)

We know:-

$$\begin{aligned}\text{Apparent power } S &= \frac{V_m I_m}{2} = \frac{100 \times 7.071}{2} = \\ &= V_{rms} \times I_{rms} \\ &= 70.71 \times 7.071 \\ &= 500 \text{ VA}\end{aligned}$$

$$\begin{aligned}\text{The power } P &= \frac{V_m I_m}{2} \times \cos \theta_z \\ &= S \cos \theta_z \\ &= 500 \times \cos 30^\circ \\ &= 433.012 \text{ W}\end{aligned}$$

$$\begin{aligned}\text{The reactive power } Q &= \frac{V_m I_m}{2} \times \sin \theta_z \\ &= S \times \sin \theta_z \\ &= 500 \times \sin 30^\circ \\ &= 250 \text{ VAR}\end{aligned}$$

$$\text{The power factor p.f} = \cos \theta = \frac{P}{S} = \frac{433.012}{500} = 0.866 \text{ W/VA}$$

$$\text{The reactive factor r.f} = \sin \theta = \frac{Q}{S} = \frac{250}{500} = 0.5 \text{ VAR/VA}$$

problem: 10.6:—

⇒ An electrical device is rated 5kVA, 100V at 0.6 power factor lag. What is the impedance of the device in rectangular coordinates?

⇒ is given:—

$$S = 5000 \text{ VA}$$

$$p.f. = 0.6 \text{ Watt/VA}$$

$$V = 100 \text{ V}$$

$$\theta_v = 0^\circ$$

We know:—

$$S = \frac{V_m I_m}{2} = V_{rms} I_{rms} = VI$$

$$\therefore I = \frac{S}{V} = \frac{5000}{100} = 50 \text{ A}$$

$$p.f. = \cos \theta$$

$$\therefore \theta = \cos^{-1}\left(\frac{p.f.}{1}\right) = \cos^{-1}(0.6) = 53.13^\circ$$

Since the p.f. is lagging, the circuit is inductive. that means.

I lags V by 53.13°

$$\therefore \text{Polar form of } V = V_{rms} \angle 0^\circ = 100 \angle 0^\circ$$

$$\therefore \text{Polar form of } I = I_{rms} \angle -53.13^\circ = 50 \angle -53.13^\circ$$

Now

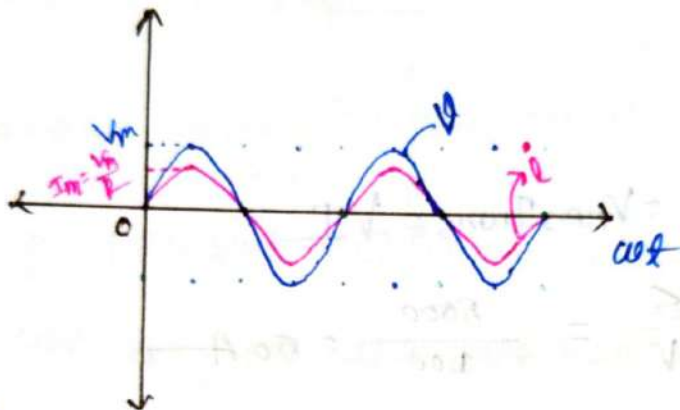
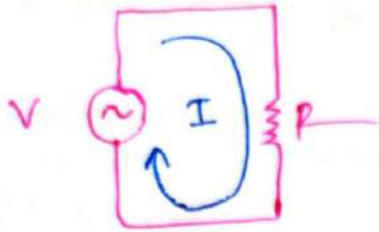
$$\text{Impedance } Z = \frac{V}{I} = \frac{100 \angle 0^\circ}{50 \angle -53.13^\circ} = 2 \angle 53.13^\circ$$

$$\therefore \text{Rectangular form } Z = (1.2 + j1.6) \Omega$$

Ans:

Resistive Circuit:-

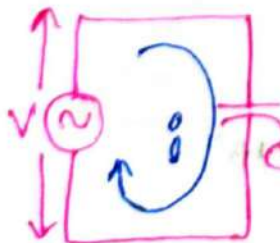
⇒ For resistive circuit, voltage and current in phase.
(zero phase difference or $\theta = 0^\circ$)



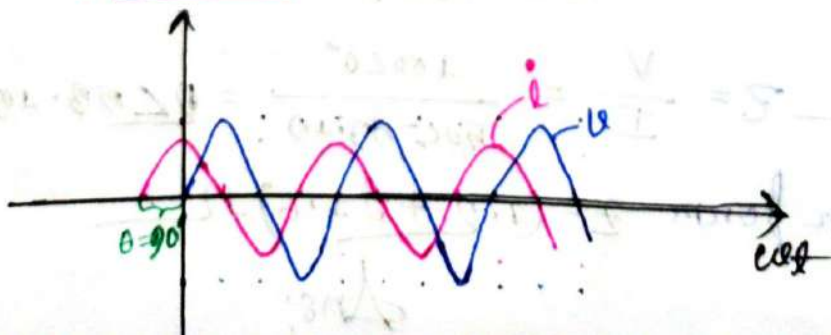
here, $V = V_m \sin(\omega t + 0^\circ)$
 $I = I_m \sin(\omega t + 0^\circ)$

pure Capacitive Circuit

⇒ For a pure capacitive branch I leads V by 90° .

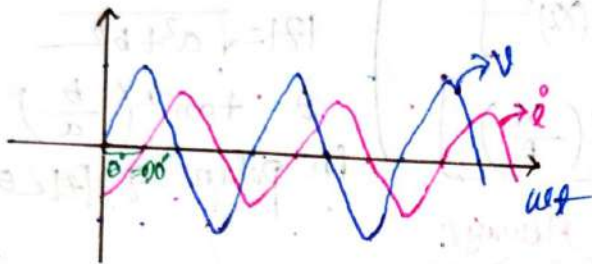
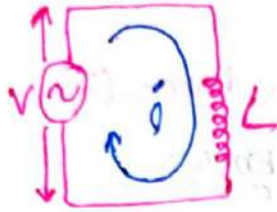


here, $V = V_m \sin(\omega t + \theta)$
 $I = I_m \sin(\omega t + \theta)$
 $\theta = 90^\circ$



pure inductive circuit:-

⇒ For a pure inductive branch I lags Voltage by 90°

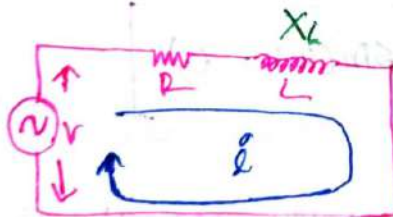


here,

$$V = V_m \sin(\omega t + 0^\circ)$$

$$I = I_m \sin(\omega t - 90^\circ)$$

R-L circuit:-



⇒ The ohmic value of inductor and capacitor is considered as reactance (X) in ohm

$$\therefore \text{Instantaneous equation } V = V_m \sin(\omega t + 0^\circ)$$

$$\therefore \text{Polar form } V = V_m \angle 0^\circ \text{ (ref. volt)}$$

$$= \frac{V_m}{\sqrt{2}} \angle 0^\circ$$

Reactance for inductive branch:-

$$\boxed{\text{Inductive Reactance } X_L = 2\pi f L \Omega}$$

Total ohmic entity is represented as impedance (Z)

$$\boxed{\therefore \text{Impedance } Z = R + jX_L \Omega}$$

$$\boxed{\therefore \text{Value of impedance } |Z| = \sqrt{R^2 + (X_L)^2}}$$

$$\boxed{\therefore \text{Angle of impedance } \theta = \tan^{-1}\left(\frac{X_L}{R}\right)}$$

Note:- $0 < \theta < 90^\circ$ Always

$$\boxed{\text{Current } I = \frac{V}{Z} = \frac{V_{rms} \angle 0^\circ}{|Z| \angle \theta} \text{ A}}$$

$$= \frac{V_{rms}}{|Z|} \angle (0^\circ - \theta)$$

$$= I_{rms} \angle -\theta$$

$$\boxed{\therefore \text{Instantaneous equation } i = I_m \sin(\omega t - \theta)}$$

Note:- Impedance Angle are exactly opposite to the current angle.

we know:-

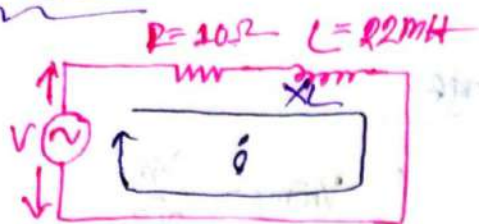
$$Z = a + jb$$

$$|Z| = \sqrt{a^2 + b^2}$$

$$\theta = \tan^{-1}\left(\frac{b}{a}\right)$$

in polar $Z = |Z| \angle \theta$

#problem: 1



⇒ For the circuit above, if the supply voltage is 220V rms, 50Hz, then determine.

- i) Reactance of inductor.
- ii) Impedance.
- iii) Current.
- iv) Sketch the voltage and current.

⇒ In given:

$$R = 10\Omega$$

$$L = 22mH$$

$$V = 220\angle 0^\circ \text{ (ref) Volt [or } V = V_{rms} \angle \theta]$$

$$f = 50Hz$$

$$V_{rms} = 220V$$

We know:

(i)

$$\begin{aligned} \text{Reactance of inductor } X_L &= 2\pi f L \\ &= 2\pi \times 50 \times 22 \times 10^{-3} \\ &= 6.911\Omega \end{aligned}$$

(ii)

$$\begin{aligned} \text{Impedance } Z &= R + jX_L \\ &= 10 + j6.911 \end{aligned}$$

(iii)

$$\begin{aligned} \text{Current } I &= \frac{V}{Z} = \frac{V_{rms} \angle 0^\circ}{|Z| \angle \theta} \\ &= \frac{220}{12.15} \angle (0^\circ - 34.64^\circ) \\ &= 18.106 \angle -34.64^\circ \end{aligned}$$

$$\begin{aligned} |Z| &= \sqrt{(10)^2 + (6.911)^2} = 12.15 \\ \theta &= \tan^{-1}\left(\frac{6.911}{10}\right) = 34.64^\circ \end{aligned}$$

(iv)

we know:-

Instantaneous equation for voltage:-

$$V(t) = V_m \sin(\omega t + \theta)$$

$$\therefore V(t) = 220\sqrt{2} \sin(\omega t + 0^\circ)$$

$$V_{rms} = \frac{V_m}{\sqrt{2}}$$

$$\therefore V_m = V_{rms} \times \sqrt{2} \\ = 220\sqrt{2}$$

Instantaneous equation for current:-

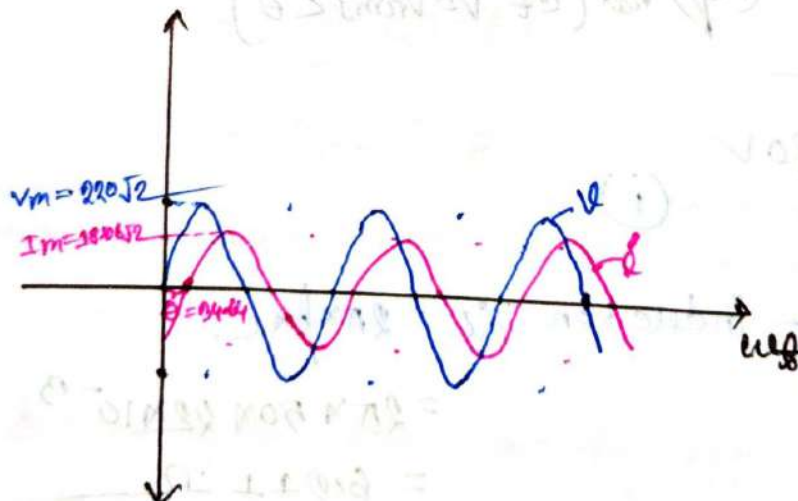
$$i(t) = I_m \sin(\omega t - \theta)$$

$$\therefore i(t) = 18.106\sqrt{2}(\omega t - 34.64^\circ)$$

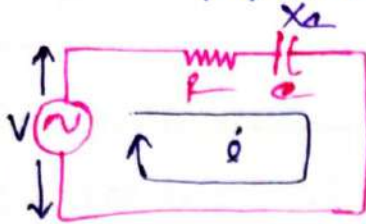
$$I_{rms} = \frac{I_m}{\sqrt{2}}$$

$$\therefore I_m = I_{rms} \times \sqrt{2} \\ = 18.106\sqrt{2}$$

So, sketch is:-



R-C circuit



∴ Instantaneous equation $i = I_m \sin(\omega t - \theta)$

∴ in polar form $V = V_{rms} \angle 0^\circ$ (ref. volt)
 $= \frac{V_m}{\sqrt{2}} \angle 0^\circ$

∴ Reactance for Capacitive branch

$$\text{Capacitive reactance } X_c = \frac{1}{2\pi f C} \Omega$$

Now total ohmic entity is represented as Impedance (Z)

$$\therefore \text{Impedance } Z = R - jX_c \Omega$$

$$\text{magnitude } |Z| = \sqrt{R^2 + (X_c)^2} \Omega$$

$$\text{Angle of Impedance } \theta = \tan^{-1}\left(\frac{-X_c}{R}\right)$$

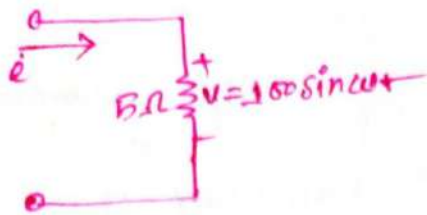
$$\text{Current } I = \frac{V}{Z} = \frac{V_{rms} \angle 0^\circ}{|Z| \angle -\theta}$$

$$= \frac{V_{rms}}{|Z|} \angle 0 - (-\theta) \quad I$$

$$= I_{rms} \angle \theta$$

$$\therefore \text{Instantaneous equation } i = I_m \sin(\omega t + \theta)$$

Problem 1.1:



- (a) Find the current i_R
 (b) Sketch the waveform for i_R and v_R

Given:

$$R = 5 \Omega$$

$$V = 100 \sin \omega t + 0^\circ$$

$$\therefore V_m = 100$$

$$\therefore V_{rms} = \frac{V_m}{\sqrt{2}} = \frac{100}{\sqrt{2}} = 70.71 \text{ V}$$

$$\therefore V = V_{rms} \angle 0^\circ$$

$$= 70.71 \angle 0^\circ$$

We know:

For a pure resistive load:

$$Z = R$$

$$= 5 \Omega$$

$$\therefore |Z| = \sqrt{R^2} = 5$$

$$\theta = \tan^{-1} \left(\frac{0}{5} \right) = 0^\circ$$

$$\therefore \text{Current } I_R = \frac{V_R}{Z_R} = \frac{V_{rms} \angle \theta}{|Z| \angle \theta}$$

$$= \frac{70.71 \angle 0^\circ}{5 \angle 0^\circ}$$

$$= 14.142 \angle 0^\circ$$

$$\therefore i_R = I_m \sin(\omega t + \theta)$$

$$= \sqrt{2} \times 14.142 \sin(\omega t + 0^\circ) \text{ A}$$

Ans:

$$\left. \begin{aligned} I_m &= I_{rms} \sqrt{2} \\ &= \sqrt{2} \times 14.142 \end{aligned} \right\}$$

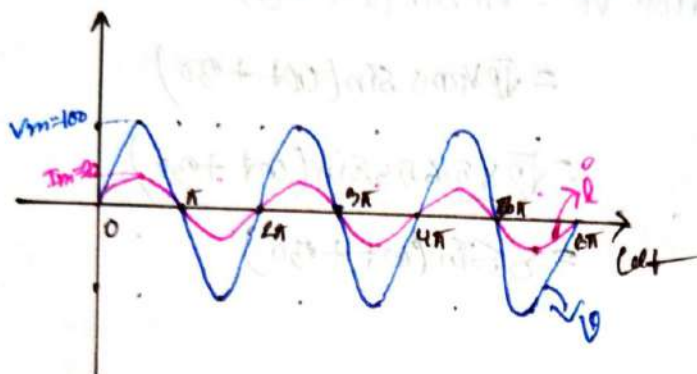
ii

we have:
mon

$$i_p = 20 \sin \omega t$$

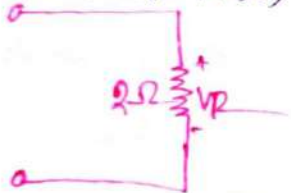
$$V_L = 100 \sin(\omega t)$$

1. Sketch is:
mon



problem 15.2
mon

$$i_p = 4 \sin(\omega t + 30^\circ)$$



- i Find the voltage V_p
- ii Sketch the wave forms of V_p and i_p

=> Is given:
mon

$$i_p = 4 \sin(\omega t + 30^\circ) = \frac{4}{\sqrt{2}} \angle 30^\circ$$

$$R = 2 \Omega$$

here, $I_m = 4$
 $\theta_i = 30^\circ$

we know:
mon

For a pure resistive load
mon
 $\theta_v = \theta_i = 30^\circ$

$$\therefore \theta_z = \theta_v - \theta_i = 30^\circ - 30^\circ = 0^\circ$$

$$\text{Impedance } Z = R = 2 \Omega$$

$$\text{in polar } Z = |Z| \angle \theta$$

$$= 2 \angle 0^\circ$$

$$|Z| = \sqrt{(2)^2}$$

$$= 2$$

Now, we know:-

$$\begin{aligned}
 V = IZ &= I_{\text{rms}} \angle \theta_i \times |Z| \angle \theta_z \\
 &= 2\sqrt{2} \angle 30^\circ \times 2 \angle 0^\circ \\
 &= 2\sqrt{2} \times 2 \angle 30^\circ + 0 \\
 &= 5.656 \angle 30^\circ
 \end{aligned}$$

$\therefore$ Instantaneous equation $V_P = V_m \sin(\omega t + \theta)$

$$= I_{\text{rms}} \sin(\omega t + 30^\circ)$$

$$= \sqrt{2} \times 5.656 \sin(\omega t + 30^\circ)$$

$$= 8 \sin(\omega t + 30^\circ)$$

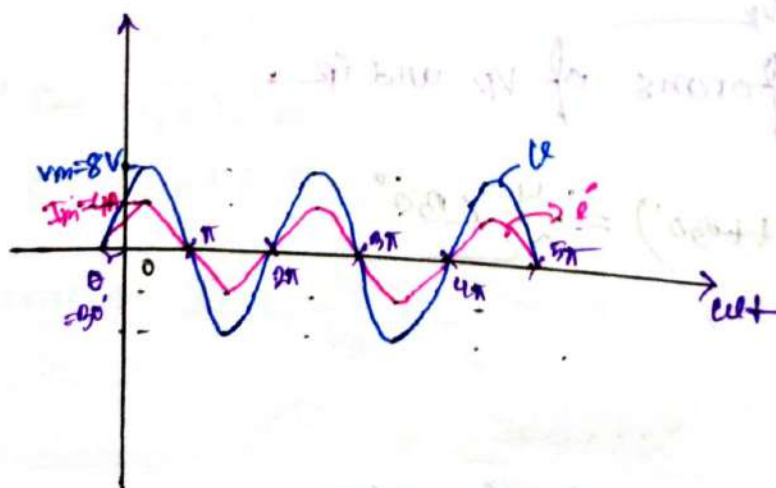
ii

we have:-

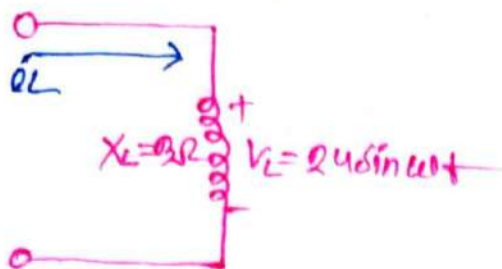
$$V_P = 8 \sin(\omega t + 30^\circ)$$

$$I_P = 4 \sin(\omega t + 30^\circ)$$

$\therefore$ Sketch:-



problem 15.2:



(a) Find the current i_L .

(b) Sketch the v_L and i_L curves.

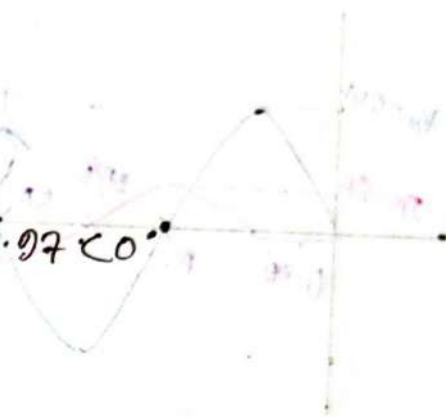
⇒ in given:

$$V_L = 24 \sin \omega t = \frac{24}{\sqrt{2}} \angle 0^\circ = 16.97 \angle 0^\circ$$

$$X_L = j3 \Omega$$

$$\therefore V_m = 24$$

$$\theta_v = 0^\circ$$



We know:

(a)

For pure inductive circuit:-

impedance $Z = jX_L$

$$\therefore Z = j3 \Omega$$

$$\therefore Z = 3 \angle 90^\circ$$

$$\left\{ \begin{array}{l} |Z| = \sqrt{3^2} = 3 \\ \theta = \tan^{-1}\left(\frac{3}{0}\right) \\ = 90^\circ \end{array} \right.$$

$$\therefore \text{Current } I_L = \frac{V_L}{Z_L}$$

$$= \frac{16.97 \angle 0^\circ}{3 \angle 90^\circ}$$

$$= 5.656 \angle -90^\circ$$

$$\therefore i_L = I_m \sin(\omega t + \theta)$$

$$= \sqrt{2} \times 5.656 \sin(\omega t - 90^\circ)$$

Ans:-

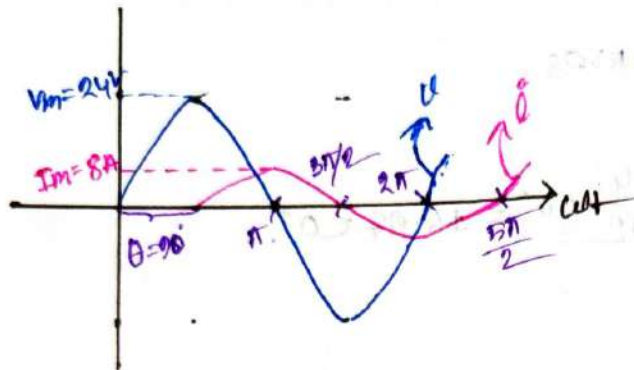
(b)

We have,

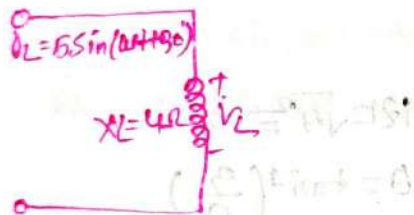
$$i_L = 8 \sin(\omega t - 90^\circ)$$

$$v_L = 24 \sin(\omega t + 0^\circ)$$

$\therefore$ Sketch:



#problem: 15.41-



(a) Find the voltage v_L

(b) sketch the v_L and i_L curves

$\Rightarrow$ In given:

$$i_L = 5 \sin(\omega t + 30^\circ) = \frac{5}{\sqrt{2}} \angle 30^\circ$$

$$I_m = 5A$$

$$X_L = 4\Omega$$

$$\theta_i = 30^\circ$$

We know,

(a)

For pure inductive load:

$$\text{Impedance } Z = +jX_L \Omega \left(\theta = \tan^{-1}\left(\frac{4}{0}\right) = 90^\circ \right)$$

$$Z = |Z| \angle \theta$$

$$= 4 \angle 90^\circ$$

$$|Z| = \sqrt{4^2} = 4$$

Now,

$$\begin{aligned} V &= IZ = \frac{5}{\sqrt{2}} \angle 30^\circ \times 4 \angle 0^\circ \\ &= \left(\frac{5}{\sqrt{2}} \times 4 \right) \angle (0^\circ + 30^\circ) \\ &= 14.14 \angle 30^\circ \end{aligned}$$

$$\begin{aligned} \therefore v(t) &= V_m \sin(\omega t + \theta) \\ &= \sqrt{2} \times 14.14 \sin(\omega t + 30^\circ) \\ &= 20 \sin(\omega t + 30^\circ) \end{aligned}$$

Ans:-

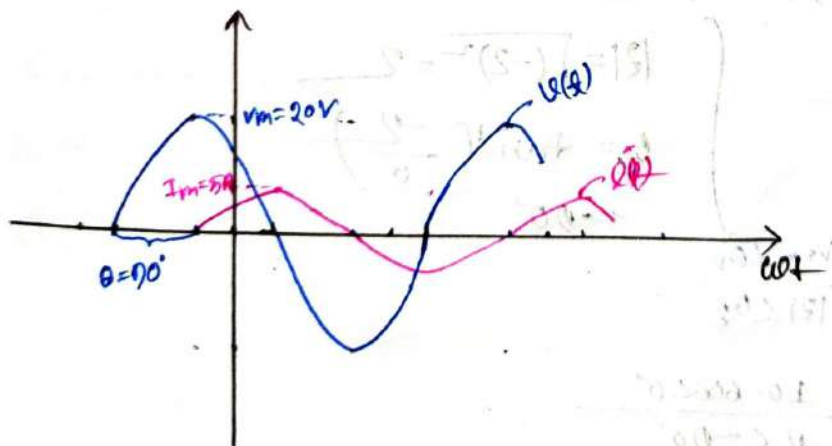
(b)

We have:-

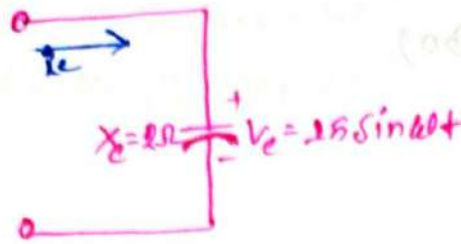
$$v(t) = 20 \sin(\omega t + 30^\circ)$$

$$i(t) = 5 \sin(\omega t + 0^\circ)$$

$\therefore$ Sketch:-



problems 15.50



(a) Find the current i_e .

(b) Sketch the v_e and i_e .

⇒ In given:—

$$X_e = 2 \Omega$$

$$v_e = 15 \sin 40t$$

$$\therefore V_e = \frac{15}{\sqrt{2}} \angle 0^\circ = 10.606 \angle 0^\circ$$

$$V_m = 15 \text{ V}$$

$$\theta_v = 0^\circ$$

We know:—

For pure conductive loads:—

$$Z = -jX_e$$

$$= -j2 \Omega$$

$$Z = 2 \angle -90^\circ$$

$$|Z| = \sqrt{(-2)^2} = 2$$

$$\theta_Z = \tan^{-1}\left(\frac{-2}{0}\right)$$

$$= -90^\circ$$

$$\therefore I_e = \frac{V_e}{Z} = \frac{V_m \angle \theta_v}{|Z| \angle \theta_Z}$$

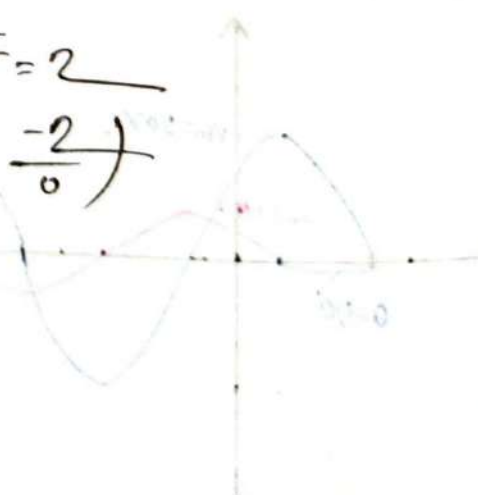
$$= \frac{10.606 \angle 0^\circ}{2 \angle -90^\circ}$$

$$= 5.303 \angle 0 - (-90^\circ)$$

$$= 5.303 \angle 90^\circ$$

$$\therefore i(t) = I_m \sin(\omega t + \theta)$$

$$= \sqrt{2} \times 5.303 \sin(\omega t + 90^\circ)$$

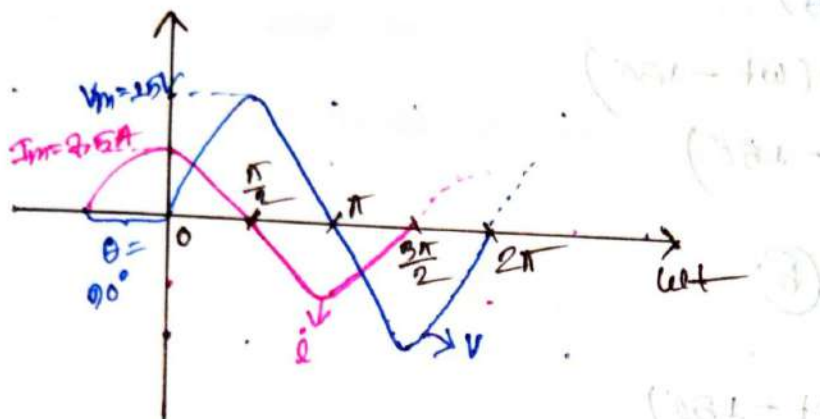
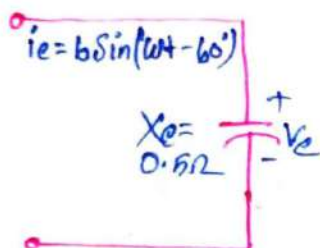


(b)

In given:

$$v_e = 15 \sin(\omega t + 0^\circ)$$

$$i_e = 7.5 \sin(\omega t + 90^\circ)$$

∴ Sketch:# problems 13.6:① Find the voltage v_s .② Sketch the v_e and i_e curves.⇒ In given:

$$X_e = 0.5\Omega$$

$$i_e = 6 \sin(\omega t - 60^\circ) = \frac{6}{\sqrt{2}} \angle -60^\circ$$

$$\theta_i = -60^\circ$$

$$I_m = 6 \text{ A}$$

We know:

(a)

For a pure conductive load:

$$Z = -jX_e$$

$$= -j0.5\Omega$$

$$Z = |Z| \angle \theta$$

$$= 0.5 \angle -90^\circ$$

$$\left(\begin{array}{l} |Z| = \sqrt{(-0.5)^2} = 0.5 \\ \theta = -\tan^{-1}\left(\frac{0.5}{0}\right) \\ = -90^\circ \end{array} \right.$$

$$\therefore V_e = Z_0 I_e$$

$$= 0.5 \angle -90^\circ \times 4.242 \angle -60^\circ$$

$$= (0.5 \times 4.242) \angle -60^\circ - 90^\circ$$

$$= 2.121 \angle -150^\circ$$

$$\therefore V_e = V_m \sin(\omega t + \theta)$$

$$= \sqrt{2} (2.121) \sin(\omega t - 150^\circ)$$

$$= 3 \sin(\omega t - 150^\circ)$$

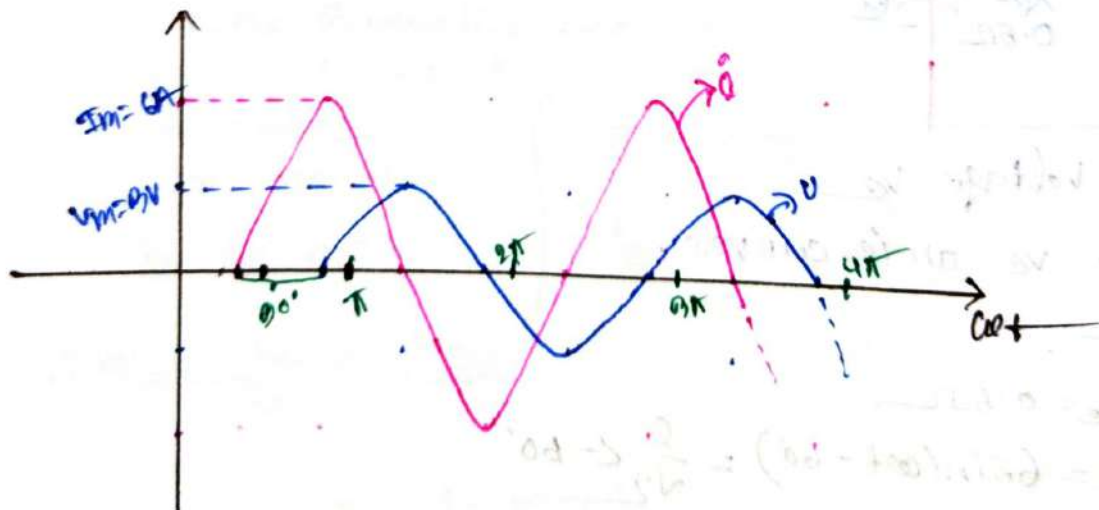
(b)

we have:

$$V(t) = 3 \sin(\omega t - 150^\circ)$$

$$i(t) = 6 \sin(\omega t - 60^\circ)$$

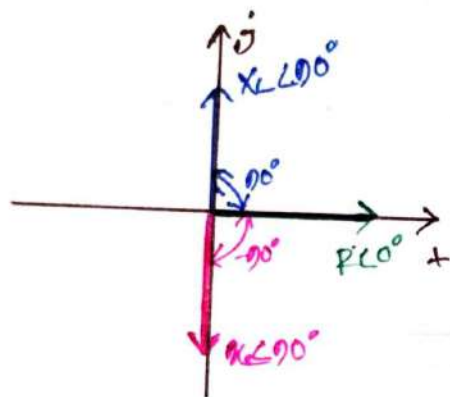
i. Sketch:



Impedance diagram:-

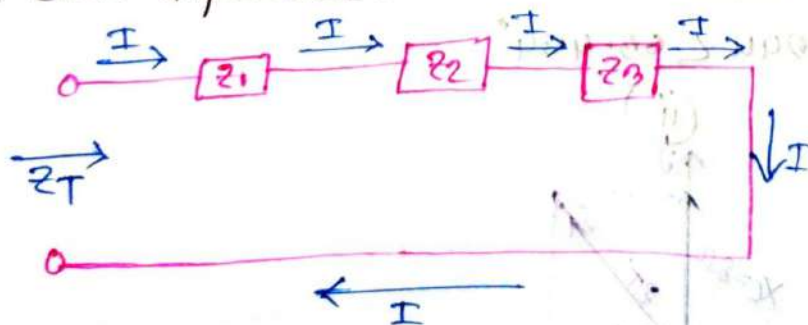
For any network

- ① resistance will Always appear on the positive real axis
- ② Inductive reactance on the positive imaginary axis,
- ③ Capacitive reactance on the negative imaginary axis.



Series Configuration:-

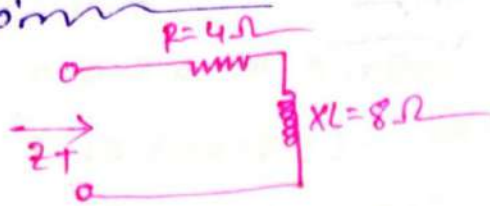
⇒ For instance, the total impedance of a system is the sum of the individual impedances. And The Current I is the same through each impedance.



$$\therefore Z_T = Z_1 + Z_2 + Z_3$$

Note:- The overall properties of series ac circuit are the same as those for DC circuit.

problem: 15.2:-



- (i) Find Z_T
- (ii) Draw the Impedance Diagram.

⇒ In given:-

$$R = 4\Omega$$

$$X_L = 8\Omega$$

we know:-

For Resistive Load:-

$$Z = R$$

$$\therefore Z_1 = 4\Omega$$

For Inductive Load:-

$$Z = jX_L$$

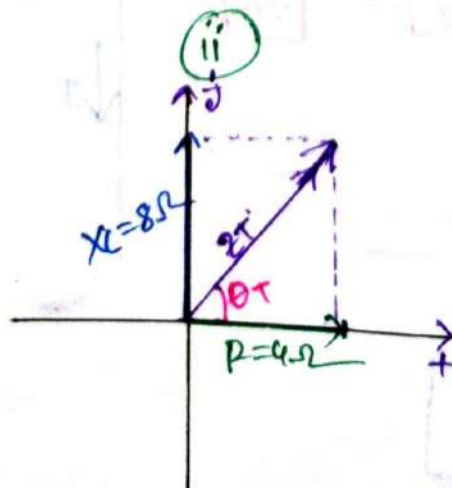
$$\therefore Z_2 = j8\Omega$$

Total Impedance:-

$$Z_T = Z_1 + Z_2$$

$$Z = (4 + j8)\Omega$$

$$\text{OR } Z_T = 8.044 \angle 63.434^\circ$$



problem 8

⇒ A voltage $141.4 \sin 314t$ V is applied to R-L series circuit which consists $R = 3 \text{ ohms}$, $L = 12.7 \text{ mH}$

(a) Draw the circuit diagram.

(b) Calculate, (i) the inductive reactance

(ii) the impedance

(iii) the current

(iv) voltage drop across the resistance and inductance

(v) the power factor, reactive factor,

(vi) the power, reactive power, apparent power.

(vii) Verify the KVL

(c) write the instantaneous expression of current, voltage drop across the resistor and across the inductance.

(d) Draw the impedance diagram, phasor diagram and power triangle.

⇒ In given:

$$V(t) = 141.4 \sin 314t \text{ V} \quad V_{\text{rms}} = \frac{141.4}{\sqrt{2}}$$

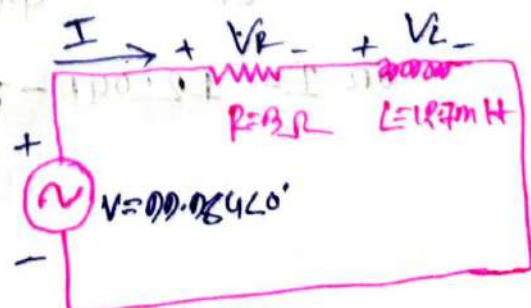
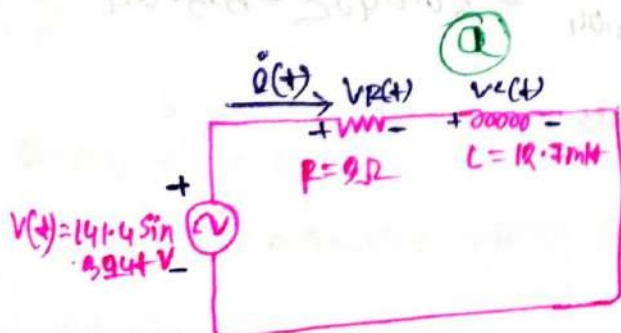
$$\therefore V_m = 141.4 \text{ V}$$

$$\omega = 314 \text{ rad/s}$$

$$R = 3 \Omega$$

$$L = 12.7 \text{ mH}$$

$$\therefore V = 100.084 \angle 0^\circ$$



(b)

(i)

we know,

$$X_L = 2\pi f \times L = \omega L = 314 \times 12.7 \times 10^{-3} \\ = 3.987 \Omega$$

(ii)

we know,

$$Z = Z_1 + Z_2 \quad \text{--- (1)}$$

For Resistor R:

$$Z_1 = R = 3 \Omega$$

For Inductance L:

$$Z_2 = +jX_L = +j3.987 \Omega$$

∴ Impedance,

$$Z_T = R + jX_L \\ = 3 + j3.987 \Omega$$

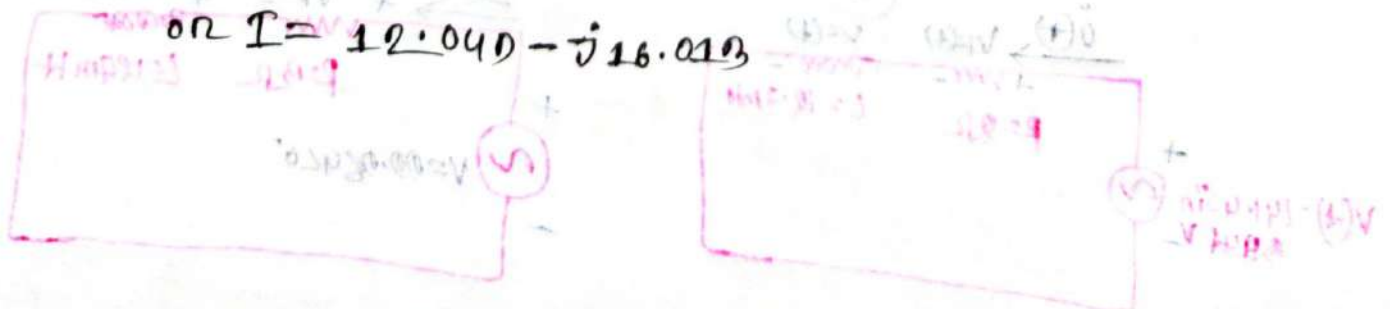
$$\text{or } Z_T = 4.987 \angle 53.04^\circ \Omega$$

(iii)

we know,

$$I = \frac{V}{Z_T} = \frac{100 \angle 0^\circ}{4.987 \angle 53.04^\circ} = 20.04 \angle -53.04^\circ$$

$$\text{or } I = 12.04 - j16.013$$



(iv)

Voltage Drop:-Across the Resistance $V_R = IZ_R$

$$= (20.040 \angle -53.04^\circ) \times (3 \angle 0^\circ)$$

$$= 60.12 \angle -53.04^\circ \text{ V}$$

Across the Inductor $V_L = IZ_L$

$$= (20.040 \angle -53.04^\circ) \times (9.087 \angle 90^\circ)$$

$$= 182.800 \angle 36.96^\circ \text{ V}$$

(v)

we know:-

$$p.f. = \cos \theta = \cos \theta_2 = \cos 53.04 = 0.601 \rightarrow \text{lagging}$$

$$r.f. = \sin \theta = \sin \theta_2 = \sin 53.04 = 0.799$$

(vi)

we know:-

$$\text{the power} = S \cos \theta_2 = 1204.721 \text{ W}$$

$$\text{The reactive power} = S \sin \theta_2 = 963.67 \times \sin 53.04 = 768.05 \text{ Var}$$

$$\text{The Apparent power} = VI = 22.984 \times 20.04 = 4603.67 \text{ VA}$$

(vii)

Verify KVL:-

$$V = V_R + V_L$$

$$22.984 \angle 0^\circ = 60.12 \angle -53.04 + 182.800 \angle 36.96^\circ$$

$$22.984 = 36.144 - j48.039 + 148.84 + j48.039$$

$$\therefore 22.984 = 22.987$$

(verified)

(c)

instantaneous expression:-

For Current:-

$$\begin{aligned} i(t) &= \sqrt{2} I_{rms} \sin(\omega t + \theta) \\ &= \sqrt{2} \times 20.040 \sin(314t - 53.04^\circ) \\ &= 28.34 \sin(314t - 53.04^\circ) \text{ A} \end{aligned}$$

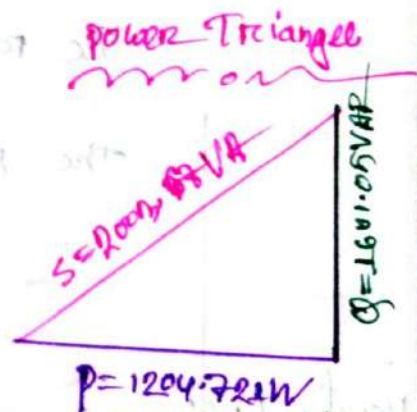
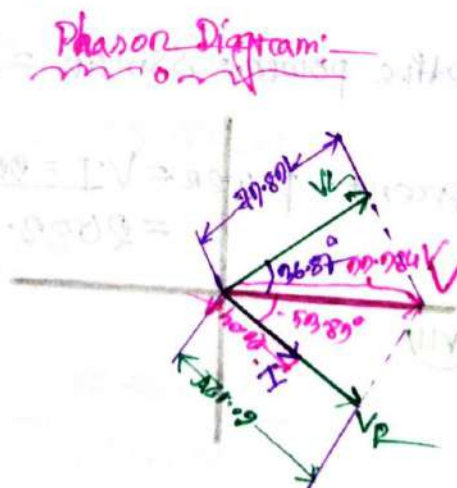
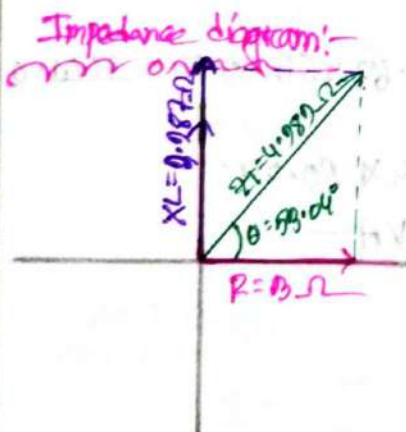
For Voltage drop across the resistor:-

$$\begin{aligned} V_R &= I Z_R \\ &= (20.040 \angle -53.04^\circ) \times (3 \angle 0^\circ) \text{ V} \end{aligned}$$

For Voltage drop across the Inductor:-

$$\begin{aligned} V_L &= I Z_L \\ &= (20.040 \angle -53.04^\circ) \times (9.087 \angle 90^\circ) \text{ V} \end{aligned}$$

(d)



#problems 1

⇒ A voltage $141.4 \sin 314t$ V is applied to RC series circuit which consists $R = 3 \Omega$ $C = 796.18 \mu F$.

- (a) Draw the circuit diagram
- (b) Calculate
 - (i) The capacitive reactance
 - (ii) The impedance
 - (iii) The current
 - (iv) Voltage drop across the resistance and capacitance
 - (v) The power factor and the reactive factor.
 - (vi) the power, reactive power, Apparent power
 - (vii) Verify KVL
- (c) Write the instantaneous equation of current
- (d) Draw the impedance diagram, phasor diagram, power triangle.

⇒ In given:

$$V(t) = 141.4 \sin 314t \text{ V} \quad \Rightarrow V = \frac{141.4}{\sqrt{2}} \angle 0^\circ$$

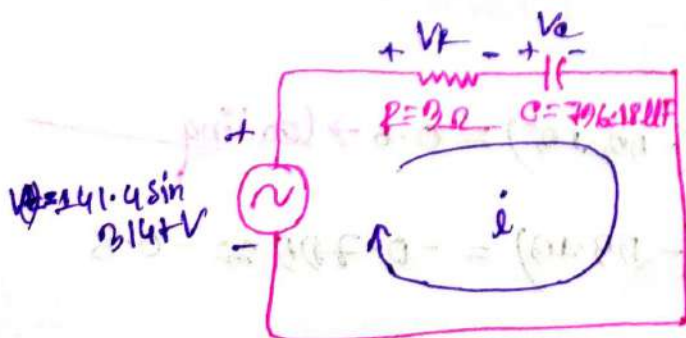
$$\therefore V_m = 141.4 \text{ V}$$

$$\omega = 314 \text{ rad/s}$$

$$R = 3 \Omega$$

$$C = 796.18 \times 10^{-6} \text{ F}$$

(a)



(b)

(i)

we know:-

$$X_c = \frac{1}{\omega C} = \frac{1}{100} = \frac{1}{314 \times 706 \times 18 \mu} \\ = 0.00024 \Omega$$

we know:-

(ii)

$$Z_T = Z_1 + Z_2 \Omega$$

$$\therefore Z_1 = R = 3 \Omega$$

$$Z_2 = -jX_c = -j4 \Omega$$

$$\therefore Z_T = 3 + (-j4) \Omega \\ = 3 - j4 \Omega$$

$$\text{OR } Z_T = 5 \angle -53.13^\circ \Omega$$

(iii)

we know:-

$$I = \frac{V}{Z_T} = \frac{100 \angle 0^\circ}{5 \angle -53.13^\circ}$$

$$= 20 \angle 53.13^\circ$$

$$\approx 20 \angle 53.13^\circ \text{ A}$$

(iv)

we know:-

$$P.f = \cos \theta_2 = \cos(-53.13^\circ) = 0.6 \rightarrow \text{leading}$$

$$\text{r.f} = \sin \theta_2 = \sin(-53.13^\circ) = -0.799 \approx -0.8$$

Note: (i) Impedance is not leading,

(ii) Impedance Angle is not lagging

Voltage Drop:

(V)

Across the resistance: $V_R = I Z_R = (20 \angle 53.13^\circ) \times (3 \angle 0^\circ)$
 $= 60 \angle 53.13^\circ \text{ V}$
 $= 36 + j48 \text{ V}$

Across the capacitance: $V_C = I Z_C = (20 \angle 53.13^\circ) \times (4 \angle -90^\circ)$
 $= 80 \angle -36.87^\circ \text{ V}$
 $= 64 - j48 \text{ V}$

(VI)

We know:

The apparent power $S = VI = 100 \times 20 = 2000 \text{ VA}$

The power $P = S \cos \theta = 2000 \times 0.6 = 1200 \text{ W}$

The reactive power $Q = S \sin \theta = 2000 \times -0.8 = -1600 \text{ VAR}$

(VII)

We know:

$V = V_R + V_C$

$100 = (36 + j48) + (64 - j48)$

$100 = 100 \text{ V}$

(proved)



(c)

we know:-

Instantaneous Expression:-

For Current:-

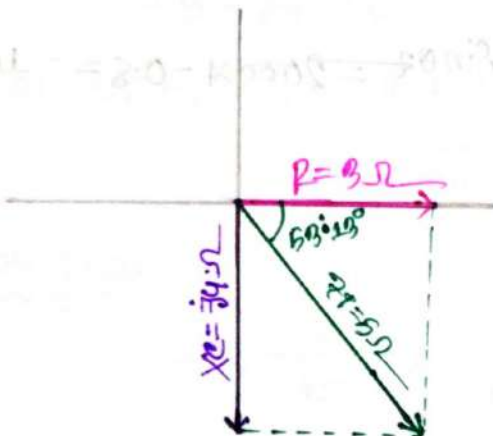
$$i(t) = \sqrt{2} I_{rms} \sin(\omega t + \theta)$$

$$= \sqrt{2} \times 20 \sin(214t + 53.13^\circ)$$

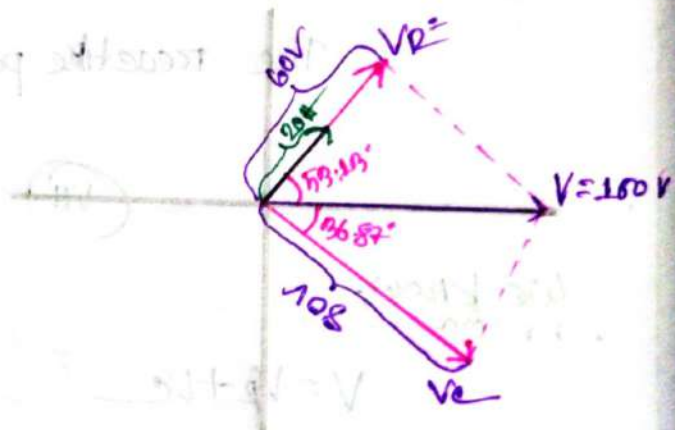
$$= 28.28 \sin(214t + 53.13^\circ)$$

(d)

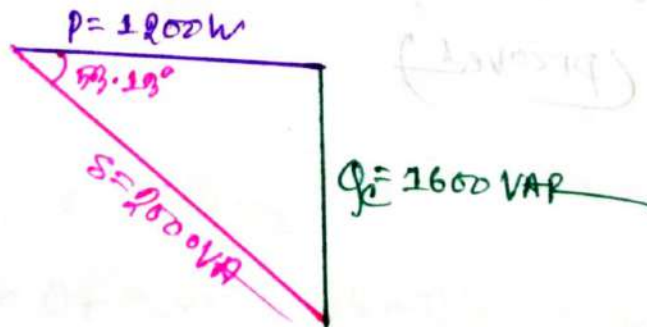
Impedance Diagram:-



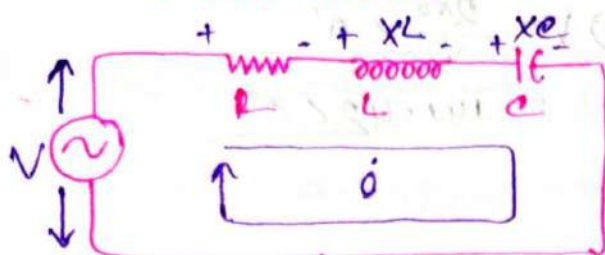
Phasor Diagram:-



Power Triangle:-



R-L-C Circuit



$\therefore$ Impedance $Z = R + jX_L - jX_C$

$$Z = R + j(X_L - X_C) \Omega$$

Magnitude of impedance $|Z| = \sqrt{R^2 + (X_L - X_C)^2}$

Angle of impedance $|Z| = \tan^{-1} \left(\frac{X_L - X_C}{R} \right)$

Current $I = \frac{V}{Z}$ A reactive power $Q = Q_L - Q_C$

problem:

If $R = 20 \Omega$, $L = 0.056 \text{ H}$, $C = 50 \mu\text{F}$ and Applied voltage $200 \sin 377t$

(a) Draw the circuit Diagram.

(b) Calculate (i) the capacitive reactance

(ii) the impedance

(iii) Voltage drop Across the each reactance and Resistance

(iv) the current

(v) the power factor, reactive factor.

(vi) the power, reactive power, apparent power.

(vii) verify KVL.

(c) Write the instantaneous expression for the current and total power.

(d) Draw the impedance Diagram, phasor Diagram, power triangle.

⇒ In given —
now

$$V(t) = 200 \sin 377t \Rightarrow V = \frac{200}{\sqrt{2}} \angle 0^\circ$$

$$= 141.42 \angle 0^\circ$$

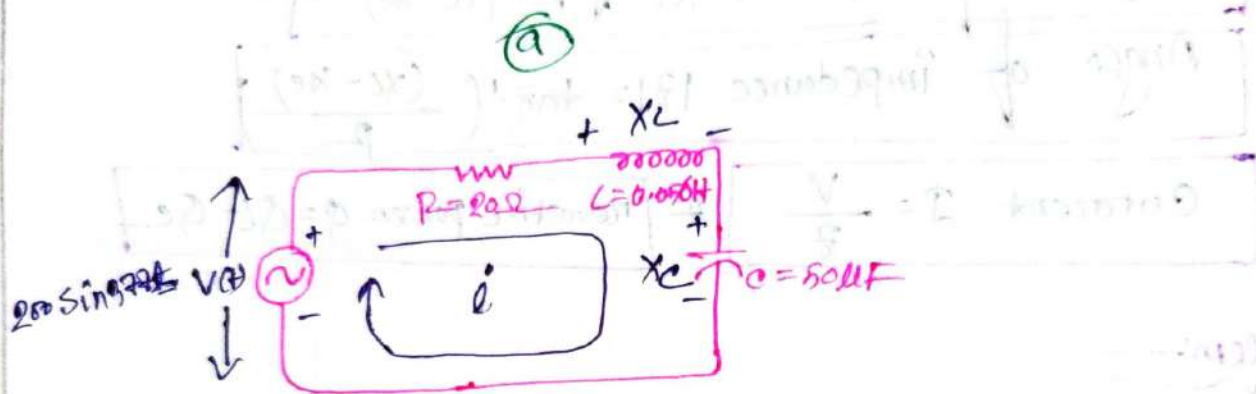
$$V_m = 200$$

$$\theta_v = 0^\circ$$

$$R = 20 \Omega$$

$$L = 0.056 \text{ H}$$

$$C = 50 \mu\text{F}$$



(b)

(i)

We know —
now

$$X_C = \frac{1}{2\pi f C} = \frac{1}{\omega C} = \frac{1}{377 \times 50 \times 10^{-6}} = 53.050 \Omega$$

$$X_L = 2\pi f L = \omega L = 377 \times 0.056 = 21.112 \Omega$$

(ii)

We know —
now

$$Z_1 = 20 \Omega$$

$$Z_2 = jX_L = j21.112 \Omega$$

$$Z_3 = -jX_C = -j53.050 \Omega$$

$$\therefore Z_T = 20 + j21.112 - j53.050$$

$$= 20 + j(21.112 - 53.050) = 20 - j31.938 \Omega$$

$$= 32.683 \angle -57.044^\circ \Omega$$

iii

We know,
now

$$I = \frac{V}{Z_T} = \frac{141.42 \angle 0^\circ}{97.653 \angle 57.044^\circ}$$

$$= 1.447 \angle -57.044^\circ \text{ A}$$

iv

Voltage Drop:
now

Across the resistor

$$V_R = I Z_R$$

$$= (1.447 \angle -57.044^\circ) \times (20 \angle 0^\circ)$$

$$= 28.94 \angle -57.044^\circ \text{ V}$$

$$= 15.82 + j16.35 \text{ V}$$

Across the capacitor $V_C = I Z_C$

$$= (1.447 \angle -57.044^\circ) \times (59.05 \angle -90^\circ)$$

$$= 85.44 \angle -147.094^\circ \text{ V}$$

$$= -67.134 - j42.041 \text{ V}$$

Across the inductor $V_L = I Z_L$

$$= (1.447 \angle -57.044^\circ) \times (12.112 \angle 90^\circ)$$

$$= 17.53 \angle 32.956^\circ \text{ V}$$

$$= 14.71 + j9.42 \text{ V}$$

v

We know,
now

$$P.f = \cos \theta = \cos(-57.044^\circ) = 0.53 \rightarrow \text{leading}$$

$$P.f = \sin \theta = \sin(-57.044^\circ) = -0.84$$

(VI)

We know—
now

$$\text{Apprent power } S = VI = 141.42 \times 0.852 = 120.60 \text{ VA}$$

$$\text{The power } P = S \cos \theta_2 = 120.60 \times 0.852 = 102.72 \text{ W}$$

$$\text{The Reactive power } Q = S \sin \theta_2 = 120.60 \times 0.527 = 63.56 \text{ VAR}$$

(VII)

We know—
now

$$V = V_R + V_L + V_C$$

$$141.42 = 80.84 + j63.56 + 168.67 - j108.64 + j42.04 - 67.134$$

$$141.42 = 141.403 \text{ V}$$

(Verified)

(C)

We know—
now

Intendenceous expression:—
now

For the current i
now

$$i(t) = \sqrt{2} I_{rms} \sin(\omega t + \theta)$$

$$= \sqrt{2} \times 0.752 \sin(377t + 57.044)$$

$$= 1.06 \sin(377t + 57.044) \text{ A}$$

For total power:

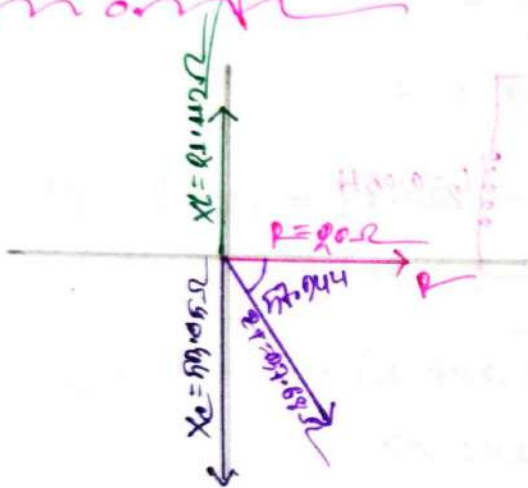
$$P(t) = P - P \cos 2\omega t + Q \sin 2\omega t$$

$$= 281.22 - 281.22 \cos 754t - 445.70 \sin 754t$$

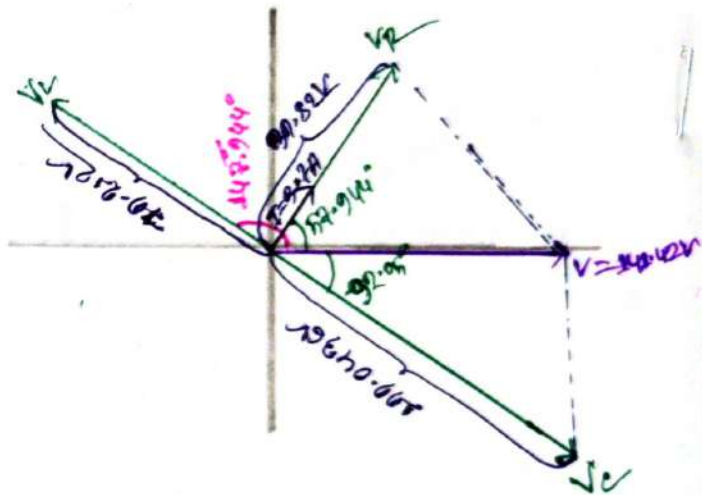
Ans:

(D)

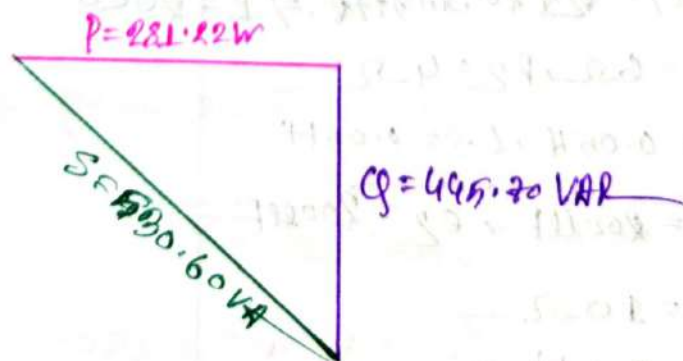
Impedance Diagram:



Phasor Diagram:



power Triangle



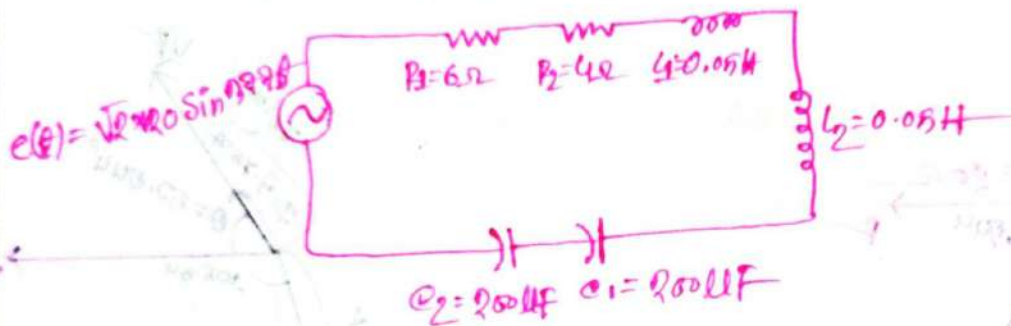
Voltage Divider Rule: (VDR)

$$V_R = \frac{Z_R}{Z_T} \times E = \frac{Z_R}{Z_T} \times V$$

$E = V = \text{total supplied voltage}$

$Z_T = \text{total impedance}$

problems as is:



- Calculate V_R , V_L , V_C and I
- Calculate Average power delivered by the circuit.
- Find V_R and V_C using VDR

in given:

$$e(t) = \sqrt{2} \times 10 \sin 377t \Rightarrow E = 20 \angle 0^\circ$$

$$R_1 = 6\Omega, R_2 = 4\Omega$$

$$L_1 = 0.05H, L_2 = 0.05H$$

$$C_1 = 200\mu F, C_2 = 200\mu F$$

$$\therefore R_T = 10\Omega$$

$$L_T = 0.1H$$

$$C_T = \left(\frac{1}{200} + \frac{1}{200} \right)^{-1} = 100\mu F$$

we know—
mon

(a)

$$X_L = \omega L = 377 \times 0.1 = 37.7 \Omega$$

$$X_C = \frac{1}{\omega C} = \frac{1}{377 \times 40 \mu} = 26.525 \Omega$$

$$Z_T = 10 + j(37.7 - 26.525) \\ = 10 + j11.18$$

$$\text{or } Z_T = 14.099 \angle 48.188^\circ \Omega$$

$$\therefore I = \frac{V}{Z_T} = \frac{20 \angle 0^\circ}{14.099 \angle 48.188^\circ} \\ = 1.333 \angle -48.188^\circ \text{ A}$$

$$V_R = I Z_R = (1.333 \angle -48.188^\circ) \times (10) \\ = 13.33 \angle -48.188^\circ \text{ V}$$

$$V_L = I Z_L = (1.333 \angle -48.188^\circ) \times (37.7 \angle 90^\circ) \\ = 50.141 \angle 41.811^\circ \text{ V}$$

$$V_C = I Z_C = (1.333 \angle -48.188^\circ) \times (26.525 \angle -90^\circ) \\ = 35.411 \angle -138.188^\circ \text{ V}$$

(b)

we know—
mon

$$\text{Average power } P = VI \cos \theta_2$$

$$= 20 \times 1.333 \times \cos(48.188^\circ)$$

$$= 17.77 \text{ W}$$

$$P.F. = \cos \theta_2 = \cos(48.188^\circ)$$

$$= 0.6667 \rightarrow \text{lagging}$$

we know:

$$V_R = \frac{Z_R}{Z_T} \times E$$

$$V_R = \frac{Z_R}{Z_T} \times E$$

$$= \frac{10 \angle 0^\circ}{15 \angle 48.188^\circ} \times 20 \angle 0^\circ$$

$$= \frac{200 \angle 0^\circ}{15 \angle 48.188^\circ}$$

$$= 13.33 \angle -48.188^\circ \text{ V}$$

$$V_C = \frac{Z_C}{Z_T} \times E$$

$$= \frac{26.526 \angle -90^\circ}{(15 \angle 48.188^\circ)} \times 20 \angle 0^\circ$$

$$= \frac{530.52 \angle -90^\circ}{15 \angle 48.188^\circ}$$

$$= 35.368 \angle -138.188^\circ \text{ V}$$

problem :-
mon

⇒ P-L-C Series circuit has an applied voltage of
 $V(t) = 385 \sin 100\pi t$ with $R = 6.5 \Omega$ $L = 60 \text{ mH}$ and
 $C = 30 \mu\text{F}$ Determine.

- (i) The reactances.
- (ii) The impedance.
- (iii) the current
- (iv) voltage drop for all component
- (v) verify KVL
- (vi) The power factor, reactive factor
- (vii) The Apparent, real, reactive powers.
- (viii) V_R, V_L Find using KVR

⇒ In given:-

$$V(t) = 385 \sin 100\pi t \Rightarrow V = 272.23 \angle 0^\circ$$

$$\omega = 100\pi$$

$$R = 6.5 \Omega$$

$$L = 60 \text{ mH}$$

$$C = 30 \mu\text{F}$$

(a)

we know:-

$$X_L = \omega L = 100\pi \times 60 \times 10^{-3} = 18.849 \Omega$$

$$X_C = \frac{1}{\omega C} = \frac{1}{100\pi \times 30 \times 10^{-6}} = 106.103 \Omega$$

(b)

$$\begin{aligned} Z_T &= 6.5 + j(18.849 - 106.103) \\ &= 6.5 - j87.254 \Omega \\ &= 87.40 \angle -85.730^\circ \Omega \end{aligned}$$

iii

We know:

$$I = \frac{V}{Z_T} = \frac{272.23 \angle 0^\circ}{87.49 \angle -85.73^\circ}$$

$$= 3.11 \angle 85.73^\circ$$

iv

Using Ohm's Law:

$$V_P = I Z_P$$

$$= (3.11 \angle 85.73^\circ) \times (6.5 \angle 0^\circ)$$

$$= 20.22 \angle 85.73^\circ \text{ V}$$

$$= 1.901 + j20.16 \text{ V}$$

$$V_L = (3.11 \angle 85.73^\circ) \times (18.84 \angle 90^\circ)$$

$$= 58.61 \angle 175.74^\circ \text{ V}$$

$$= -58.44 + j4.353 \text{ V}$$

$$V_E = (3.11 \angle 85.73^\circ) \times (106.103 \angle -90^\circ)$$

$$= 329.08 \angle -4.26^\circ \text{ V}$$

$$= 329.068 - j24.511$$

Using VDR:

$$V_P = \frac{Z_P \times E}{Z_T} = \frac{(6.5 \angle 0^\circ) \times (272.23 \angle 0^\circ)}{87.49 \angle -85.74^\circ}$$

$$= 20.22 \angle 85.74^\circ \text{ V}$$

$$V_L = \frac{Z_L \times E}{Z_T} = \frac{(18.84 \angle 90^\circ) \times (272.23 \angle 0^\circ)}{87.49 \angle -85.74^\circ}$$

$$= 58.61 \angle 175.74^\circ \text{ V}$$

$$V_C = \frac{Z_C \times I}{Z_T} = \frac{(322.9 \angle -4.26)(272.23 \angle 0^\circ)}{(82.50 \angle -85.74)^\circ}$$

$$= 930.112 \angle -4.26^\circ \text{ V}$$

(V)

we know:
now

$$V = V_R + V_L + V_C$$

$$272.23 = (1.501 + j20.46) + (-58.44 + j4.35) + (322.663 - j24.511)$$

$$272.23 = 272.123 \text{ V}$$

(verified)

(VI)

we know:
now

$$\text{pf} = \cos \theta_2 = \cos(-85.739) = 0.074 \rightarrow \text{leading}$$

$$\text{r.f} = \sin \theta_2 = \sin(-85.739) = -0.092$$

(vii)

we know:
now

$$\text{The Apparent power } S = VI = 272.23 \times 3.11 = 846.63 \text{ VA}$$

$$\text{The real power } P = S \cos \theta = 846.63 \times \cos(-85.739) = 62.10 \text{ W}$$

$$\text{The reactive power } Q = S \sin \theta = 846.63 \times -0.092 = -84.00 \text{ VAR}$$

Parallel Circuits

(i)

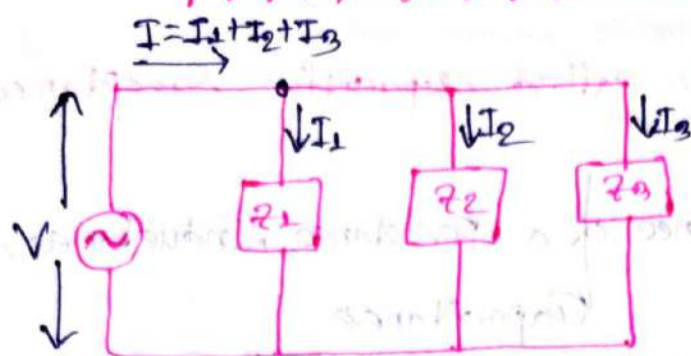
(ii)

The effective power of the circuit is $P = 2000 \times 0.807 = 1614 \text{ W}$
 The total power $P = 2000 \times 0.807 = 1614 \text{ W}$
 The effective power is $P = 2000 \times 0.807 = 1614 \text{ W}$

$$P_{\text{eff}} = P_1 + P_2 + P_3 = (1000 \times 0.807) + (1000 \times 0.807) + (1000 \times 0.807) = 2421 \text{ W}$$

$$V = \frac{P_{\text{eff}}}{I} = \frac{2421}{1.5} = 1614 \text{ V}$$

P-L-e in Parallel:-



$$\therefore \text{total impedance: } \frac{1}{Z_T} = \frac{1}{Z_1} + \frac{1}{Z_2} + \frac{1}{Z_3}$$

Current Divider Rule (CDR)

$$I_k = \frac{Z_T}{Z_k} \times I$$

Z_T = total impedance

I = total entering current

Admittance:-

$\Rightarrow$ Admittance exactly inversely proportional to the impedance. denote by Y .

$$\therefore Y = \frac{1}{Z} \text{ Siemens/mho (S)}$$

$$Y = G \mp jB \text{ S/V}$$

$G = \frac{1}{R} = \text{Conductance}$

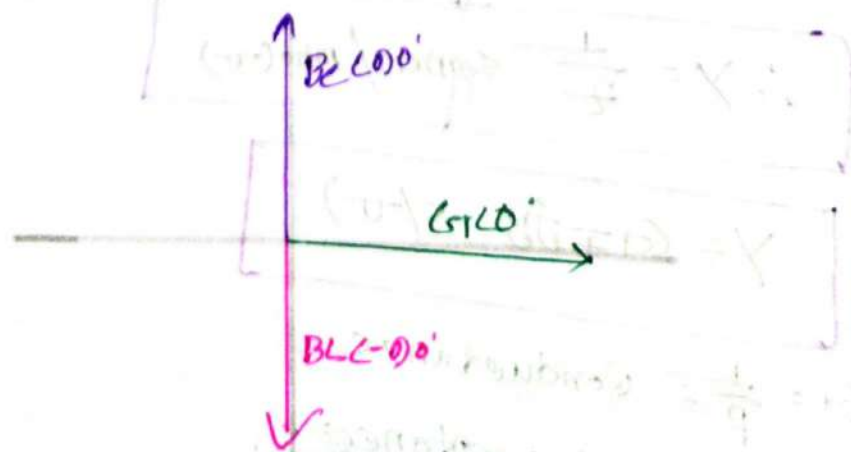
$B = \frac{1}{X} = \text{Susceptance}$

- Note - (i) B_L or b_L is called Inductive Susceptance
 (ii) B_C or b_C is called Capacitive Susceptance.

Admittance of a Resistance, Inductance, Capacitance

- (1) Admittance of resistance $Y_R = \frac{1}{Z_R} = G \angle 0^\circ \text{ (S)}$
- (2) Admittance of inductance $Y_L = \frac{1}{Z_L} = B_L \angle -90^\circ$
- (3) Admittance of capacitance $Y_C = \frac{1}{Z_C} = B_C \angle 90^\circ$
- (4) Admittance of R-L Series Branch $Y_{RL} = \frac{1}{Z_{RL}}$
- (5) Admittance of R-C " " " $Y_{RC} = \frac{1}{Z_{RC}}$
- (6) " " " R-C-L " " " $Y_{RCL} = \frac{1}{Z_{RCL}}$

Admittance Diagram:



#problems R-L-C circuit: in parallel

⇒ A parallel circuit with three Branches has a voltage applied of 230V rms, 50Hz supply with the following impedances

Branch 1: A series R-L branch with $R_1 = 8\Omega$ & $L = 100\text{mH}$

Branch 2: A series R-C branch with $R_2 = 12\Omega$ & $C = 33\mu\text{F}$

Branch 3: An inductor of $L = 220\mu\text{H}$

Determine:—

- ① The total impedance, impedance of each Branch,
- ② Susceptance of branch 3,
- ③ total current.
- ④ the branch currents using CDR
- ⑤ verify KCL.
- ⑥ All power and their factor.
- ⑦ Vector Diagram and power triangle,

①

⇒ In given:—

$$V = 230 \angle 0^\circ$$

$$f = 50\text{Hz}$$

$$\begin{aligned} B1; Z_1 &= 8 + jX_L \\ &= 8 + j2\pi f \times 100\text{m} \\ &= 8 + j2\pi \times 50 \times 100 \times 10^{-3} \\ &= 8 + j31.41\Omega \end{aligned}$$

$$B1; R = 8\Omega$$

$$L = 100\text{mH}$$

$$B2; R = 12\Omega$$

$$C = 33\mu\text{F}$$

$$B3; L = 220\mu\text{H}$$

$$\begin{aligned} B2; Z_2 &= 12 - jX_C \\ &= 12 - j\frac{1}{2\pi f \times C} \\ &= 12 - j\frac{1}{2\pi \times 50 \times 33 \times 10^{-6}} \\ &= 12 - j96.45\Omega \end{aligned}$$

$$\begin{aligned}
 \text{Q3; } Z_{T3} &= +jXL \\
 &= j \times 2\pi fL \\
 &= j \times 2\pi \times 50 \times 220 \times 10^{-6} \\
 &= j60.115 \times 10^{-3} \Omega
 \end{aligned}$$

∴ Total Impedance:-

$$\begin{aligned}
 Z_T &= \left(\frac{1}{Z_{T1}} + \frac{1}{Z_{T2}} + \frac{1}{Z_{T3}} \right)^{-1} \\
 &= \left(\frac{1}{8 + j31.41} + \frac{1}{12 - j06.48} + \frac{1}{j60.115 \times 10^{-3}} \right)^{-1} \Omega \\
 &= 4.232 \times 10^{-5} + j0.060021 \Omega \\
 &= 0.060 \angle 80.76^\circ
 \end{aligned}$$

(2)

We know:-

Susceptance of branch 3

$$\begin{aligned}
 B &= \frac{1}{XL} \\
 &= \frac{1}{XL} \\
 &= \frac{1}{60.115 \times 10^{-3}} \\
 &= 14.468 \text{ S}
 \end{aligned}$$

(3)

we know:-

$$I = \frac{V}{Z_T} = \frac{230 \angle 0^\circ}{0.060 \angle 89.98^\circ}$$

$$= 3833.333 \angle -89.98^\circ \text{ A}$$

$$= 2.327 - j3833.33 \text{ A}$$

(4)

we know:-

$$\text{CDR } I_a = \frac{Z_T}{Z_a} \times I$$

$$\therefore B1; I_1 = \frac{Z_T}{Z_1} \times I$$

$$= \frac{0.060 \angle 89.98^\circ}{0.0241 \angle 75.71^\circ} \times 3833.333 \angle -89.98^\circ$$

$$= 9506 \angle -14.23^\circ \text{ A}$$

$$= 9175.1 - j6.876 \text{ A}$$

$$B2; I_2 = \frac{Z_T}{Z_2} \times I$$

$$= \frac{0.060 \angle 89.98^\circ}{0.07100 \angle -82.00^\circ} \times 3833.333 \angle -89.98^\circ$$

$$= 2.36 \angle +82.00^\circ \text{ A}$$

$$= 0.2916 + j2.3410 \text{ A}$$

$$B3; I_3 = \frac{Z_T}{Z_3} \times I$$

$$= \frac{0.060 \angle 89.98^\circ}{0.060 \angle 0^\circ} \times 3833.333 \angle -89.98^\circ$$

$$= 3833.333 \angle -89.98^\circ \text{ A}$$

$$= -j3833.33 \text{ A}$$

For KCL:-

$$I = I_1 + I_2 + I_3$$

$$2.327 - j0.333.33 = 1.751 - 6.875j + 0.2216 + j2.3410 - j0.333.33$$

$$2.327 - j0.333.33 = (1.751 + 0.2216) + j(-6.875 + 2.3410 - 0.333.33)$$
$$= 2.0426 - j0.337.8631 \text{ A}$$

$$\therefore 2.327 - j0.333.33 \approx 2.327 - j0.333.33 \text{ A}$$

Verified!

(6)

we know:-

$$\text{Apparent power } S = VI$$

$$= 230 \times 3.33.33 \text{ VA}$$

$$= 766.66 \times 10^3 \text{ VA}$$

$$= 766.66 \text{ mVA}$$

$$\text{Real power } P = S \cos \theta$$

$$= 766.66 \text{ m} \times \cos(80.96)$$

$$= 5.352 \times 10^4 \text{ W}$$

$$\text{reactive power } Q = S \sin \theta$$

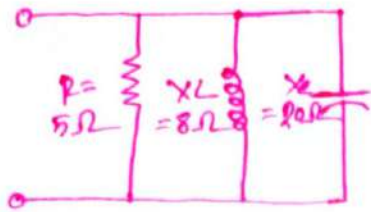
$$= 766.66 \text{ m} \times \sin(80.96)$$

$$= 0.766.66 \text{ mVAR}$$

$$\text{P.f} = \cos \theta = \cos(80.96) = 6.981 \times 10^{-4}$$

$$\text{R.f} = \sin \theta = \sin(80.96) = 0.9999$$

problems —
mon



- (i) Find the total impedance
- (ii) sketch the impedance diagram
- (iii) Find the Admittance for each parallel branch
- (iv) Calculate the total admittance.
- (v) Find conductance of branch 1
- (vi) Find Susceptance of branch 2, 3
- (vii) sketch the admittance diagram

is given —
ref.

$$Z_1 = 5 \Omega$$

$$Z_2 = j8 \Omega$$

$$Z_3 = -j20 \Omega$$

$$\therefore Z_T = \left(\frac{1}{Z_1} + \frac{1}{Z_2} + \frac{1}{Z_3} \right)^{-1}$$

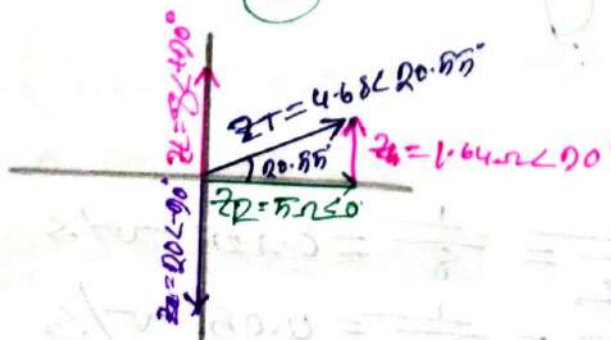
$$= \left(\frac{1}{5} + \frac{1}{j8} - \frac{1}{j20} \right)^{-1}$$

$$= \left(\frac{1}{5} - \frac{j}{40} \right)^{-1}$$

$$= 4.0383 + j1.6403$$

$$= 4.68 \angle 20.55^\circ$$

(b)



(a)

we know:-

$$Y = \frac{1}{Z}$$

$$\therefore Y_1 = \frac{1}{Z_1} = \frac{1}{5} = 0.2 \text{ u/s}$$

$$Y_2 = \frac{1}{Z_2} = \frac{1}{-j8} = +0.125j \text{ u/s}$$

$$Y_3 = \frac{1}{Z_3} = \frac{1}{-j20} = 0.05j \text{ u/s}$$

(b)

we know:-

$$Y_T = \frac{1}{Z_T}$$

$$\therefore Y_T = \frac{1}{4.983 + j1.643}$$

$$= 0.2000 - j0.074j$$

$$= (0.2 - j0.074) \text{ u/s}$$

$$= (0.213 \angle -20.54^\circ)$$

(c)

we know:-

$$G = \frac{1}{R}$$

$$\therefore G = \frac{1}{5} = 0.2 \text{ u/s}$$

(d)

we know:-

$$B = \frac{1}{X}$$

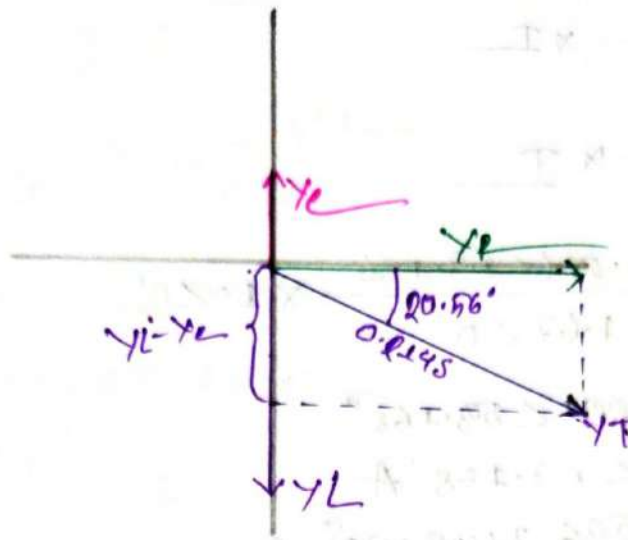
For B2:-

$$B_L = \frac{1}{X_L} = \frac{1}{8} = 0.125 \text{ u/s}$$

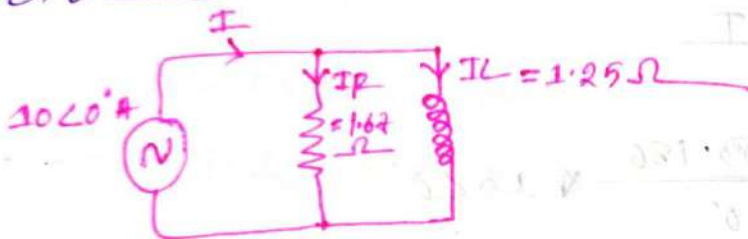
For B3:-

$$B_C = \frac{1}{X_C} = \frac{1}{20} = 0.05 \text{ u/s}$$

(f)



problem:



(i) Find the admittance

(ii) Find the current I_R , I_L

⇒ In given:

(1)

$$I = 10 \angle 0^\circ \text{ A}$$

$$R = 1.67 \Omega$$

$$X_L = 1.25 \Omega$$

$$\therefore Z_1 = R = 1.67 \Omega$$

$$Z_2 = X_L = j \times 1.25 \Omega$$

$$\therefore Z_T = \left(\frac{1}{Z_1} + \frac{1}{Z_2} \right)^{-1}$$

$$= \left(\frac{1}{1.67} + \frac{1}{j \times 1.25} \right)^{-1}$$

$$= 0.5996 + j0.8011 \Omega$$

$$\therefore Y = \frac{1}{Z_T} = 0.5088 - j0.8 \text{ S}$$

we know:-
~~mon~~

$$I_m = \frac{Z_T}{Z_R} \times I$$

$$\therefore I_p = \frac{Z_T}{Z_R} \times I$$

$$= \frac{1.006 \angle 53.186}{1.67 \angle 0^\circ} \times 10 \angle 0^\circ$$

$$= 5.001 \angle 53.168^\circ$$

$$\approx 6 \angle 53.168^\circ \text{ A}$$

$$= 3.506 + j4.802 \text{ A}$$

$$\therefore I_c = \frac{Z_T}{Z_R} \times I$$

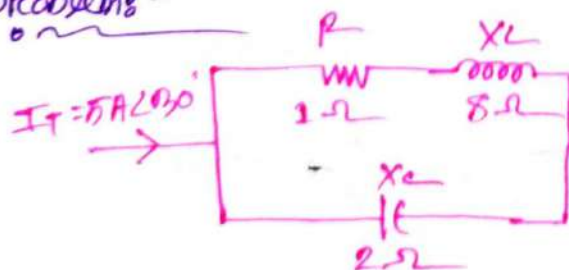
$$= \frac{1.006 \angle 53.186}{1.25 \angle 0^\circ} \times 10 \angle 0^\circ$$

$$= 8.0048 \angle -36.814^\circ \text{ A}$$

$$\approx 8 \angle -36.814^\circ \text{ A}$$

Ans:-

problems -
~~mon~~



\$\Rightarrow\$ Find the current through each parallel branch
 $\Rightarrow$ In given:-

$$Z_1 = 1 \Omega$$

$$Z_2 = +j8 \Omega$$

$$\therefore Z' = 1 + j8 \Omega$$

$$Z_3 = -j2 \Omega$$

$$\therefore Z_T = \left(\frac{1}{Z'} + \frac{1}{Z_3} \right)^{-1}$$

$$= \left(\frac{1}{1+j8} - \frac{1}{2j} \right)^{-1}$$

$$= 0.408 - j2.648 \Omega$$

$$= 2.65 \angle -87.66^\circ \Omega$$

$$I_{Z_3} = \frac{Z_T}{Z_3} \times I$$

$$= \frac{2.65 \angle -87.66^\circ}{2 \angle -90^\circ} \times (5 \angle 30^\circ)$$

$$= 6.625 \angle 32.34^\circ \text{ A}$$

$$= 5.594j0.543 \text{ A}$$

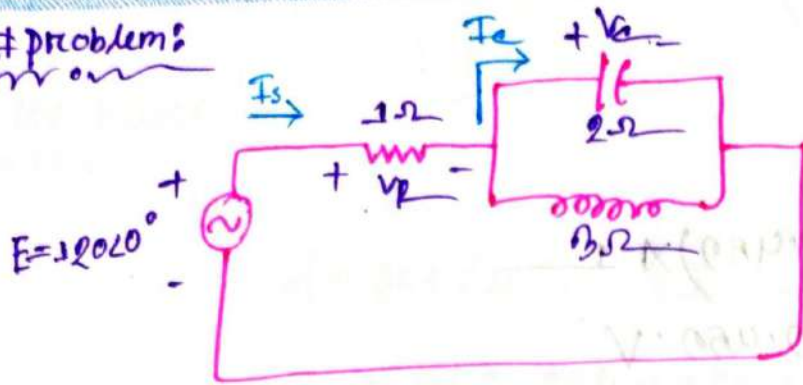
$$I_{Z'} = \frac{Z_T}{Z'} \times I$$

$$= \frac{2.65 \angle -87.66^\circ}{8.062 \angle -82.87^\circ} \times 5 \angle 30^\circ$$

$$= 1.643 \angle -140.53^\circ \text{ A}$$

$$= -1.268 - j1.044 \text{ A}$$

problem:



- Calculate Z_T
- Determine I_s
- Calculate V_R and V_L
- Find I_L
- Compute the power delivered.
- Find P.f

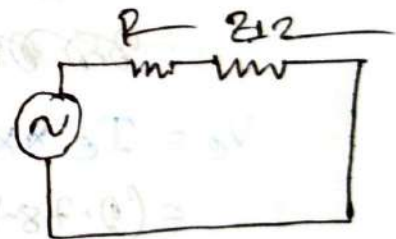
⇒ In given:

$$E = 120 \angle 0^\circ ; E = 120V$$

$$R = 1\Omega$$

$$X_C = -j2\Omega$$

$$X_L = j3\Omega$$



$$\therefore Z_T = \left(\frac{1}{X_C} + \frac{1}{X_L} \right)^{-1} = \left(\frac{1}{-j2} + \frac{1}{j3} \right)^{-1} = -j6\Omega$$

$$\therefore Z_T = (1 - j6)\Omega$$

$$= 1 - j6\Omega$$

we know

$$I_s = \frac{E}{Z_T} = \frac{120}{1 - j6} = 3.24 + j19.459 \text{ A}$$

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③

we know
more

$$V_p = I_s R_s$$

$$= (3.26 + j10.45) \times 1$$

$$= 3.26 + j10.45 \text{ V}$$

④

$$I_c = \frac{Z_{12}}{X_c} \times I_s$$

$$= \frac{-j6}{-2j} \times (3.26 + j10.45)$$

$$= 0.78 + j58.35 \text{ A}$$

$$V_c = I_c \times X_c$$

$$= (0.78 + j58.35) \times -2j$$

$$= 116.7 - j10.56 \text{ V}$$

Ans:-

⑤

we know
more

$$p.f. = \cos \theta$$

$$= \cos(-80.53^\circ)$$

$$= 0.164 \rightarrow \text{leading}$$

$$Z_T = 6.08 \angle -80.53^\circ$$

(B)

we know
mon

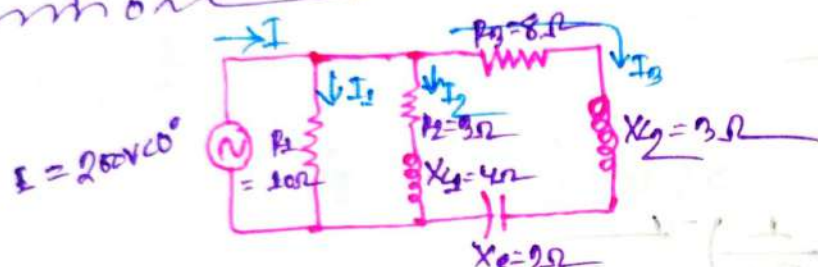
$$P_{del} = I_s^2 R$$

$$= (3.26 + j10.457)^2$$

$$= -368.025 + j26.82j$$

Ans:

#problem
mon



- Compute I .
- Find I_1, I_2, I_3 .
- Verify Kirchhoff's current law.
- Find the total impedance; Admittance.

⇒ In given:
mon

$$E = 200V \angle 0^\circ / E = 200V$$

$$R_1 = 10\Omega$$

$$R_2 = 3\Omega$$

$$R_3 = 8\Omega$$

$$X_{L1} = 4\Omega$$

$$X_{L2} = 3\Omega$$

$$X_c = 2\Omega$$

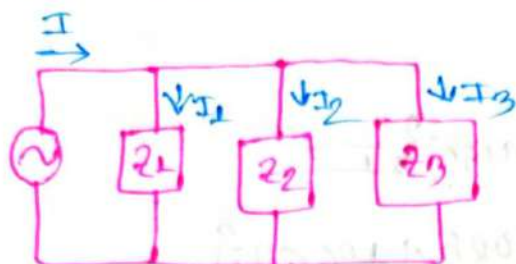
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Per Draw the circuit:-



$$Z_1 = 10 \Omega$$

$$Z_2 = 3 + 4j \Omega$$

$$Z_3 = (8 + j3 - 9j) \Omega$$

$$= (8 - j6) \Omega$$

$$\therefore Z_T = \left(\frac{1}{Z_1} + \frac{1}{Z_2} + \frac{1}{Z_3} \right)^{-1} \Omega$$

$$= \left(\frac{1}{10} + \frac{1}{3 + 4j} + \frac{1}{8 - j6} \right)^{-1} \Omega$$

$$= (3 + j) \Omega$$

(a)

we know:-

$$I = \frac{E}{Z_T} = \frac{200}{3 + j} = (60 - 20j) A$$

(b)

we know:-

$$I_1 = \frac{Z_T}{Z_1} \times I = \frac{3 + j}{10} \times (60 - 20j) = 20 A$$

$$I_2 = \frac{Z_T}{Z_2} \times I = \frac{3 + j}{3 + 4j} (60 - j20) = 24 - 32j A$$

$$I_3 = \frac{Z_T}{Z_3} \times I = \frac{3 + j}{8 - 6j} (60 - j20) = 16 + 12j A$$

(c)

$$I = I_1 + I_2 + I_3$$

$$60 - 20j = 20 + 24 - 32j + 16 + 12j$$

$$60 - 20j = 60 - 20j$$

(proved)

(d)

we know

$$X_i = \left(\frac{1}{Z_T} \right) \cdot v/s$$

$$= \left(\frac{1}{3+j} \right) \cdot v/s$$

$$= 0.3 - 0.1j \cdot v/s$$

$$= 0.3 - j(0.1) \cdot v/s$$

Ans:-

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Network Theorems!

Superposition Theorem:

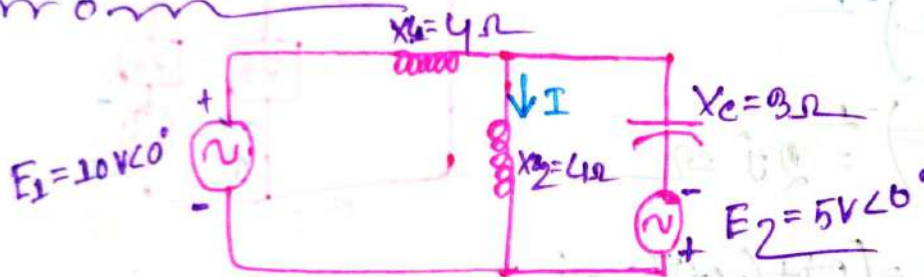
⇒ The logic that was used for DC circuit is the same for the AC circuits.

Deactivating Source rules:

① Voltage Source are short circuited.

② Current Source are open circuited.

Problem 1801:



① Determine the current I .

⇒ Is given:

$$E_1 = 10\angle 0^\circ = 10V$$

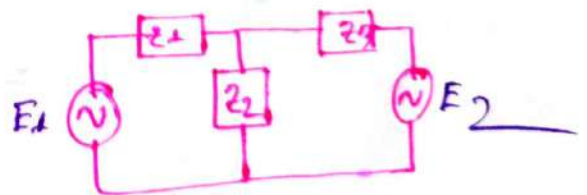
$$Z_1 = j\omega X_1 = j4\Omega$$

$$Z_2 = j\omega X_2 = j4\Omega$$

$$Z_3 = -j\omega X_3 = -j3\Omega$$

$$E_2 = 5\angle 0^\circ = 5V$$

Re Draw the circuit:



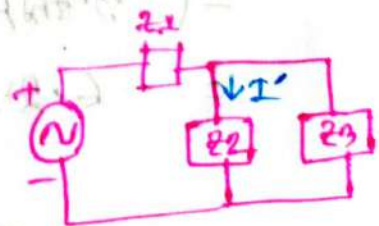
For E_1 only:

we know:-

$$Z_{23} = \left(\frac{1}{Z_2} + \frac{1}{Z_3} \right)^{-1}$$

$$= \left(\frac{1}{j4} + \frac{1}{-j3} \right)^{-1} = -12j\Omega$$

$$\therefore Z_T = (Z_1 + Z_{23}) = 4j - 12j = -8j\Omega$$



Current: $I_s = \frac{E_1}{Z_T} = \frac{10}{-8j} = 1.25j \text{ A}$

$$\therefore I' = \frac{Z_{23}}{Z_2} \times I_s = \frac{-12j}{4j} \times 1.25j$$

$$= -3.75j \text{ A} \rightarrow \text{clockwise}$$

For E_2 only:

We know:

$$Z_{12} = \left(\frac{1}{Z_1} + \frac{1}{Z_2} \right)^{-1}$$

$$= \left(\frac{1}{j4} + \frac{1}{j4} \right)^{-1} = 2j \Omega$$

$$Z_T = (Z_2 + Z_3) = [2j + (-j)] = j \Omega$$

$$I_s = \frac{E_2}{Z_T} = \frac{5}{-j} = 5j \text{ A}$$

$$\therefore I'' = \frac{Z_{12}}{Z_2} \times I_s = \frac{2j}{4j} \times 5j$$

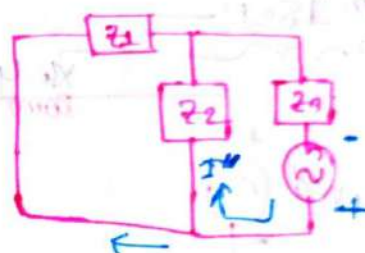
$$= 1.5j \text{ A} \rightarrow \text{counter clockwise}$$

$$\therefore I = I' - I''$$

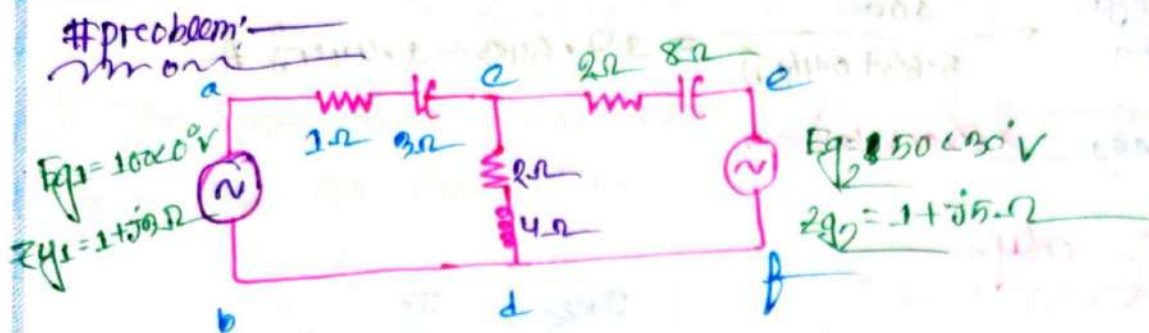
$$= (-3.75j \text{ A} - 1.5j \text{ A})$$

$$= -5.25j \text{ A}$$

Ans:



#problem:-



(i) Calculate the current in branch 'ac'.

Given:-

$$E_{g1} = 100 \angle 0^\circ V = 100 V$$

$$E_{g2} = 50 \angle 30^\circ = 43.301 + j25 V$$

$$Z_{g1} = 1 + j3 \Omega$$

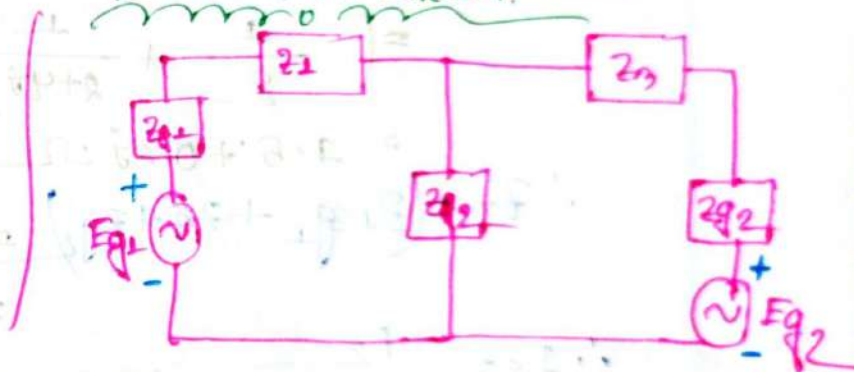
$$Z_{g2} = 1 + j5 \Omega$$

$$Z_1 = 1 - j3 \Omega$$

$$Z_2 = 2 + j4 \Omega$$

$$Z_3 = 2 - j8 \Omega$$

Re draw the circuit:-



Consider E_{g1} only:-

We know:-

$$Z_{eq} = Z_3 + Z_{g2} = (2 - j8) + (1 + j5) \Omega$$

$$= 3 - j3 \Omega$$

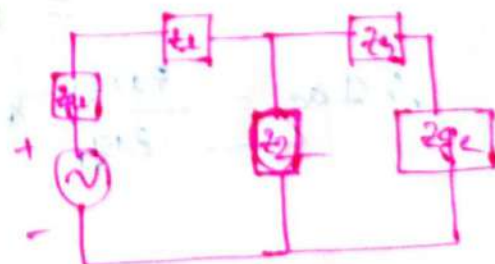
$$Z_{eq} = \left(\frac{1}{Z_2} + \frac{1}{Z_{eq2}} \right)^{-1} = \left(\frac{1}{2 + j4} + \frac{1}{3 - j3} \right)^{-1} \Omega$$

$$= 0.60 + j0.461 \Omega$$

$$Z_T = (Z_{g1} + Z_1 + Z_{eq})$$

$$= 1 + j3 + 1 - j3 + 0.60 + j0.461$$

$$= 2.60 + j0.461 \Omega$$



$$I_s = \frac{E_1}{Z_T} = \frac{100}{5.63 + 0.461j} = 17.643 - 1.444j \text{ A}$$

$$I_s = I_{ae1} \rightarrow \text{clockwise}$$

Consider E_2 only

we know

$$\begin{aligned} Z_{12} &= Z_1 + Z_{11} \\ &= (1 - 3j + 1 + 3j) \Omega \\ &= 2 \Omega \end{aligned}$$

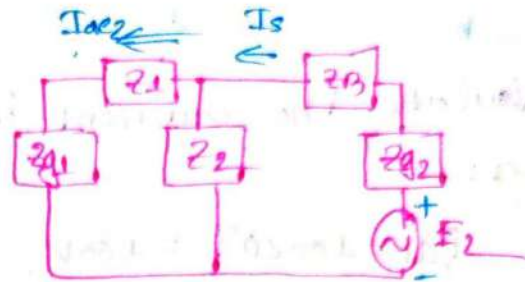
$$\begin{aligned} Z_{12} &= \left(\frac{1}{Z_{11}} + \frac{1}{Z_2} \right)^{-1} \\ &= \left(\frac{1}{2} + \frac{1}{2 + 4j} \right)^{-1} \\ &= 1.5 + 0.5j \Omega \end{aligned}$$

$$\begin{aligned} \therefore Z_T &= (Z_{12} + Z_3 + Z_{22}) = (1.5 + 0.5j + 2 - 8j + 1 + 5j) \\ &= (4.5 - 2.5j) \Omega \end{aligned}$$

$$\begin{aligned} \therefore I_s &= \frac{E_2}{Z_T} = \frac{40 \angle 30^\circ + 25j}{4.5 - 2.5j} \text{ A} \\ &= 5 + 8.33j \text{ A} \end{aligned}$$

$$\begin{aligned} \therefore I_{ae2} &= \frac{Z_{12}}{Z_{11}} \times I_s = \frac{1.5 + 0.5j}{2} \times (5 + 8.33j) \text{ A} \\ &= 1.667 + 7.5j \rightarrow \text{counter clock wise} \end{aligned}$$

$$\begin{aligned} \therefore \text{Original } I_{ae} &= I_{ae1} - I_{ae2} \\ &= (17.643 - 1.444j) - (1.667 + 7.5j) \text{ A} \\ &= 15.976 - 8.944j \text{ A} \end{aligned}$$



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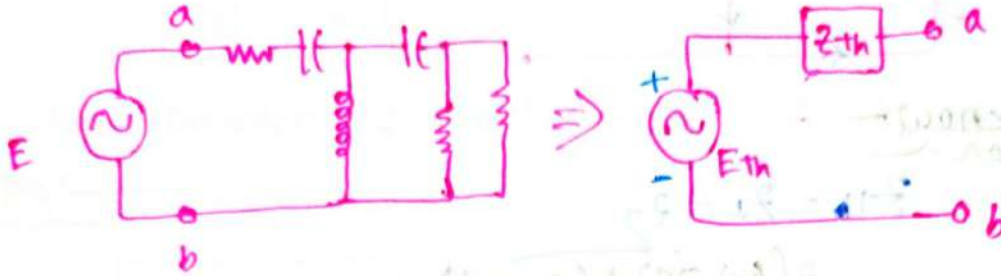
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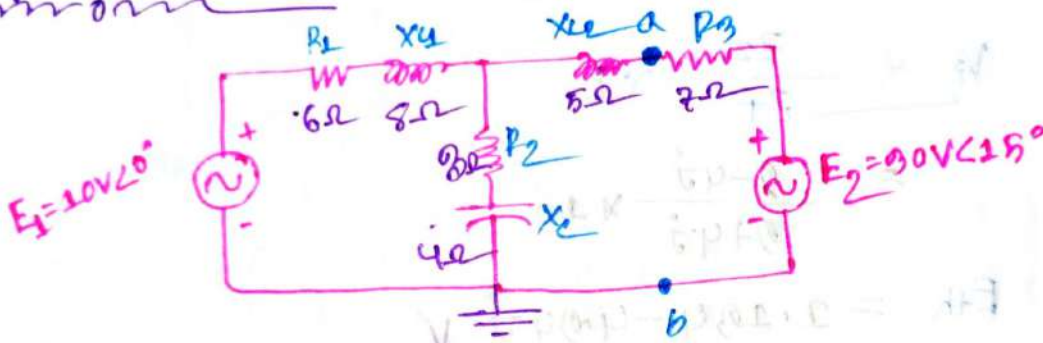
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THEVENIN'S Theorem:—

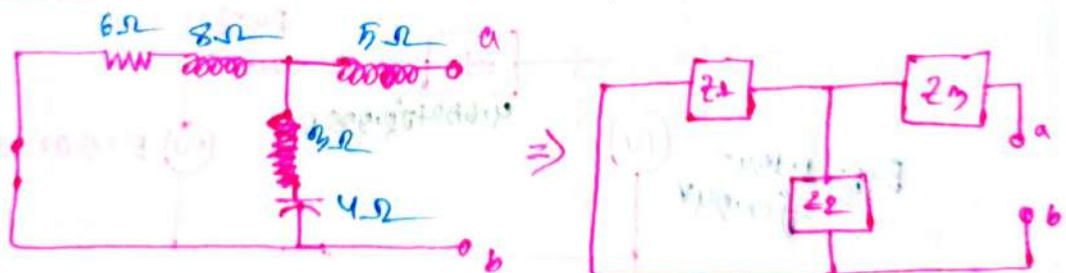
⇒ The logic that was used for DC current is the same for the AC currents.



problem 1:—



⇒ Find the thevenin equivalent of terminal a and b
 ⇒ For Z_{th} —



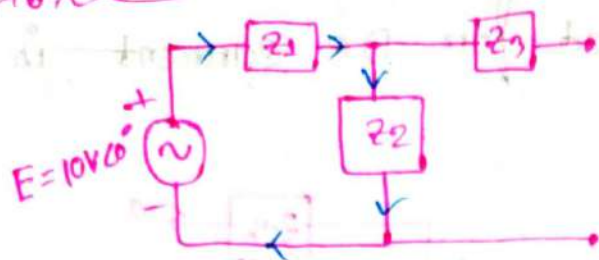
$$Z_1 = 6 + j8 \Omega$$

$$Z_2 = 3 - j4 \Omega$$

$$Z_3 = j5 \Omega$$

$$\begin{aligned} \therefore Z_{Th} &= \left(\frac{1}{Z_1} + \frac{1}{Z_2} \right)^{-1} + Z_3 \\ &= \left(\frac{1}{6 + j8} + \frac{1}{3 - j4} \right)^{-1} + j5 \\ &= 4.639 + j2.98 \Omega \end{aligned}$$

For E_{th} —



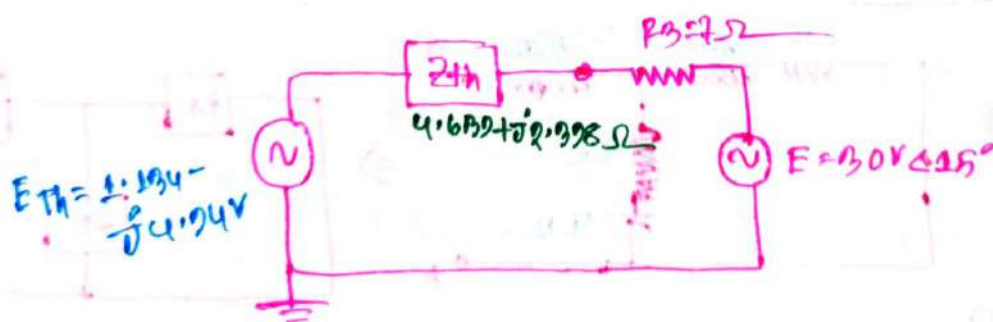
⇒ we know —

$$\begin{aligned} Z_{th} &= Z_1 + Z_2 \\ &= (6 + j8) + (3 - j4) \\ &= 9 + j4 \, \Omega \end{aligned}$$

$$\begin{aligned} V_{Z_2} &= \frac{Z_2}{Z_1 + Z_2} \times E \\ &= \frac{3 - j4}{9 + j4} \times 10 \end{aligned}$$

$$E_{th} = 1.134 - j0.94 \, V$$

So, the thevenin equivalent —



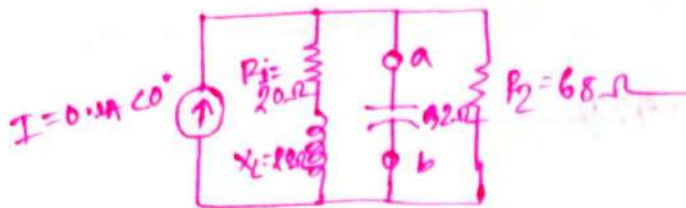
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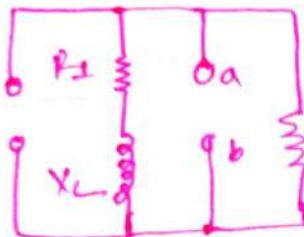
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Problem: _____
mon



① Find the thevenin Equivalent for the circuit.

⇒ For Z_{th} : _____
mon

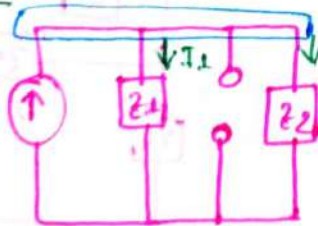


$$Z_{th} = \left(\frac{1}{R_1 + X_L} + \frac{1}{R_2} \right)^{-1}$$

$$= \left(\frac{1}{20 + 20j} + \frac{1}{68} \right)^{-1}$$

$$= 18.035 + j11.955 \Omega$$

For E_{th} : _____
mon



$$Z_1 = 20 + 20j \Omega$$

$$Z_2 = 68 \Omega$$

We know _____
mon

$$I_1 = \frac{Z_{th}}{Z_1} \times I = \left(\frac{18.035 + j11.955}{20 + 20j} \right) \times 0.1$$

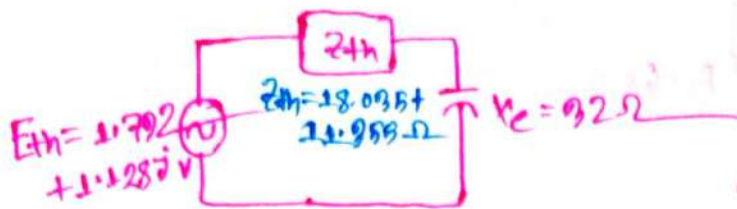
$$= 0.073 - j0.0166 A$$

$$E_{th} = V_{Z_1} = I_1 Z_1$$

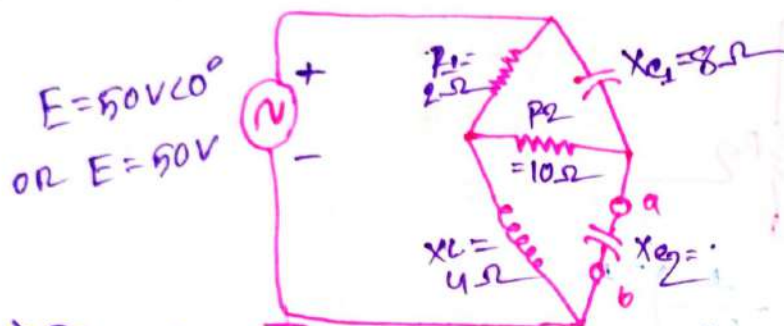
$$= (0.073 - j0.0166) \times (20 + 20j)$$

$$= 1.702 + j1.28 V$$

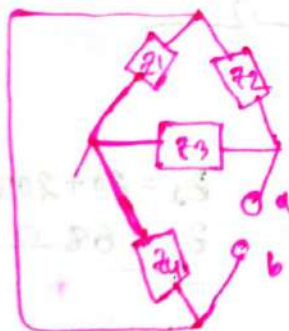
∴ the thevenin Equivalent —



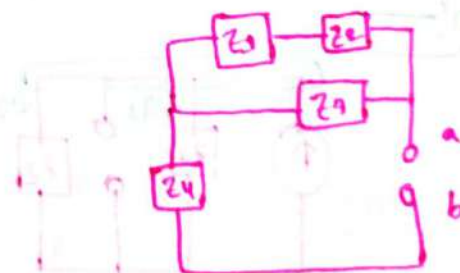
Problem 2:



⇒ Find the thevenin equivalent for the terminal a and b
 ⇒ For Z_{th} —



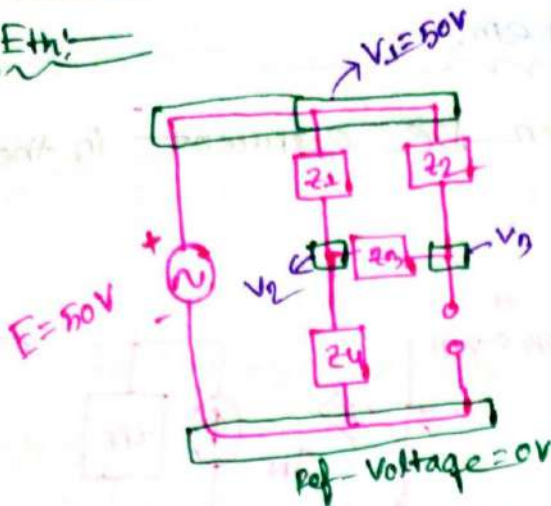
⇒



$$\begin{aligned} Z_1 &= 2 \Omega \\ Z_2 &= -j8 \Omega \\ Z_3 &= 10 \Omega \\ Z_4 &= j4 \Omega \end{aligned}$$

$$\begin{aligned} \therefore Z_{th} &= \left(\frac{1}{Z_1 + Z_2} + \frac{1}{Z_3} \right)^{-1} + Z_4 \\ &= \left(\frac{1}{2 - j8} + \frac{1}{10} \right)^{-1} + j4 \\ &= 4.230 + j0.1538 \Omega \end{aligned}$$

For E_{th} :-



$$\therefore V_2 \Rightarrow \frac{V_2 - 0}{Z_4} + \frac{V_2 - V_3}{Z_3} + \frac{V_2 - 50}{Z_1} = 0$$

$$\Rightarrow \frac{V_2}{j4} + \frac{V_2 - V_3}{10} + \frac{V_2 - 50}{2} = 0$$

$$\Rightarrow \frac{5V_2 + 2j(V_2 - V_3) + 1j(V_2 - 50)}{20j} = 0$$

$$\Rightarrow (5V_2 + j12V_2) - j2V_3 = 500j \quad \dots \dots \dots (1)$$

For V_3 :-

$$\frac{V_3 - 50}{Z_2} + \frac{V_3 - V_2}{Z_3} = 0$$

$$\Rightarrow \frac{V_3 - 50}{-j8} + \frac{V_3 - V_2}{10} = 0$$

$$\Rightarrow \frac{-5(V_3 - 50) + 4j(V_3 - V_2)}{40j} = 0$$

$$\Rightarrow -j4V_2 \{-5V_3 + 4V_3j\} = -250$$

$$\Rightarrow -j4V_2 + V_3(-5 + j4) = -250 \quad \dots \dots \dots (11)$$

$\therefore$

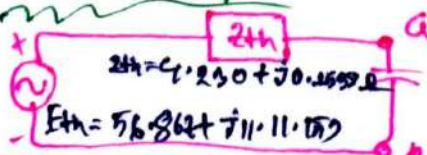
$$V_2 = 42.018 + j10.742 \text{ V}$$

$$V_3 = 56.867 + j11.159 \text{ V}$$

$$6.867 + j11.159$$

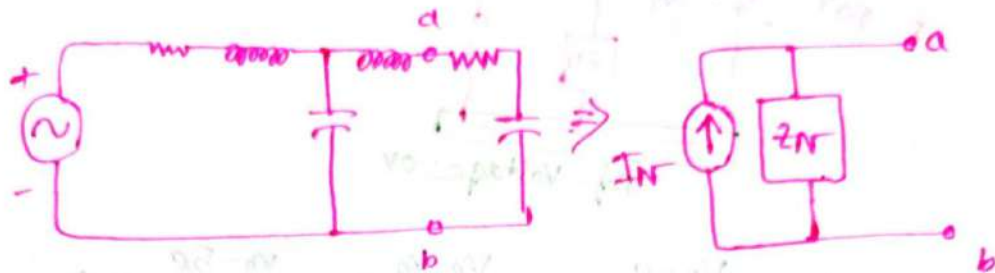
$$\therefore E_{th} = V_3 - 0 =$$

the thevenin equivalent:-

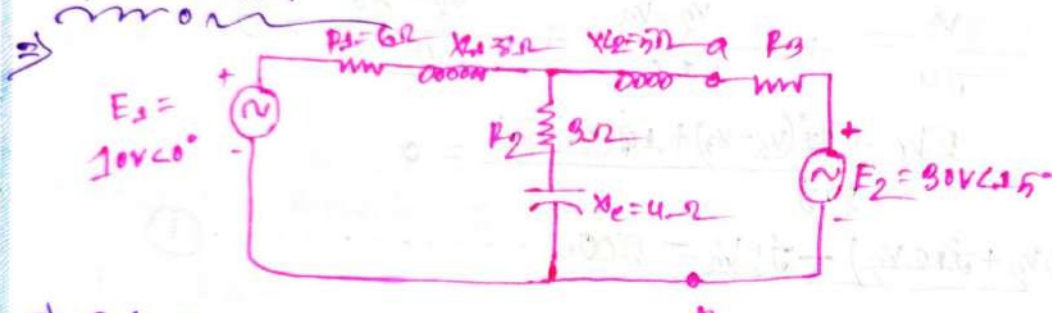


Norton's Theorem:—

⇒ The logic that was used for DC current is the same for AC current.

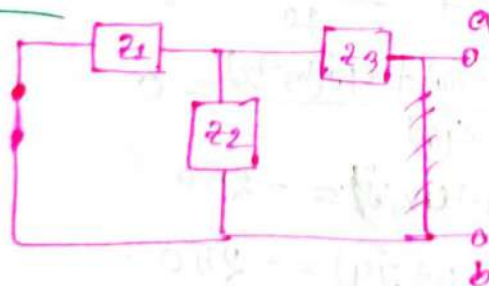


problem: 1:—



⇒ Calculate Norton's Equivalent for terminal a and b.

⇒ For Z_N :—



$$Z_1 = 6 + 8j \Omega$$

$$Z_2 = 3 - 4j \Omega$$

$$Z_3 = 5j \Omega$$

$$\begin{aligned} \therefore Z_N &= \left(\frac{1}{Z_1} + \frac{1}{Z_2} \right)^{-1} + Z_3 \\ &= \left(\frac{1}{6 + 8j} + \frac{1}{3 - 4j} \right)^{-1} + 5j \\ &= 4.639 + j2.038 \Omega \end{aligned}$$

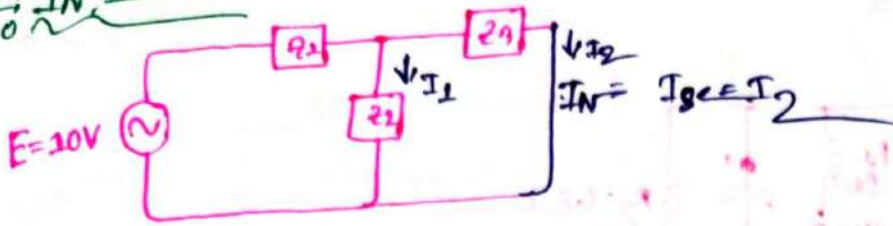
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TIME : _____

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For I_N mon



We have mon

$$Z_N = 4.639 + j2.938 \Omega$$

We know mon

$$Z_T = \left(\frac{1}{Z_3} + \frac{1}{Z_2} \right)^{-1} + Z_1$$

$$= \left(\frac{1}{5j} + \frac{1}{3-4j} \right)^{-1} + (6+8j)$$

$$= (13.5 + 10.5j) \Omega$$

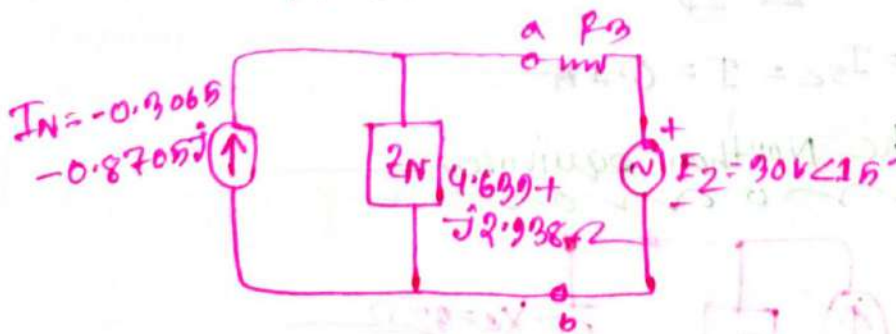
$$I_s = \frac{E}{Z_T} = \frac{10}{13.5 + 10.5j} = 0.461 - 0.358j \text{ A}$$

$$\therefore I_2 = \frac{\left(\frac{1}{Z_2} + \frac{1}{Z_3} \right)^{-1}}{(Z_3)} \times I_s$$

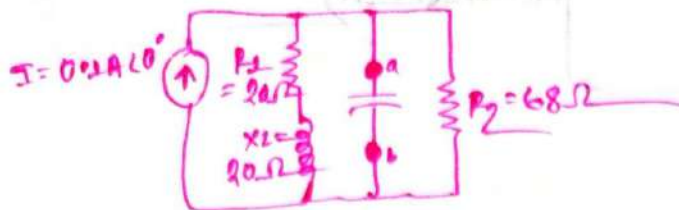
$$= \frac{7.5 + 2.5j}{5j} \times (0.461 - 0.358j)$$

$$I_N = I_s = I_2 = -0.3065 - 0.8705j \text{ A}$$

So, the Norton equivalent:

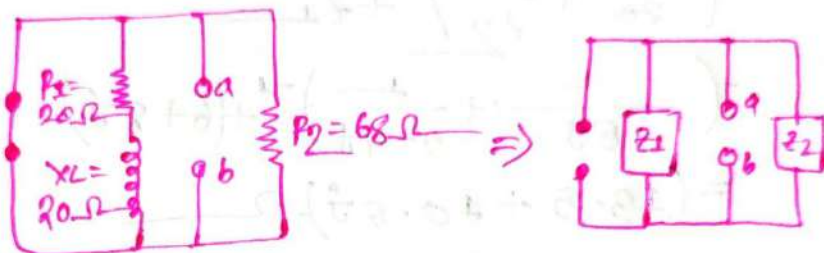


#problem: 21—



① Find the Norton equivalent for terminal a and b.

⇒ For Z_N !—



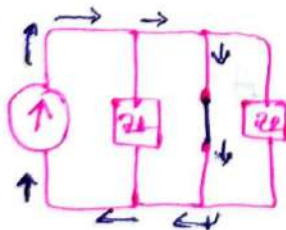
$$Z_1 = 20 + 20j \, \Omega$$

$$Z_2 = 68 \, \Omega$$

we know

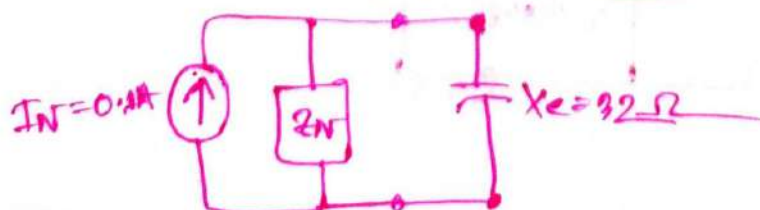
$$Z_N = \left(\frac{1}{Z_1} + \frac{1}{Z_2} \right)^{-1} = \left(\frac{1}{20 + 20j} + \frac{1}{68} \right)^{-1} = (18.095 + 11.955j) \, \Omega$$

For I_N !—

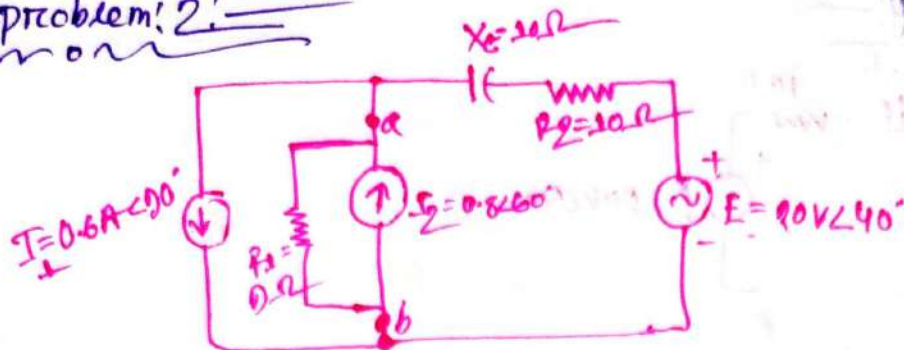


$$\therefore I_N = I_{sc} = I = 0.1 \, A$$

so, The the Norton equivalent!—



#problem 2:



① Find the Norton equivalent for the terminal a and b.

⇒ In given:

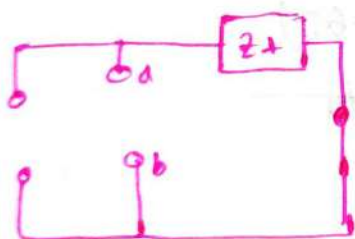
$$I_1 = 0.6A \angle 90^\circ = 0.6jA$$

$$X_L = -10j\Omega$$

$$R_L = 10\Omega$$

$$E = 90 \angle 40^\circ = 15.320 + j2.8585j$$

For Z_N :

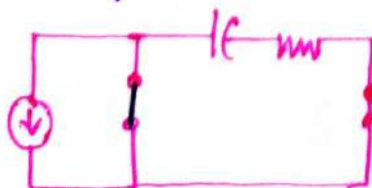


$$\therefore Z_N = Z_1 = 10 - 10j\Omega$$

For I_N :

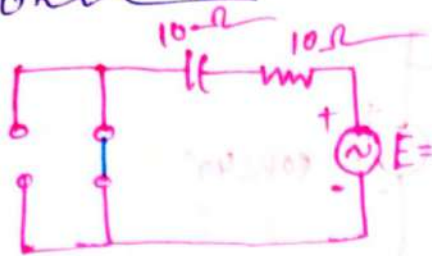
using Super position principle:

Consider only I_1 :



$$\therefore I_N = I_1 = 0.6jA \rightarrow \text{counter clockwise.}$$

Consider E only:—



$$Z_T = Z_1 = 10 - 10j \Omega$$

$$\therefore I_{sc} = I_N' = I_3 = \frac{E}{Z_T}$$

$$= \frac{15.320 + 12.8685j}{10 - 10j}$$

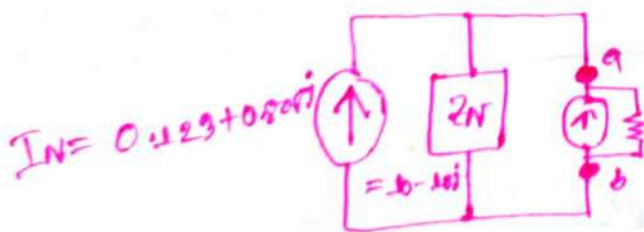
$$= 0.123 + 1.408j \text{ A} \rightarrow \text{clockwise.}$$

$$\therefore I_N = I_N'' - I_N'$$

$$= (0.123 + 1.408j) - (0.6j)$$

$$= 0.123 + 0.808j \text{ A}$$

So, the Norton equivalent:—



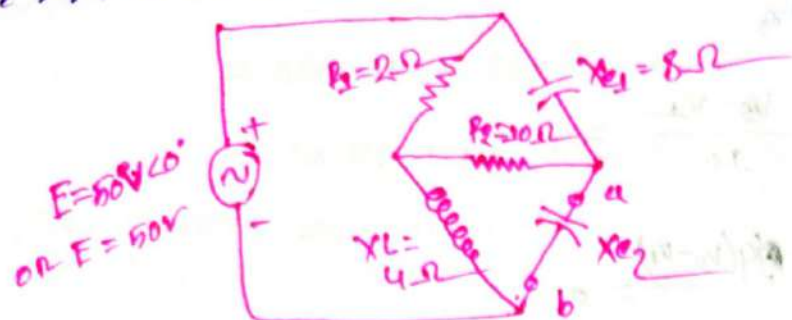
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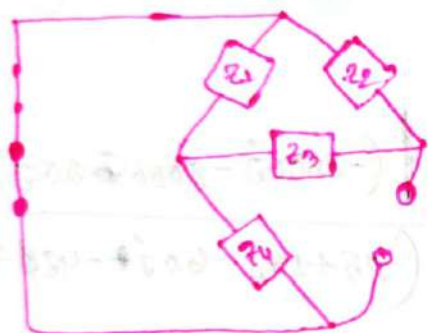
DATE : / /

#problem! no!

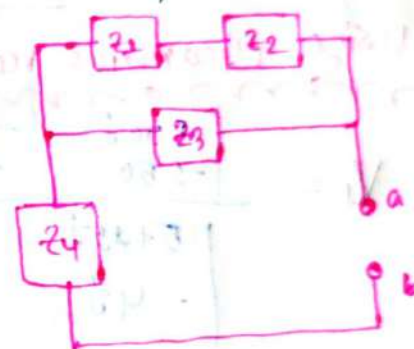


① Find the Norton equivalent circuit for the terminals a and b.

For Z_N :



$\Rightarrow$



here,

$$Z_1 = 2\Omega$$

$$Z_2 = -j8\Omega$$

$$Z_3 = 10\Omega$$

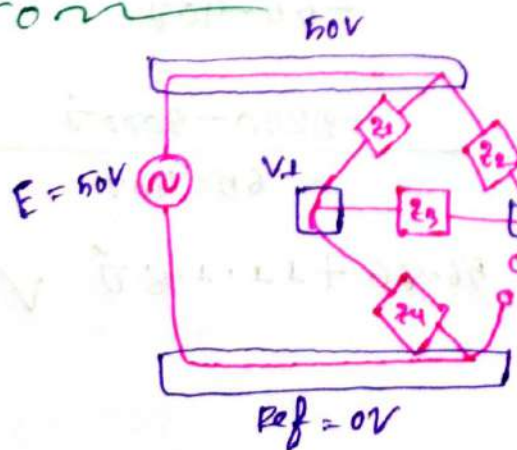
$$Z_4 = j4\Omega$$

$$\therefore Z_N = \left(-\frac{1}{Z_1 + Z_2} + \frac{1}{Z_3} \right)^{-1} + Z_4$$

$$= \left(\frac{1}{2 + (-j8)} + \frac{1}{10} \right)^{-1} + 4j$$

$$= 4.930 + 0.159j \Omega$$

For E_{th} :



$$\frac{V_1 - 50}{Z_1} + \frac{V_1 - V_2}{Z_3} + \frac{V_1 - 0}{Z_4} = 0$$

$$\Rightarrow \frac{V_1 - 50}{2} + \frac{V_1 - V_2}{10} + \frac{V_1}{j4} = 0$$

$$\Rightarrow \frac{10j(V_1 - 50) + 2j(V_1 - V_2) + 5V_1}{20j} = 0$$

$$\Rightarrow V_1(5 + 12j) - 22V_2 = j500 \quad \text{--- (1)}$$

Now $\frac{V_2 - 50}{Z_2} + \frac{V_2 - V_1}{Z_3} = 0$

$\Rightarrow \frac{V_2 - 50}{-j8} + \frac{V_2 - V_1}{10} = 0$

$\Rightarrow \frac{-5(V_2 - 50) + j4(V_2 - V_1)}{40j} = 0$

$\Rightarrow -5V_2 + j4V_2 + (-j4V_1) = -250$

$\Rightarrow -j4V_1 + V_2(-5 + j4) = -250 \dots \dots \dots \textcircled{1}$

Using Cramer's rule:

$$V_1 = \frac{\begin{vmatrix} 500j & -2j \\ -250 & -5 + 4j \end{vmatrix}}{\begin{vmatrix} 5 + 12j & -2j \\ -4j & -5 + 4j \end{vmatrix}} = \frac{(-2500j - 2000 - 500j)}{(-25 + 20j - 60j - 48 + 8)}$$

$$= \frac{-2500j - 2000 - 500j}{-65 - 40j}$$

$$= 42.018 + 19.742j \text{ V}$$

$$= \frac{-1250 - 3000j + 2000j}{-65 - 40j}$$

$$= \frac{-9250 - 1000j}{-65 - 40j}$$

$$= 56.86 + 11.158j \text{ V}$$

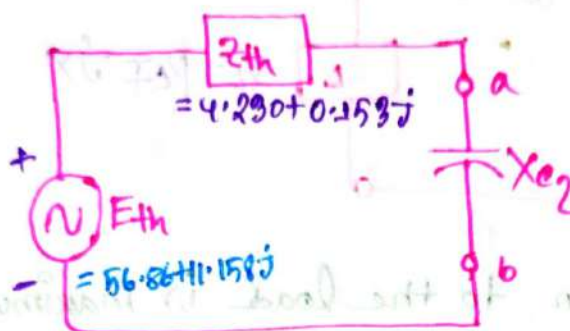
$$V_2 = \frac{\begin{vmatrix} 5 + 12j & 500j \\ -4j & -250 \end{vmatrix}}{\begin{vmatrix} 5 + 12j & -2j \\ -4j & -5 + 4j \end{vmatrix}}$$

So, $E_{th} = V_2 - 0$

$$= 56.86 + 11.158j - 0$$

$$= 56.86 + 11.158j$$

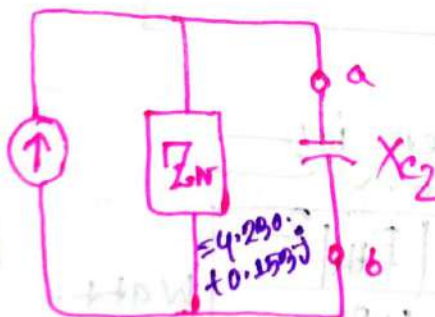
∴ the thevenin equivalent:



Now the Norton equivalent:

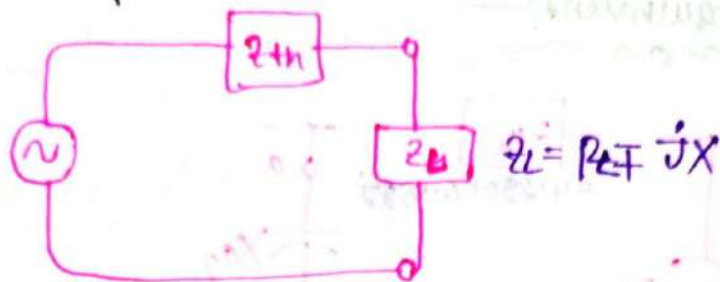
$$I_N = \frac{E_{th}}{Z_{th}}$$

$$= \frac{56.86 + 11.158j}{4.290 + 0.159j}$$



#Maximum Power Transfer theorem:

⇒ For Maximum power, Load impedance (Z_L) should be the Complex Conjugate of thevenin or Norton's impedance.



∴ The power transfer to the load is maximum

$$\text{if } R_L = R_{th} \text{ \& } X_L = -X_{th}$$

∴

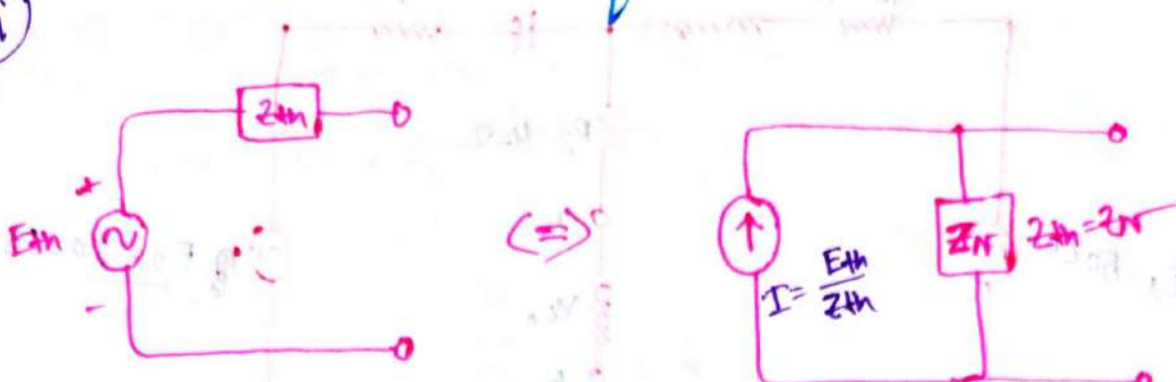
$$Z_L = R_L + jX$$

∴ The maximum power is:—

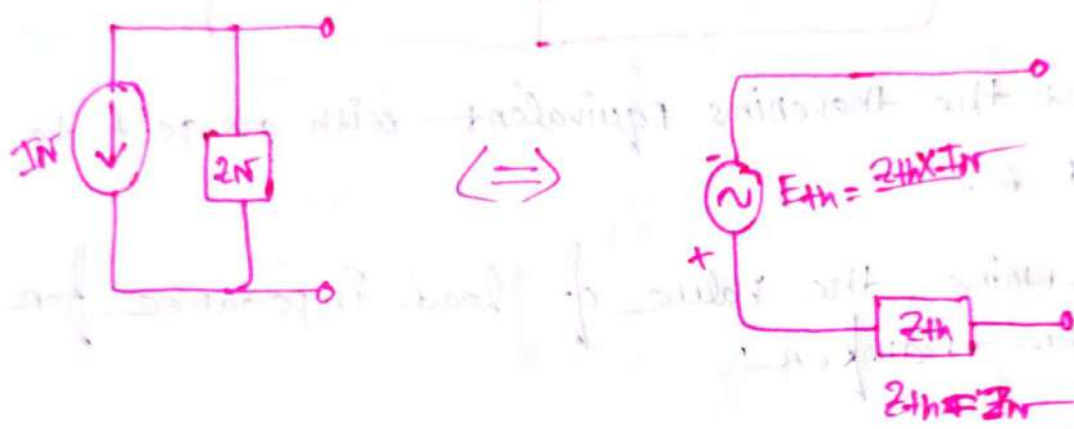
$$P_L = \frac{|E_{th}|^2}{4R_{th}} \text{ Watt.}$$

#Source Transformation

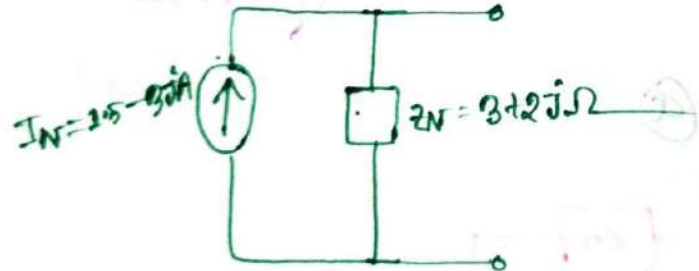
(i)



(ii)

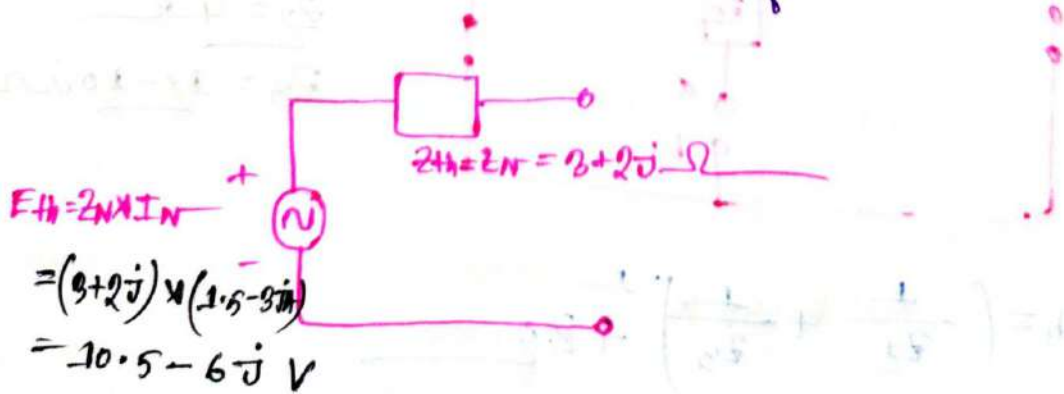


Ex:-



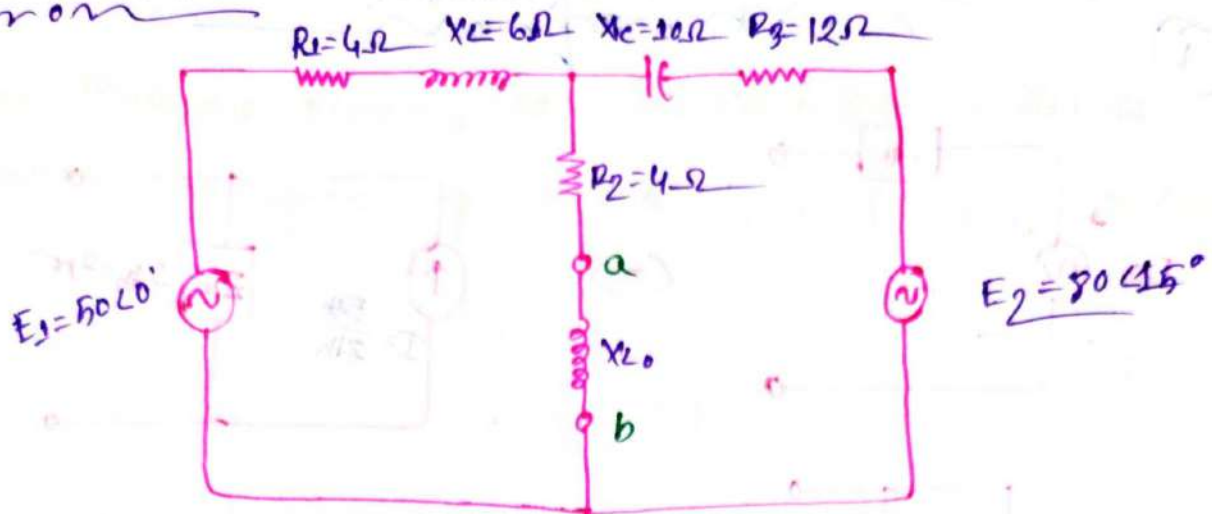
⇒ Convert the circuit into a voltage source.

⇒



Ans:-

Problem 1: _____

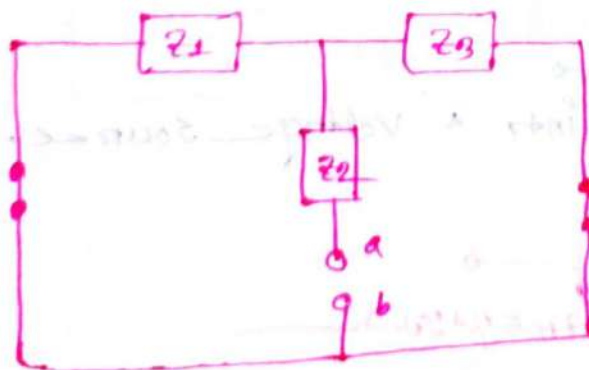


- Find the thevenin's Equivalent with respect to terminal a and b.
- Determine the value of load impedance for maximum power transfer.
- Determine the power consumed by load.

⇒ In given: _____

⇒ For Z_{th} : _____

①



$$Z_1 = 4 + 6j \Omega$$

$$Z_2 = 4 \Omega$$

$$Z_3 = 12 - 10j \Omega$$

$$\therefore Z_{th} = \left(\frac{1}{Z_1} + \frac{1}{Z_3} \right)^{-1} + Z_2$$

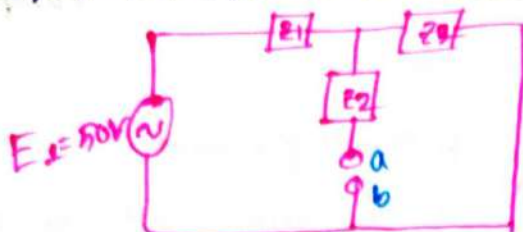
$$= \left(\frac{1}{4 + 6j} + \frac{1}{12 - 10j} \right)^{-1} + 4$$

$$= 9.882 + 3.4705j \Omega$$

For E_{th} :—

Using Super position principle :—

Consider E_1 only :—



We know :—

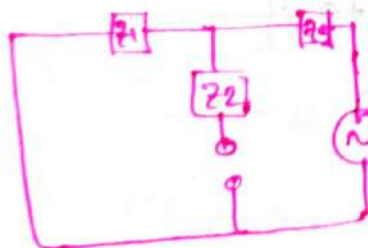
$$V_{Z2} = \frac{Z_2}{Z_1 + Z_2} \times E_1$$

$$= \frac{4 + 6j}{(12 - 10j) + (4 + 6j)} \times 50$$

$$= 42.647 - 20.588j \text{ Volt}$$

$$\therefore E_{ab1} = V_{Z2} = 42.647 - 20.588j \text{ V}$$

Consider E_2 only :—



$$E_2 = 80 \angle 15^\circ = 77.27 + 20.705j \text{ V}$$

We know :—

$$V_{Z2} = \frac{Z_2}{Z_1 + Z_2} \times E_2$$

$$= \frac{4 + 6j}{4 + 6j + 12 - 10j} \times (77.27 + 20.705j)$$

$$= 2.837 + 34.861j \text{ Volt}$$

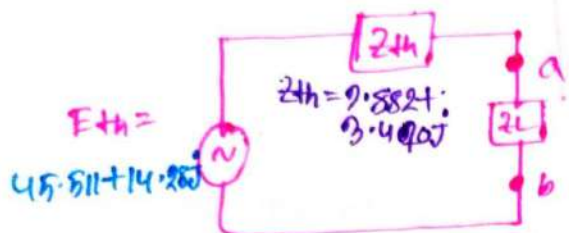
$$\therefore E_{ab2} = 2.837 + 34.861j \text{ Volt}$$

Now — $E_{ab} = E_{oc} = E_{th} = E_{ab1} + E_{ab2}$

$$= 42.648 - j20.58 + 2.838 + 34.863j$$

$$= 45.511 + j14.281 \text{ Volt}$$

For thevenin equivalent! —



we know —

For maximum power transfer —

$\Rightarrow Z_L$ should be the complex conjugate of Z_{th}

$$\therefore Z_L = 0.882 - j3.470 \Omega$$

$$R_L = R_{th} = 0.882$$

$$X_L = -X_{th} = -3.470 \Omega$$

we know —

power consumed by load =

$$P = \frac{|E_{th}|^2}{4R_{th}}$$

$$= \frac{(47.698)^2}{4 \times 0.882}$$

$$= 52.558 \text{ W}$$

$$|E_{th}| = \sqrt{(45.511)^2 + (14.28)^2}$$

$$= 47.698 \text{ V}$$

Ans! —

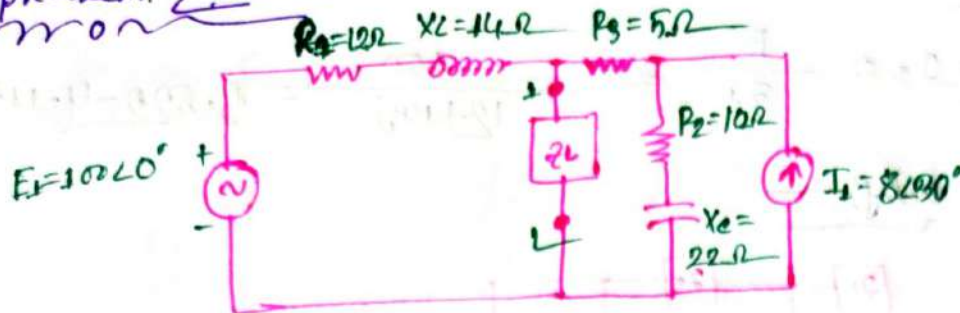
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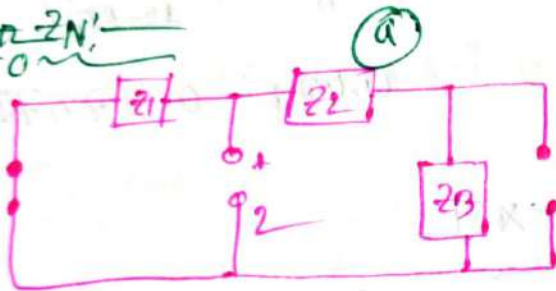
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#problem: 2:



- Find the Norton equivalent with respect to terminal 1 and 2.
- Determine the value of load impedance for maximum power transfer.
- Determine the power consumed by load.

⇒ In given for Z_N :



$$Z_1 = 12 + 14j \Omega$$

$$Z_2 = 5 \Omega$$

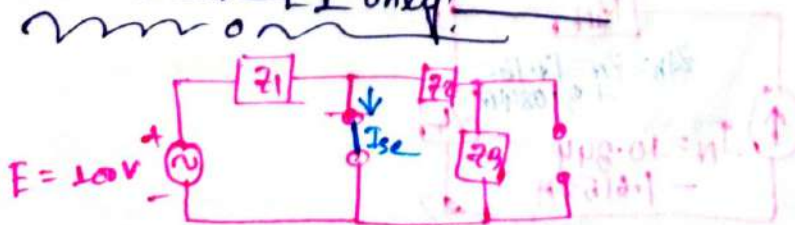
$$Z_3 = 10 - 22j \Omega$$

$$\begin{aligned} \therefore Z_N &= \left(\frac{1}{Z_2 + Z_3} + \frac{1}{Z_1} \right)^{-1} \\ &= \left(\frac{1}{5 + 10 - 22j} + \frac{1}{12 + 14j} \right)^{-1} \\ &= 17.160 + 9.084j \Omega \end{aligned}$$

For I_N :

Using Superposition principle:

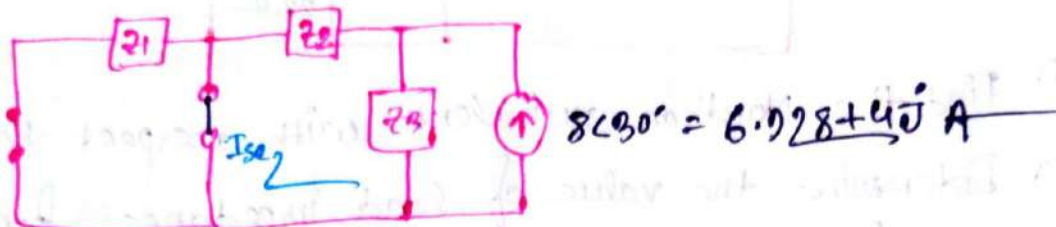
Consider E only:



We know,

$$I_{se} = I_s = \frac{E}{Z_1} = \frac{100}{12 + j4} = 3.522 - 4.117j \text{ A}$$

Consider I_1 only:



We know,

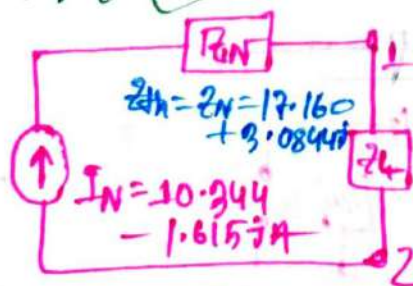
$$Z_T = \left(\frac{1}{Z_2} + \frac{1}{Z_3} \right)^{-1} = \left(\frac{1}{5} + \frac{1}{10 - j2} \right)^{-1} = 4.471 - 0.7757j \Omega$$

$$\begin{aligned} I_{se2} &= \frac{Z_T}{Z_2} \times I \\ &= \frac{4.471 - 0.7757j}{5} \times (6.028 + 4j) \\ &= 6.8155 + 2.5010j \text{ A} \end{aligned}$$

$$\therefore I_N = I_{se1} + I_{se2}$$

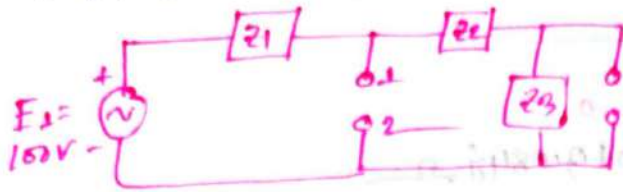
$$\begin{aligned} &= (3.522 - 4.117j + 6.8155 + 2.5010j) \text{ A} \\ &= 10.344 - 1.615j \text{ A} \end{aligned}$$

So, the Norton Equivalent:



For E_{th} :

Consider E_1 only:



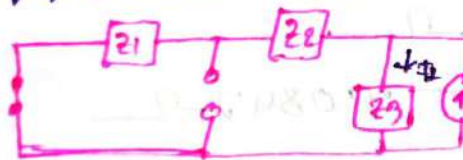
We know:

$$E'_{12} = \frac{Z_2 + Z_3}{Z_1 + Z_2 + Z_3} \times E_1$$

$$= \frac{(5 + 10 - 22j)\Omega}{(12 + 14j + 5 + 10 - 22j)\Omega} \times 150$$

$$= 73.26 - 59.773j \text{ V}$$

Consider I_1 only:



We know:

$$I_2 = \left(\frac{Z_T}{Z_1 + Z_2} \right) \times I$$

$$= \left(\frac{18.6355 - 9.145j}{12 + 14j + 5} \right) \times (6.028 + 4j) \text{ A}$$

$$= 6.480 - 2.2408j \text{ A}$$

$$Z_T = \left(\frac{1}{Z_1 + Z_2} + \frac{1}{Z_3} \right)^{-1}$$

$$= \left(\frac{1}{12 + 14j + 5} + \frac{1}{10 - 22j} \right)^{-1}$$

$$= -9.145j + 18.6355\Omega$$

$$V_{21} = E''_{12} = I_2 \times Z_1$$

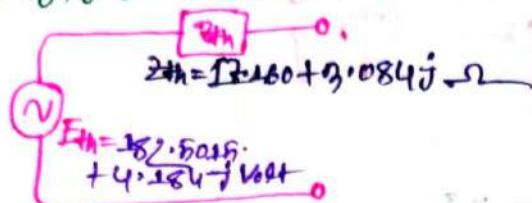
$$= (6.480 - 2.2408j) \times (12 + 14j)$$

$$= 109.2415 + 63.0577j$$

$$\therefore E_{12} = E'_{12} + E''_{12}$$

$$= 182.5015 + 4.184j$$

$\therefore$ thevenin equivalent:



(b)

We know:

For maximum power transfer:

$\Rightarrow Z_L$ should be the complex conjugate of Z_{th}

$$\therefore Z_L = 17.160 - 3.084j \Omega$$

$$R_L = 17.160 \Omega$$

$$X_L = -X_{th} = -3.084j \Omega$$

(c)

We know:

power consumed by the load:

$$P_L = \frac{|E_{th}|^2}{4R_{th}}$$

$$= \frac{(17.434)^2}{4 \times 17.160}$$

$$= 4.428 \text{ W}$$

$$|E_{th}| = \sqrt{(182.5015)^2 + (4.184)^2}$$

$$= 17.434 \text{ V}$$

Ans

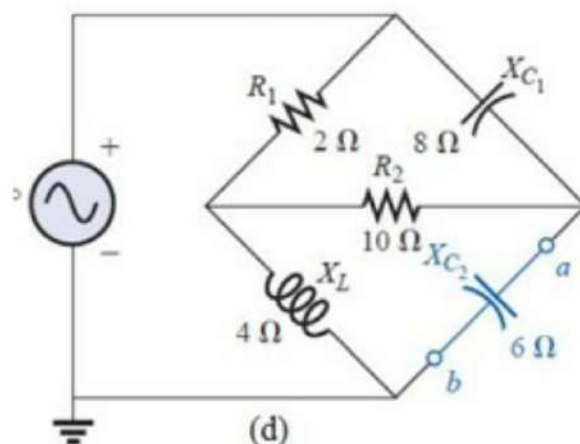
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TIME : _____ DATE : / /

Methods of Analysis Geoog

1. Find the Norton's equivalent circuit for the portions of the networks of following figure external to the elements between points a and b. Also determine the load impedance for a and b for maximum power transfer & find the maximum power transferred to the load. The supply voltage is **1.2** volts (assume initial angle to be zero) and frequency is 50 Hz.

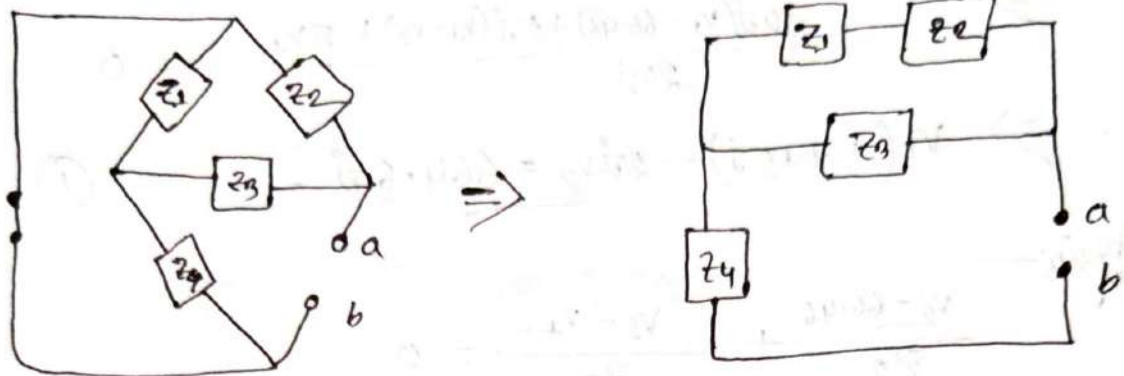


2. For the circuit above, determine the current across the capacitor in terminals a & b using- (i) node voltage or nodal analysis and (ii) mesh current analysis. (**Hint:** same methods of nodal & mesh analysis as in DC circuits in mid term, except that all voltage, currents & impedance has to be considered as complex entities with magnitude & phase angle.)

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Answer: the photo

i



Here:-

$$Z_1 = 2 \Omega$$

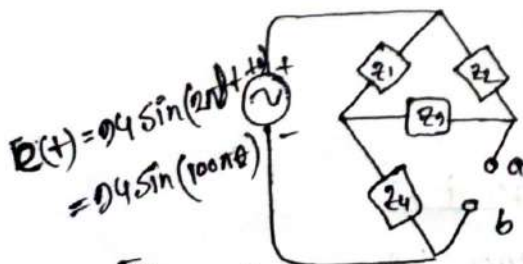
$$Z_2 = -8j \Omega$$

$$Z_3 = 10 \Omega$$

$$Z_4 = 4\sigma\Omega$$

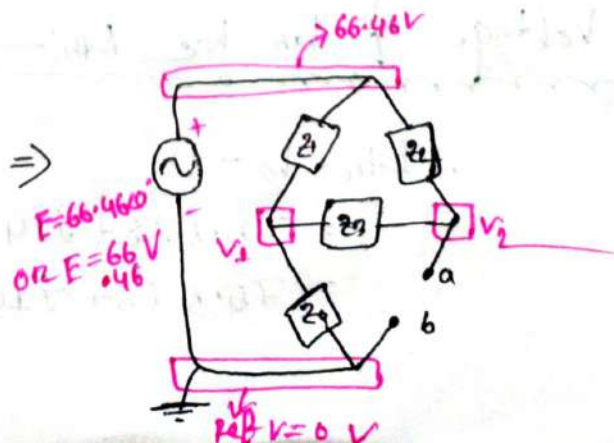
$$\begin{aligned}\text{So, the } Z_N = Z_{Th} &= \left(\frac{1}{Z_1 + Z_2} + \frac{1}{Z_3} \right)^{-1} + Z_4 \\ &= \left(\frac{1}{2 + (-8j)} + \frac{1}{10} \right)^{-1} + 4j \\ &= 4.2307 + 0.1508j \, \Omega\end{aligned}$$

For Eth: —



$$\therefore E = \frac{F_m}{\sqrt{2}} \angle 0^\circ$$

$$= \frac{0.4}{\sqrt{2}} \angle 0^\circ = 66.46 \angle 0^\circ$$



Using Nodal Analysis:-

$$\frac{V_1 - 66.46}{Z_1} + \frac{V_1 - V_2}{Z_3} + \frac{V_1 - 0}{Z_4} = 0$$

$$\Rightarrow \frac{V_1 - 66.46}{2} + \frac{V_1 - V_2}{10} + \frac{V_1}{j4} = 0$$

$$\Rightarrow \frac{10j(V_1 - 66.46) + 2j(V_1 - V_2) + 5V_1}{20j} = 0$$

$$\Rightarrow V_1(5 + 12j) - 2jV_2 = 664.6j \quad \text{--- (i)}$$

Again:-

$$\frac{V_2 - 66.46}{Z_2} + \frac{V_2 - V_1}{Z_3} = 0$$

$$\Rightarrow \frac{V_2 - 66.46}{-j8} + \frac{V_2 - V_1}{10} = 0$$

$$\Rightarrow \frac{-5(V_2 - 66.46) + 4j(V_2 - V_1)}{40j} = 0$$

$$\Rightarrow -4jV_1 + V_2(-5 + 4j) = -332.3 \quad \text{--- (ii)}$$

Now solve eq (i) and (ii) $\Rightarrow$

$$\therefore V_1 = 57.047 + j26.242 \text{ V}$$

$$V_2 = 75.588 + j14.832 \text{ V}$$

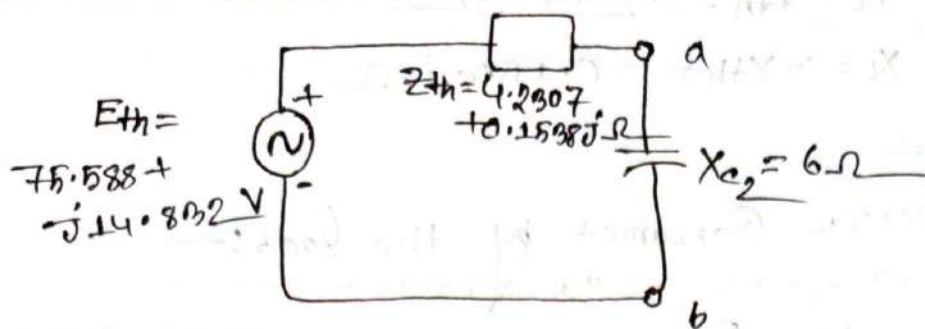
So, Voltage of $V_{ab} = V_{oc} = E_{th}$:-

$$\therefore E_{th} = V_2 - 0$$

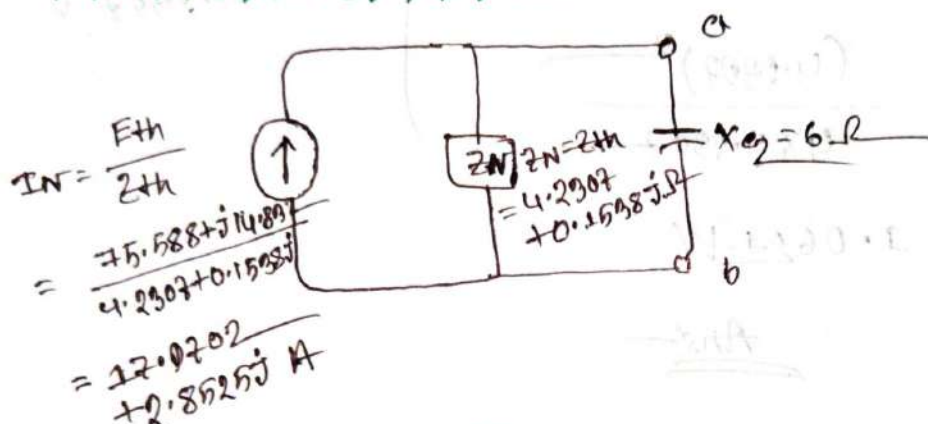
$$= 75.588 + j14.832 + 0$$

$$= 75.588 + j14.832 \text{ V}$$

∴ The thevenin equivalent!—



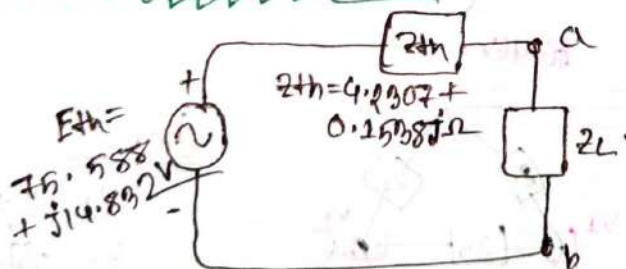
So, the Norton equivalent!—



ii

We have!—

The thevenin equivalent!—



We know!—

⇒ For maximum power transfer!—

Z_L should be the Complex Conjugate of Z_{th}

$$\therefore Z_L = 4.2307 - 0.1538j \Omega$$

∴ The load impedance for a and b for maximum power transfer $Z_L = 4.2307 - 0.1538j \Omega$

Now,

here,

$$R_L = R_{th} = 4.2307 \Omega$$

$$X_L = -X_{th} = -0.1538 j \Omega$$

Now, we know,

power consumed by the load:-

$$P_L = \frac{|E_{th}|^2}{4R_{th}}$$

$$= \frac{(4.2307)^2}{4 \times 4.2307}$$

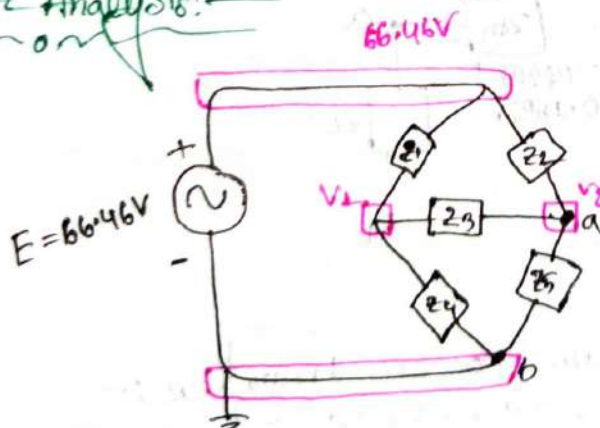
$$= 1.0621 W$$

$$|E_{th}| = \sqrt{(4.2307)^2 + (0.1538)^2} = 4.2307 V$$

Ans:-

Ans: to the Q no. 2

NODAL Analysis:-



here

$$Z_1 = 2 \Omega$$

$$Z_2 = -8j \Omega$$

$$Z_3 = 10 \Omega$$

$$Z_4 = 4j \Omega$$

$$Z_5 = -6j \Omega$$

Now,

$$\frac{V_1 - 66.46}{Z_1} + \frac{V_1 - V_2}{Z_3} + \frac{V_1 - 0}{Z_4} = 0$$

$$\Rightarrow \frac{10j(V_1 - 66.46) + 2j(V_1 - V_2) + 5V_1}{20j} = 0$$

$$\Rightarrow V_1(5 + 12j) - 2jV_2 = 664.6j \quad \text{--- (1)}$$

Now:

$$\frac{V_2 - 66.46}{Z_2} + \frac{V_2 - V_1}{Z_3} + \frac{V_2 - 0}{Z_5} = 0$$

$$\Rightarrow \frac{V_2 - 66.46}{-8j} + \frac{V_2 - V_1}{10} + \frac{V_2}{-6j} = 0$$

$$\Rightarrow \frac{-15(V_2 - 66.46) + 12j(V_2 - V_1) - 20 V_2}{120j} = 0$$

$$\Rightarrow -15V_2 + 996.9 + 12jV_2 - 12jV_1 - 20V_2 = 0$$

$$\Rightarrow -12jV_1 + V_2(-35 + 12j) = -996.9$$

$$\Rightarrow -12jV_1 + V_2(-35 + 12j) = -996.9 \quad \dots \dots \dots (11)$$

Now solve the equation (i) and (ii) $\Rightarrow$

$$\therefore V_1 = 52.881 + 21.139j \text{ V}$$

$$V_2 = 37.529 - 5.2462j \text{ V}$$

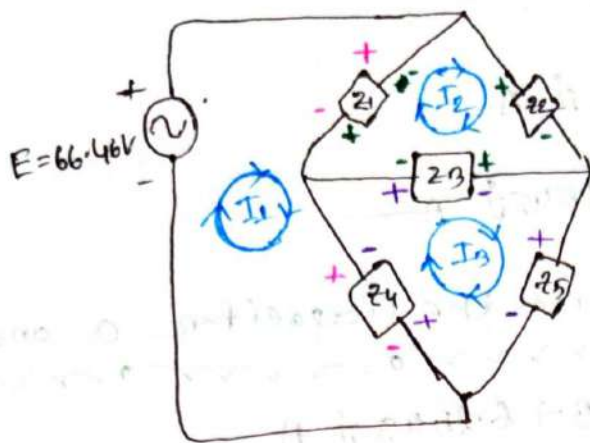
So, the voltage difference between Z_5 :

$$\begin{aligned} V_{Z_5} &= V_2 - 0 \\ &= 37.529 - 5.2462j \text{ V} \end{aligned}$$

So, the current across the capacitor in terminals a and b:

$$\begin{aligned} I_{ab} &= \frac{V_{Z_5}}{Z_5} = \frac{V_{Z_5}}{-6j} = \frac{37.529 - 5.2462j}{-6j} \\ &= 0.8743 + 6.254j \text{ A} \end{aligned}$$

(ii)

Mesh Analysis:Loop 1:

$$\Rightarrow -66.46 + Z_1(I_1 - I_2) + Z_4(I_1 - I_3) = 0$$

$$\Rightarrow -66.46 + 2(I_1 - I_2) + 4j(I_1 - I_3) = 0$$

$$\Rightarrow 2I_1 - 2I_2 + 4jI_1 - 4jI_3 = 66.46$$

$$\Rightarrow I_1(2 + 4j) - 2I_2 - 4jI_3 = 66.46 \quad \text{--- (i)}$$

Loop 2:

$$Z_2 I_2 + Z_3(I_2 - I_3) + Z_1(I_2 - I_1) = 0$$

$$\Rightarrow -8jI_2 + 10(I_2 - I_3) + 2(I_2 - I_1) = 0$$

$$\Rightarrow -8jI_2 + 10I_2 - 10I_3 + 2I_2 - 2I_1 = 0$$

$$\Rightarrow -2I_1 + I_2(12 - 8j) - 10I_3 = 0 \quad \text{--- (ii)}$$

Loop 3:

$$Z_5 I_3 + Z_4(I_3 - I_1) + Z_3(I_3 - I_2) = 0$$

$$-6jI_3 + 4j(I_3 - I_1) + 10(I_3 - I_2) = 0$$

$$-6jI_3 + 4jI_3 - 4jI_1 + 10I_3 - 10I_2 = 0$$

$$-4jI_1 - 10I_2 + I_3(10 - 2j) = 0 \quad \text{--- (iii)}$$

Now solve the equations (I), (II), (III) $\Rightarrow$

$$\therefore I_1 = 6.1589 - 6.292j \text{ A}$$

$$I_2 = -0.655 + 3.616j \text{ A}$$

$$I_3 = 0.8943 + 6.2548j \text{ A}$$

So the current across the capacitor a and b terminals

$$I_{ab} = I_3 = 0.8943 + 6.2548j \text{ A}$$

Ans:-

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#Electromagnetism
and Fundamental
Laws.

#Magnetic field:-

⇒ The area around the magnetic field where the poles experience the force of attraction and repulsion is known as magnetic field.

Note:- The alternative name of magnetic field is magnetic flux density that is denoted by B

$$\therefore B = \frac{\Phi}{A} \quad \text{Wb/m}^2 \text{ or T}$$

B = magnetic field Wb/m^2 or T

Φ = magnetic flux Wb (weber)

A = Area

m^2

#Magnetic Flux:-

⇒ Magnetic flux is the measurement of the total magnetic field which passing through a given area. denoted by Φ

$$\therefore \Phi = AB \quad \text{Wb}$$

Faraday's First Law:-

⇒ Any change in the magnetic field of a coil of wire will cause an EMF to be induced in the coil. This EMF induced is called Induced EMF and if the Conductor circuit is closed, a current will also calculate through the circuit. And the current is called Induced current.

Faraday's Second Law:-

Induced emf $\boxed{\mathcal{E} = N \times \frac{d\Phi}{dt}}$

$\mathcal{E}$ = Induced emf

N = Number of turns in coil.

$d\Phi$ = change in magnetic flux.

Lenz's Law:-

Induced emf $\boxed{\mathcal{E} = -N \frac{d\Phi}{dt}}$

Note:- The negative sign indicates that the induced emf and the change in magnetic flux have opposite signs.

Some Definitions Related to Electromagnetic Induction:-

Magnetomotive Force:- or Ampere-turns:-

⇒ The product of number of turns of a coil and current is called MMF.

$$\boxed{MMF = NI} \quad [AT]$$

Magnetic field strength (H):-

⇒ The magnetomotive force per unit length is called magnetizing force.

$$\boxed{H = \frac{MMF}{l} = \frac{NI}{l}} \quad [AT/m]$$

N = No. of turns.

I = Current.

l = length.

Absolute permeability (μ)

$$\boxed{\mu = \frac{B}{H}} \quad [H/m]$$

B = flux density.

Magnitude of Dynamic Induced EMF

$$E = Blv \sin \theta \quad [\text{Volt, V}]$$

B = flux density,

l = length of conductor.

v = conductor velocity

θ = Angle between flux line and the direction of motion of conductor.

problem: -

⇒ A conductor moves with a velocity of 15 m/s at an angle of (a) 90° , (b) 60° , (c) 30° , to a magnetic field produced between two square forced poles of side length 2 cm. If the $\Phi = 5 \mu\text{Wb}$ Find the magnitude of the induced e.m.f in each case.

⇒ In given: -

$$v = 15 \text{ m/s}$$

$$l = 2 \text{ cm} = 0.02 \text{ m}$$

$$\Phi = 5 \times 10^{-6} \text{ Wb}$$

$$A = l^2 = (0.02)^2 = 4 \times 10^{-4} \text{ m}^2$$

We know: -

$$\theta = 90^\circ$$

$$B = \frac{\Phi}{A} = \frac{5 \times 10^{-6}}{4 \times 10^{-4}} = 12.5 \times 10^{-3} \text{ T}$$

$$E = Blv \sin \theta = 12.5 \times 10^{-3} \times 0.02 \times 15 \times \sin 90^\circ = 3.75 \text{ mV}$$

(b)

$$\theta = 60^\circ$$

$$E_{60} = B l v \sin 60$$

$$= 13.75 \text{ m} \sin 60^\circ$$

$$= 11.92 \text{ m V}$$

(c)

$$\theta = 90^\circ$$

$$E_{90} = B l v \sin 90^\circ = E_{90} \sin 90^\circ$$

$$= 13.75 \text{ m} \sin 90^\circ$$

$$= 13.75 \text{ m V}$$

Magnitude of Force:-

$$F = B l I \sin \theta \quad [\text{Newton, N}]$$

problem: 2.5:-

Calculate the force experienced by the conductor of 20 cm long, carrying 50 amperes, placed at right angles to the lines of force of flux density $10 \times 10^3 \text{ wb/m}^2$

⇒ In given

$$l = 20 \text{ cm} = 20 \times 10^{-2} \text{ m}$$

$$I = 50 \text{ A}$$

$$B = 10 \times 10^3$$

$$\theta = 90^\circ$$

We know:-

$$F = B l I \sin \theta$$

$$= 50 \times 10 \times 10^3 \times 20 \times 10^{-2} \times \sin (90^\circ)$$

$$= 0.1 \text{ N}$$

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Problem: 3:—

⇒ A conductor 350 mm long carries a current of 10 A and is at right-angles to a magnetic field lying between two circular pole faces each of radius 60 mm. If the total flux between the pole faces is 0.5 mWb, Calculate the magnitude of the force exerted on the conductor.

⇒ In given:—

$$l = 350 \text{ mm} = 350 \times 10^{-3} \text{ m}$$

$$I = 10 \text{ A}$$

$$r = 60 \text{ mm} = 60 \times 10^{-3} \text{ m}$$

$$A = \pi r^2 = \pi (60 \times 10^{-3})^2 = 0.0113 \text{ m}^2$$

$$\Phi = 0.5 \times 10^{-3} \text{ Wb}$$

$$\theta = 90^\circ$$

we know:—

$$F = B l I = \left(\frac{\Phi}{A} \right) l I$$

$$= \left(\frac{0.5 \times 10^{-3}}{0.0113} \right) \times 350 \times 10^{-3} \times 10$$

$$= 0.154703 \text{ N}$$

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D.C Generator
Ch: 20/20

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#DC Machines:-

⇒ There are two types of DC Machines:-

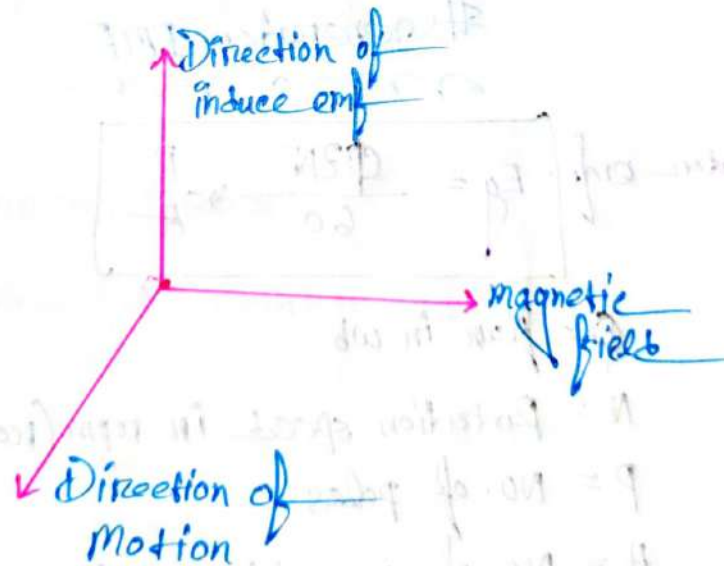
1. DC Generator

2. DC Motor

DC Generator:-

⇒ Generator is an electrical machine which converts mechanical energy into a electrical Energy. Mechanical power is provide via engine or turbines.

⇒ Fleming's right hand rule is used to determine induced current direction.

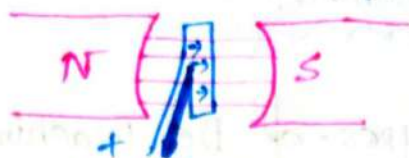


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Stationary
electromagnetic
with moving conductor coil.

$\therefore$ Induced EMF $\propto \frac{d\phi}{dt}$

$$\therefore \text{EMF} = N \frac{d\phi}{dt} = L \frac{di}{dt}$$

ϕ = flux (wb)

$\frac{d\phi}{dt}$ = rate of change of flux.

Generator EMF

$$\text{Generator emf} = E_g = \frac{\phi Z N}{60} \times \frac{P}{A}$$

ϕ = flux in wb

N = Rotation speed in rpm (rotation per minute)

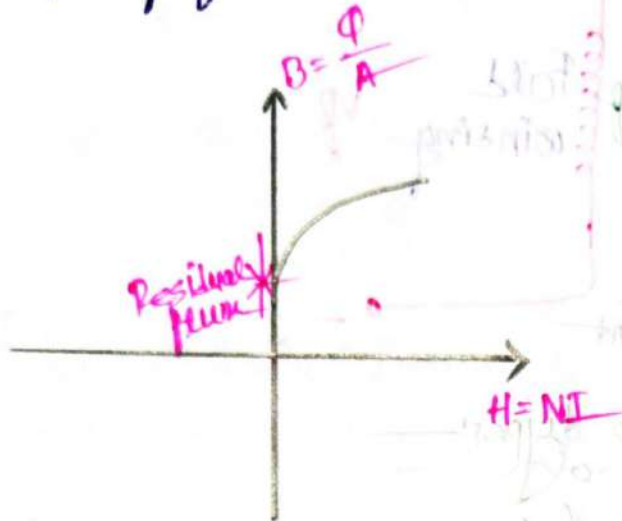
P = NO. of poles.

A = NO. of magnetic path/parallel path.

Z = NO. of total conductor.

Q For electro magnetic:-

$\Phi \propto I_f$ field current.



Q Generator of EMF:-

$$E_g = \frac{\Phi Z N}{60} \times \frac{P}{A}$$

$$= k_a \omega \Phi$$

$$= k \omega \Phi$$

ω_m = Armature rotation = $\frac{2\pi N}{60}$

$k_a = \frac{PZ}{60A}$

$k_\omega = \frac{PZ}{2\pi A}$

$I_f = I_a$ for shunt wound DC motor.

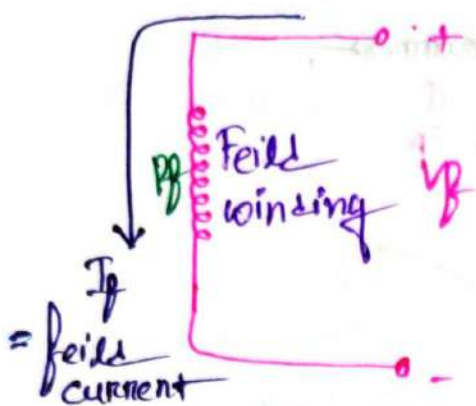
$= I$ for series wound DC motor.

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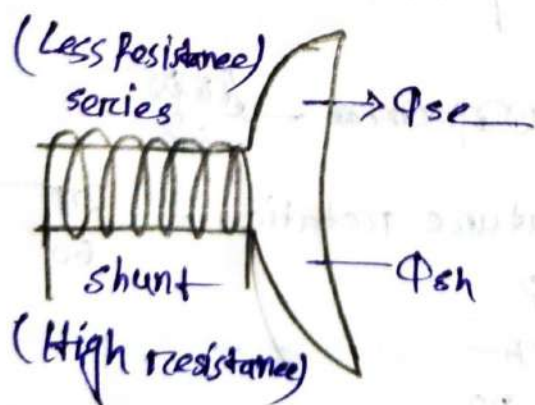


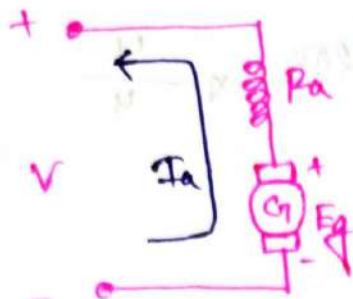
There are two types:

① Separately

② Self excited

Series Shunt Compound



Armature Separately:From: $E_g = V$

$$E_g = V + I_a R_a + \text{Brush drop}$$

problem:

For a four pole DC separately excited generator, the flux is 4 mWb , the rotational speed is 1200 rpm , the number of conductors per slot is 20 , the number of slots are 120 , the generator is lap wound. If the armature resistance is $0.2\ \Omega$ and brush drop is 2 V per brush. Consider Armature current 4 A . Determine,

- (i) The generator emf
- (ii) The terminal voltage.
- (iii) Armature copper loss.

➔ In given:

$$\Phi = 4\text{ mWb} = 4 \times 10^{-3}\text{ Wb}$$

$$P = 4$$

$$N = 1200\text{ rpm}$$

$$Z = 120 \times 20 = 2400$$

$$A = P = 4$$

For lap winding

$$A = P$$

For wave winding

$$A = 2$$

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We know:-

$$E_g = \frac{\phi Z N}{60} \times \frac{P}{A}$$

$$= \frac{4 \times 10^{-3} \times 2400 \times 1200}{60} \times \frac{4}{4}$$

$$= 102 \text{ V}$$

(ii)

We know:-

$$E_g = V + I_a R_a + \text{Brush Drop}$$

$$V = E_g - I_a R_a - \text{Brush Drop}$$

$$= 102 - 4 \times 0.2 - 4$$

$$= 187.2 \text{ V}$$

In given:-

$$R_a = 0.2 \Omega$$

$$I_a = 4 \text{ A}$$

brush drop = 2V per termin

$\therefore 4 \times 4 = 2 \times 2 \text{ V} \rightarrow \text{Both}$
24V termin

(ii)

We know:-

$$\text{Armature copper loss} = I_a^2 R_a$$

$$= (4)^2 \times 0.2$$

$$= 3.2 \text{ W}$$

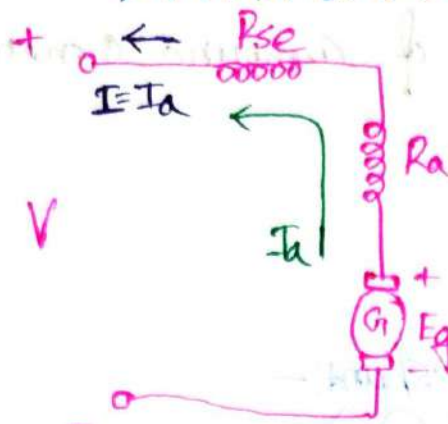
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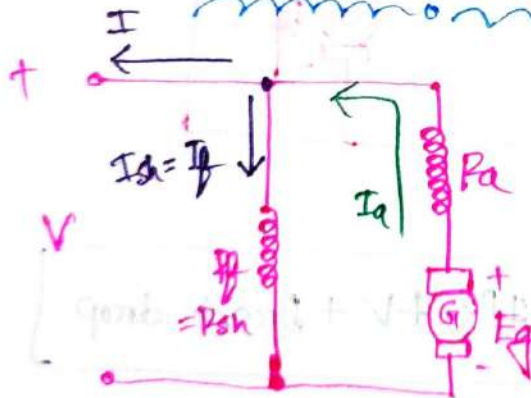
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Series Generator :



$$\therefore E_g = V + I_a R_a + I_a R_{se} + \text{Brush Drop}$$

Shunt Generator :



Shunt Generator is like a separately field winding.

$$\therefore E_g = V + I_a R_a + \text{Brush drop}$$

$$I = I_f + I_a$$

$$I_a = I - I_f$$

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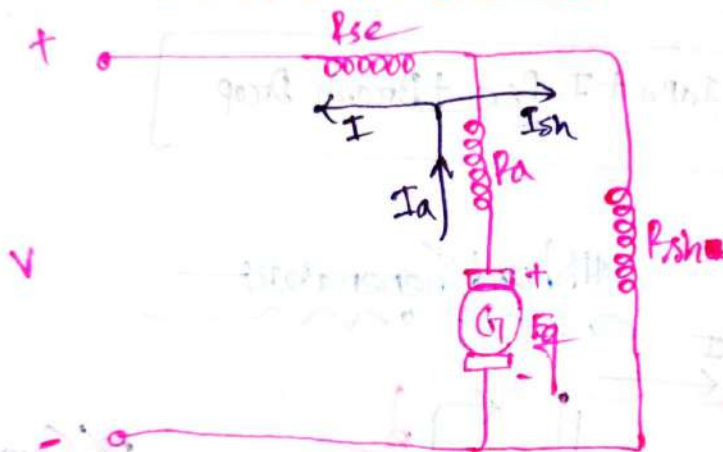
Compound Generator :-

⇒ There are two types of compound generator :-

1. Short shunt

2. Long shunt

Short shunt :-

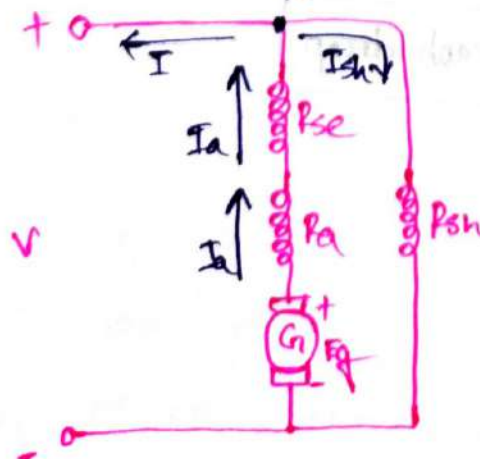


$$I_{sh} = \frac{V_{sh}}{R_{sh}}$$

$$V_{sh} = E_g - I_a R_a - BD$$

$$\therefore E_g = I_a R_a + I_f R_{se} + V + \text{Brush drop}$$

Long shunt :-



$$\therefore E_g = I_a R_a + I_f R_{se} + V + \text{Brush Drop}$$

$$I_{sh} = \frac{V}{R_{sh}}$$

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#problem: 2:-

⇒ For $E_g = 102 \text{ V}$ assuming $R_a = 0.2 \Omega$, $R_{se} = 0.2 \Omega$, $R_{sh} = 250 \Omega$
 $I_a = 4 \text{ A}$ Determine the terminal voltage for

- (i) Series generator.
- (ii) Long shunt generator.
- (iii) Short shunt generator.

⇒ In given:

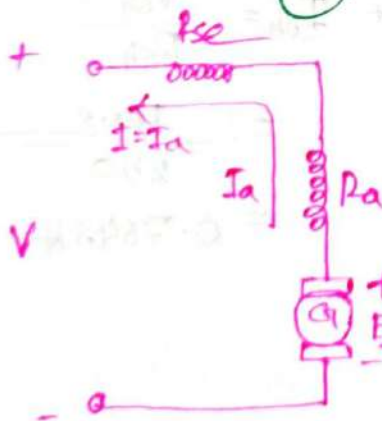
$$E_g = 102 \text{ V}, R_a = 0.2 \Omega$$

$$R_{se} = 0.2 \Omega$$

$$R_{sh} = 250 \Omega$$

$$I_a = 4 \text{ A}$$

(i)

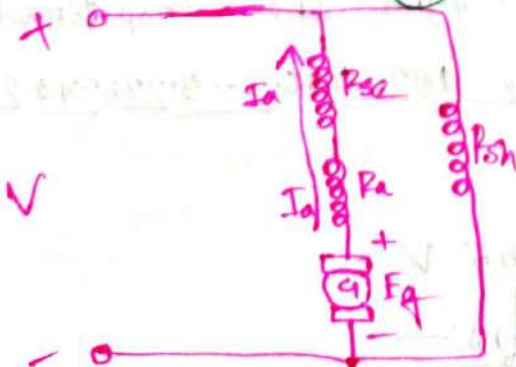


we know

$$E_g = R_a I_a + I_a R_{se} + \text{Brush drop} + V$$

$$\begin{aligned} \therefore V &= E_g - R_a I_a - I_a R_{se} - 0 \\ &= 102 - 4(0.2 + 0.2) - 0 \\ &= 100.4 \text{ V} \end{aligned}$$

(ii)

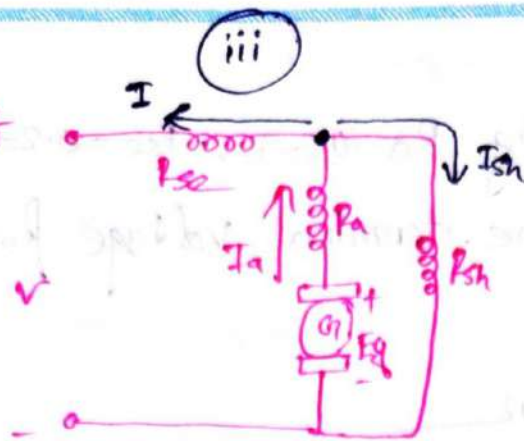


we know

$$E_g = I_a R_a + I_a R_{se} + V + \text{Brush drop}$$

$$\begin{aligned} \therefore V &= E_g - I_a R_a - I_a R_{se} - 0 \\ &= 102 - 4(0.2 + 0.2) \\ &= 100.4 \text{ V} \end{aligned}$$

Ans:-



∴ Voltage across the shunt field :—

$$V_{sh} = E_g - I_a R_a - \text{Brush drop}$$

$$= 102 - 4 \times 0.2 - 0$$

$$= 101.2 \text{ V}$$

$$\text{Shunt field current } I_{sh} = \frac{V_{sh}}{R_{sh}}$$

$$= \frac{101.2}{250}$$

$$= 0.7648 \text{ A}$$

$$\therefore I = I_a - I_{sh}$$

$$= 4 - 0.7648$$

$$= 3.2352 \text{ A}$$

$$\therefore V = V_{sh} - I R_{se} = E_g - I_a R_a - I R_{se} - \text{Drop, Brush}$$

$$= 101.2 - 3.2352 \times 0.2 = 102 - 4 \times 0.2 - 3.2352 \times 0.2 - 0$$

$$= 100.55 = 100.55 \text{ V}$$

Ans:—

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#problem: 26.8:-

⇒ A four-pole generator, having wave-wound armature winding has 51 slots, each slot containing 20 conductors. What will be the voltage generated in the machine when driven at 1500 rpm assuming the flux per pole to be 70 mWb?

⇒ In given:-

$$\Phi = 70 \text{ mWb} = 7 \times 10^{-2} \text{ Wb}$$

$$p = 4 \text{ pole}$$

$$A = 2 \dots \text{for wave wound}$$

$$Z = 51 \times 20 = 1020$$

$$N = 1500 \text{ rpm}$$

We know:-

$$E_g = \frac{\Phi Z N}{60} \times \frac{p}{A}$$

$$= \frac{7 \times 10^{-2} \times 1020 \times 1500}{60} \times \frac{4}{2}$$

$$= 178.5 \text{ V}$$

power Stages of a DC Generator:-

★ Mechanical Efficiency:-

$$\eta_m = \frac{B}{A} = \frac{E_g I_a}{\text{output of driving Engine}} \times 100\%$$

$$\text{Iron and friction losses} = A - B$$

$$= \text{mechanical power supplied} - E_g I_a$$

Electrical Efficiency:

$$\eta_e = \frac{C}{B} = \frac{VI}{E_g I_a} \times 100\%$$

C = watts available in load circuit

B = total watts generated.

$$C_u (\text{Copper}) \text{ Losses} = B - C$$

$$C_u \text{ Losses} = E_g I_a - VI$$

Overall or Commercial Efficiency:

$$\eta_c = \frac{C}{A} = \frac{VI}{A} \times 100\%$$

C = watts available in load circuit

A = mechanical power supplied.

Note: $\eta_c = \eta_m \times \eta_e$

$$\left[\frac{C}{A} = \left(\frac{C}{B} \right) \left(\frac{B}{A} \right) \right]$$

$$\left[\eta_c = \left(\eta_e \right) \left(\eta_m \right) \right]$$

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#problem: 8:-

5) A 20 kW compound generator works on full load with a terminal voltage of 230 V. The armature, series and shunt resistance are 0.1Ω , 0.2Ω , and 50Ω . Calculate the value of .

① generated e.m.f when the generator is connected in short shunt.

⇒ Is given:-

$$V = 230 \text{ volt}$$

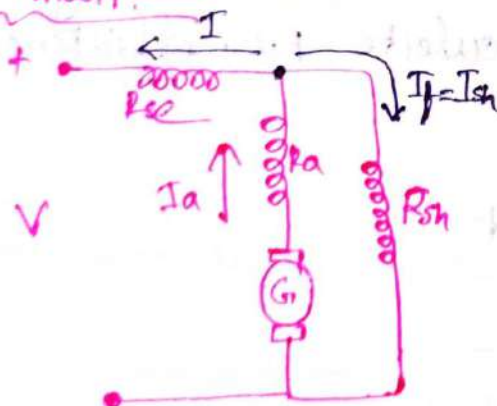
$$P = 20 \text{ kW} = 20 \times 10^3 \text{ W}$$

$$R_a = 0.1 \Omega$$

$$R_{se} = 0.2 \Omega$$

$$R_{sh} = 50 \Omega$$

For short shunt:-



we know:-

$$P = IV$$

$$\therefore I = \frac{P}{V} = \frac{20 \times 10^3}{230} = 86.956 \text{ A}$$

Now,

$$\begin{aligned} V_{sh} &= V + I R_{se} \\ &= 230 + 86.956 \times 0.2 \\ &= 247.3912 \text{ V} \end{aligned}$$

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So, Shunt field current $I_{sh} = \frac{V_{sh}}{R_{sh}} = \frac{247.9912}{50} = 4.9478 \text{ A}$

$\therefore I_a = I_{sh} + I = 4.947 + 86.9656 = 91.9134 \text{ A}$

We know:

$$E_g = V + I_a R_a + B \cdot D + I P_{se}$$

$$= 230 + 91.9134 \times 0.1 + 0 + 86.9656 \times 0.2$$

$$= 256.58446 \text{ V}$$

Ans:-

#problem 26.28:-

$\Rightarrow$ A long-shunt compound-wound generator gives 240 volts at F.L. Output of 100 A. The resistance are : armature (including brush contact) 0.1Ω , series field 0.02Ω , interpole field 0.025Ω , shunt field (including regulating resistance) 100Ω . The iron loss at F.L. is 1000 W. Windage and friction losses total 500 W. Calculate F.L. efficiency of the machine.

$\Rightarrow$ In given:-

$V = 240 \text{ volt}$

$I_L = 100 \text{ A}$

$R_a = 0.1 \Omega$

$R_{se} = 0.02 \Omega$

$R_{in} = 0.025 \Omega$

$R_{sh} = 100 \Omega$

Iron loss = 1000 W

Friction loss = 500 W

$1000 + 500 = 1500 \text{ W}$

We know:

Output = $V I_L = 240 \times 100 = 24000 \text{ W}$

$I_{sh} = \frac{V}{R_{sh}} = \frac{240}{100} = 2.4 \text{ A}$

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$$\text{Armature copper loss} = I_a^2 R_a \dots \dots \dots (1)$$

here,

$$I_a = I_L + I_{sh}$$

$$= 100 + 2.4$$

$$= 102.4 \text{ A}$$

For this circuit:-

$$\text{total } R_a = 0.1 + R_{in} + R_{se}$$

$$= 0.1 + 0.025 + 0.02$$

$$= 0.145 \Omega$$

So, Armature copper loss = $I_a^2 R_a$

$$= (102.4)^2 \times 0.145$$

$$= 1521 \text{ W}$$

$$\text{Shunt field copper loss} = I_{sh}^2 R_{sh} = I_{sh} V$$

$$= (2.4)^2 \times 100 = 5.76 \times 240$$

$$= 576 = 576 \text{ W}$$

So, total loss = $1521 + 576 + 1500 = 3597 \text{ W}$

we know:-

$$\eta = \frac{\text{output}}{\text{output} + \text{total loss}} \times 100 \%$$

$$= \frac{24000}{24000 + 3597} \times 100 \%$$

$$= 87\%$$

Ans.

Problem:-

⇒ A 4 pole de-shunt generator having wave-wound armature winding has total 14 conductors. The de generator is driven at 1500 rpm and the flux per pole is 10 mWb. Compute:-

- (i) the generating voltage. E_g
- (ii) the armature current, I_a , If the load resistance is 300Ω , the shunt field resistance is 200Ω and the armature resistance is 5Ω .

Given:-

$$P = 4$$

$$A = 2$$

$$Z = 14$$

$$N = 1500$$

$$\Phi = 10 \text{ mWb} = 10 \times 10^{-3} \text{ Wb}$$

(i)

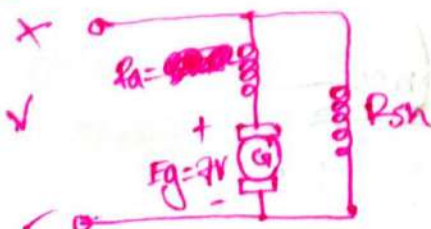
we know:-

$$E_g = \frac{ZN\Phi}{60} \times \frac{P}{A}$$

$$= \frac{14 \times 1500 \times 10 \times 10^{-3}}{60} \times \frac{4}{2}$$

$$= 7 \text{ V}$$

(ii)

For shunt:-

$$R_a = 5\Omega$$

$$R_{sh} = 200\Omega$$

$$R_L = 300\Omega$$

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We know:

$$I_a = \frac{E_g}{(R_a + R_{se} + R_{sh})} = \frac{7}{(5 + 200 + 300)} = 0.013 A$$

Problem: 2:

3) A long shunt compound dc generator delivers a load current of 100 A at 640 V and has armature, series, shunt resistance 0.05 Ω, 0.03 Ω, 250 Ω. Compute

① Armature current I_a

② The generated voltage, Assuming Brush drop 2 V

Is given:

$$R_a = 0.05 \Omega$$

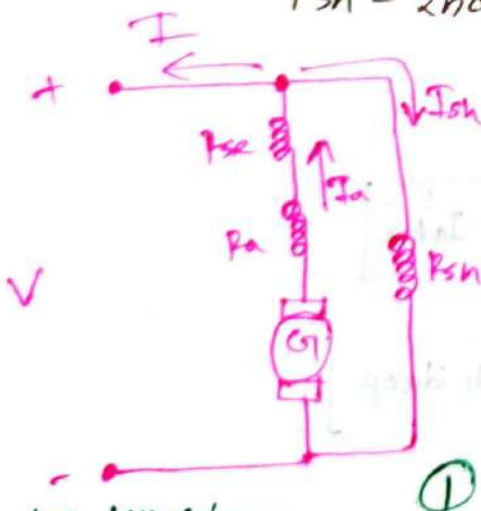
$$R_{se} = 0.03 \Omega$$

$$R_{sh} = 250 \Omega$$

$$V = 640 V$$

$$I = 100 A$$

$$B.D = 2 V$$



$$\begin{aligned} \therefore I_a &= I + I_{sh} \\ &= 100 + 2.56 \\ &= 102.56 A \end{aligned}$$

②

We know:

$$I_{sh} = \frac{V}{R_{sh}}$$

$$= \frac{640}{250}$$

$$= 2.56 A$$

$$\begin{aligned} E_g &= I_a (R_a + R_{se}) + V + B.D \\ &= 102.56 (0.05 + 0.03) + 640 + 2 V \\ &= 650.2048 V \end{aligned}$$

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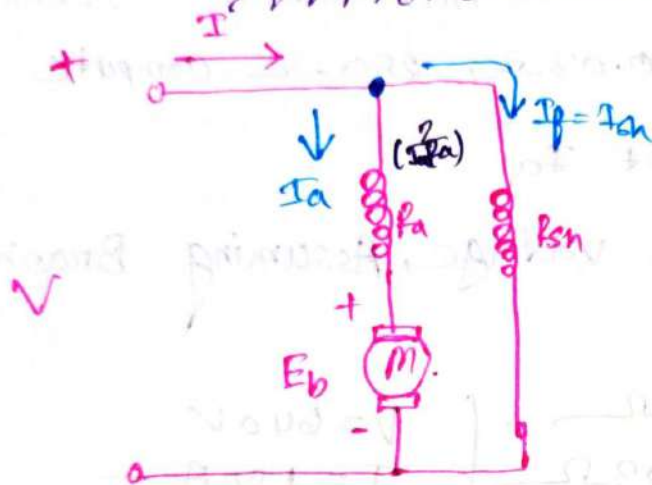
The Motor
Ch. 29

#DC Motor:—

(i) Shunt motor.

(ii) Series motor.

Shunt Motor:—



E_b = Back emf
(generated emf of rotating armature)

$$\therefore P_m = E_b I_a$$

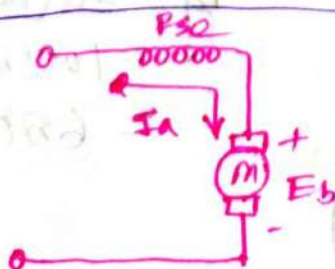
$$P_o = \omega T_a$$

$$P_m = E_b I_a = V I - I_a^2 R_a$$

$$\therefore V = I_a R_a + E_b + \text{Brush drop}$$

For Series Motor:—

$$V = E_b + B.D + I_a R_{se}$$



#problem: 11

For a DC shunt motor the flux is 10 mWb , the speed is 1400 rpm and the proportionality constant is 0.7 . If the supply voltage is 220 V , the armature resistance and field resistance is 0.5Ω and 250Ω . Determine

- ① Back emf
- ② Armature current
- ③ field current
- ④ overall efficiency
- ⑤ output power for a developed torque of 1.6 Nm

⇒ In given:

$$\Phi = 10 \text{ mWb} = 10 \times 10^{-3} \text{ Wb}$$

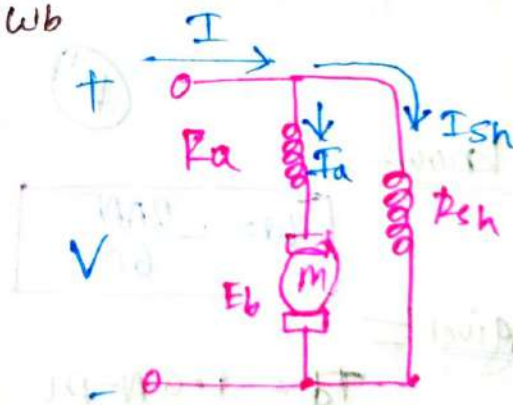
$$N = 1400 \text{ rpm}$$

$$k = 0.7$$

$$V = 220 \text{ V}$$

$$R_a = 0.5 \Omega$$

$$R_{sh} = 250 \Omega$$



①

We know:

$$E_b = \frac{\Phi Z N}{60} \times \frac{P}{A} = \Phi Z k$$

$$= 10 \times 10^{-3} \times 1400 \times 0.7$$

$$= 0.8 \text{ V}$$

②

using KVL:

$$V = E_b + I_a R_a + \text{brush drop} \quad \text{--- (1)}$$

$$\therefore V = E_b + I_a R_a + 0 \quad \text{--- (2)}$$

Armature Current $I_a = (V - E_b) / R_a$

$$= \frac{V - E_b}{R_a}$$

$$= \frac{220 - 0.8}{0.5}$$

$$= 439.2 \text{ A}$$

(iii)

we know:Field Current $I_{fh} = \frac{V}{R_{fh}}$

$$\therefore I_{fh} = \frac{220}{260}$$

$$= 0.846 \text{ A}$$

(v)

we know:

$$\omega = \frac{2\pi N}{60}$$

$$= \frac{2\pi N}{60} = 446.607 \text{ rad/s}$$

In given:

$$T_d = 1.6 \text{ N-m}$$

$$\therefore P_o = \omega T_d = 446.607 \times 1.6$$

$$= 714.5712 \text{ W}$$

 $P_o = \text{Mechanical Output power}$

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(IV)

we know:-

$$\text{Overall Efficiency} = \frac{\text{Mechanical output power}}{\text{Electrical Input power}} \times 100\%$$

$$\therefore \text{Overall Efficiency} = \frac{\omega T_d}{VI} \quad \text{--- (1)}$$

here:-

$$\omega = \frac{2\pi N}{60} = 146.607 \text{ rad/s}$$

$$T_d = 1.6 \text{ N-m}$$

$$\therefore P_{out} = \omega T_d = 146.607 \times 1.6 = 234.5712 \text{ W}$$

$$\therefore I = I_a + I_{sh}$$

$$= 420.4 + 0.88 \text{ A}$$

$$= 421.28 \text{ A}$$

$$VI = 421.28 \times 220$$

$$= 92.68 \times 10^3 \text{ W}$$

put the value in eq (1) :-

or

$$\eta = \left(\frac{146.607 \times 1.6}{92.68 \times 10^3} \right) \times 100\%$$

$$= 0.253\%$$

Power and Losses

- ① Electrical input $P_{in} = VI$
- ② power across Armature $P_m = E_b \times I_a$
- ③ Loss at armature (as copper loss) $= P_{in} - P_m$
 $= VI - E_b I_a$
- ④ Mechanical output power,
 $= (I_a)^2 \times R_a$
 $P_{out} = \omega T_b$
- ⑤ Rotary Losses (Stray loss or Iron loss and frictional loss) $= P_m - P_o$
 $= E_b I_a - \omega T_b$
- ⑥ $\omega = \frac{2\pi N}{60}$

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problem:—

⇒ A 200V D.C. shunt wound motor has an armature resistance of 0.4Ω and at a certain load has an armature current of $30A$ and runs at 1350 rev/min . If the load on the shaft of the motor is increased so that the armature current increases to $45A$, determine the speed of the motor, assuming the flux remain constant.

⇒ In given:—

$$I_{a1} = 30A$$

$$I_{a2} = 45A$$

$$N_1 = 1350 \text{ rpm}$$

$$R_a = 0.4\Omega$$

$$V = 200$$

$$\therefore E_1 = V - I_{a1} R_a$$

$$= 200 - 30 \times 0.4$$

$$= 188V$$

$$E_2 = V - I_{a2} R_a$$

$$= 200 - 45 \times 0.4$$

$$= 182V$$

∴ we know:—

$$\frac{E_1}{E_2} = \frac{N_1}{N_2}$$

$$\therefore N_2 = \frac{N_1 E_2}{E_1} = \frac{1350 \times 182}{188} = 1307 \text{ rpm}$$

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Problem: 2:

Q. A D.C. motor connected to a 440V supply takes an armature current of 150A on full load. If Armature circuit has resistance of 0.84Ω . Calculate.

① E_b .

∴ To given:

$$V = 440 \text{ V}$$

$$I_a = 150 \text{ A}$$

$$R_a = 0.84 \Omega$$

we know:

$$V = I_a R_a + E_b$$

$$\therefore E_b = V - I_a R_a$$

$$= 440 - 150 \times 0.84$$

$$= 314 \text{ V}$$

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Induction Motor:-
Ch: 34:-

Induction Motor:-

⇒ An induction motor consists of two main part

(i) Stator

(ii) Rotor.

Self-starting motor:-

⇒ Induction motor starts to run, immediately supply the power. That's why three-phase induction motor is a self-starting motor.

rotating transformer:-

⇒ The working principle of an induction motor is almost similar to a transformer except that the secondary side is rotating that's why an induction motor is also called rotating transformer.

Synchronous speed: (N_s)

⇒ The speed of rotating flux, which is produced due to the supply of three-phase voltage in the stator of an induction motor. It is called synchronous speed.

$$\therefore N_s = \frac{120f}{p} \text{ (rpm)}$$

f = frequency.

p = No. of poles

Slip-Speed: (N_{sl})

⇒ The difference between synchronous speed (N_s) and the actual rotor speed (N) is called slip speed.

$$N_{sl} = N_s - N \text{ (rpm)}$$

$$N_{sl} = s \times N_s$$

N_s = Synchronous speed

N = Actual rotor speed

s = slip.

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Slip: (s)

⇒ the per-unit value of slip-speed with respect to synchronous speed is called slip.

$$\therefore s = \frac{N_s - N}{N_s} = \frac{N_s - N}{N_s} = \left(1 - \frac{N}{N_s}\right)$$

Actual rotor speed: (N)

$$N = (1 - s) \times N_s$$

Using running Condition:-

① Frequency of Rotor Current (f_r):

$$\therefore f_r = s \times f$$

② Rotor Induced Voltage (E_r):

$$E_r = s \times E_2$$

③ Rotor Reactance (X_r):

$$X_r = s \times X_2$$

Note:- $\Rightarrow$ At standstill ($N=0$) Condition:-• The value of slip, $s = 1$ • $f_r = f$ • $E_r = E_2$ • $X_r = X_2$ # problem:-

$\Rightarrow$ A 4 pole, 3-phase induction motor operates from a supply whose frequency is 50 Hz. calculate:-

- (i) the speed at which the magnetic field of the stator is rotating.
- (ii) the speed of the rotor when the slip is 0.04.
- (iii) the frequency of the rotor currents when the slip is 0.03.
- (iv) The frequency of the rotor currents at standstill.

 $\Rightarrow$ Is given:- $p = 4$ pole $f = 50$ Hz

(i)

we know:-

$$N_s = \frac{120f}{P} = \frac{120 \times 50}{4} = 1500 \text{ rpm}$$

(ii)

we know:-

$$\begin{aligned} N &= (1 - s) N_s \\ &= (1 - 0.04) \times 1500 \\ &= 1440 \text{ rpm} \end{aligned}$$

(iii)

we know:-

$$\begin{aligned} f_r &= s \times f \\ &= 0.03 \times 50 = 1.5 \text{ Hz} \end{aligned}$$

(iv)

we know:-

For start at full:-

$$\begin{aligned} f_r &= f \\ &= 50 \text{ Hz} \end{aligned}$$

(v)

#problem 2:—
 on on

⇒ A 3- ϕ induction motor is wound for 4 poles and it supplied from 50 Hz system calculate:—

- (i) the synchronous speed.
- (ii) the rotor speed, when slip is 4%.
- (iii) rotor frequency when rotor runs at 600 rpm.

⇒ In given:—
 on on

$$P = 4 \text{ poles.}$$

$$f = 50 \text{ Hz}$$

(i)

we know:—
 on on

$$N_s = \frac{120f}{P} = \frac{120 \times 50}{4} = 1500 \text{ rpm}$$

(ii)

we know:—
 on on

$$\begin{aligned} N &= (1-s) \times N_s \\ &= (1-4\%) \times 1500 \\ &= 1440 \text{ rpm.} \end{aligned}$$

(iii)

we know:—
 on on

$$\begin{aligned} f_r &= s \times f \\ &= 0.6 \times 50 \\ &= 30 \text{ Hz} \end{aligned}$$

$$\begin{aligned} s &= \left(1 - \frac{N}{N_s}\right) \\ &= \left(1 - \frac{600}{1500}\right) \\ &= 0.6 \end{aligned}$$

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Synchronous
(Alternator)
Generator
Ch: 37

Speed and Frequency of an Alternator

①

$$\text{Frequency} : f = \frac{P}{2} \times \frac{N}{60} = \frac{PN}{120} \text{ Hz}$$

$\therefore N = \text{Actual rotor speed (rpm)}$

OR

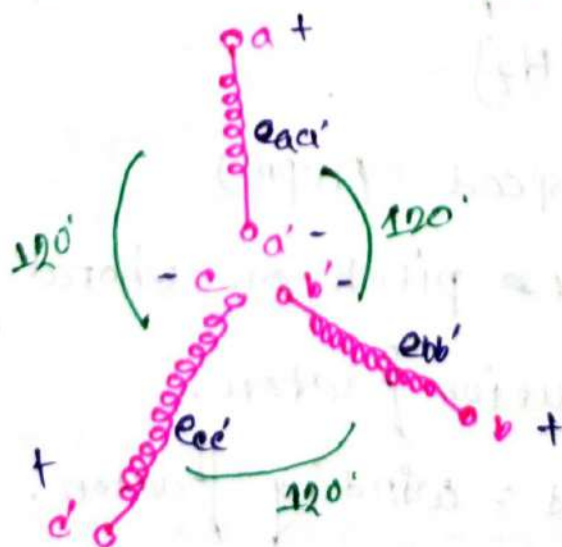
$$f = \frac{PN}{120} \text{ Hz}$$

$N = \text{Actual rotor speed (rpm)}$

$P = \text{total No. of magnetic poles.}$

$f = \text{frequency of generated emf in Hz.}$

Equation of the Induced EMF—



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$$e_{aa'} = E_{max} \sin \omega t = \sqrt{2} E_{ph} \sin \omega t \quad [V]$$

$$e_{bb'} = E_{max} \sin(\omega t - 120^\circ) = \sqrt{2} E_{ph} \sin(\omega t - 120^\circ) \quad [V]$$

$$e_{cc'} = E_{max} \sin(\omega t + 120^\circ) = \sqrt{2} E_{ph} \sin(\omega t + 120^\circ) \quad [V]$$

$$\therefore E_{ph} = 4.44 k_e k_d \phi T = 4.44 k_w \phi T \quad [V]$$

$$E_{max} = \sqrt{2} \times E_{ph} \quad [V]$$

E_{ph} = RMS value of per phase voltage

ϕ = Flux (Wb) per phase

T = Number of coils or turns/phase

$$f = \frac{PN}{120} \text{ [Hz]}$$

N = Rotor speed [rpm]

pitch or chored factor = k_e or k_p

k_d = Distribution factor.

$k_w = k_e \times k_d$ = winding factor.

$$E_L = \sqrt{3} E_{ph} = \text{Line voltage.}$$

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#problem:-

⇒ A 3 ϕ , 16 poles alternator has star-connected winding with 240 turns coil in each phase. The flux per pole is 0.03 Wb. the speed is 375 rpm. Its stator winding factor is 0.96. Find.

- (i) the frequency.
- (ii) the rms value of phase voltage, and line voltage of this generator.
- (iii) the three phase voltage as a function of time.

⇒ In given:-

$$P = 16 \text{ poles.}$$

$$T = 240$$

$$\Phi = 0.03 \text{ Wb}$$

$$N = 375 \text{ rpm}$$

$$k_w = 0.96$$

We know:-

$$f = \frac{PN}{120} = \frac{16 \times 375}{120} = 50 \text{ Hz}$$

(ii)

We know:-

$$\begin{aligned} E_{ph} &= 4.44 \times k_w \times f \times \Phi \times T \\ &= 4.44 \times 0.96 \times 50 \times 0.03 \times 240 \\ &= 1594.464 \text{ V} \end{aligned}$$

$$\begin{aligned} \therefore E_L &= \sqrt{3} \times E_{ph} \\ &= 2657.27 \text{ V} \end{aligned}$$

(iii)

We know:

$$\begin{aligned}
 E_{\max} &= \sqrt{2} \times E_{\text{ph}} \\
 &= \sqrt{2} \times 1534 \cdot 464 \\
 &= 2170 \text{ V}
 \end{aligned}$$

$$\begin{aligned}
 \omega &= 2\pi f \\
 &= 2\pi \times 50 \\
 &= 100\pi \\
 &= 314 \cdot 156 \text{ rad/s}
 \end{aligned}$$

Now,

$$e_{aa'} = E_{\max} \sin(\omega t) = 2170 \sin(314t) \text{ V}$$

$$e_{bb'} = E_{\max} \sin(\omega t - 120) = 2170 \sin(314t - 120) \text{ V}$$

$$e_{cc'} = E_{\max} \sin(\omega t + 120) = 2170 \sin(314t + 120) \text{ V}$$

Problem 2:-

⇒ If A 3 phase, 16 poles Alternator has $T = 0.6$, $\phi = 5 \times 10^{-2} \text{ Wb}$
 $N = 450 \text{ rpm}$, $k_p = 0.966$, $k_d = 0.96$ Find:-

- (i) the frequency
- (ii) The rms value of phase voltage and line voltage.
- (iii) The three phase voltage as a function of time.

⇒ In given:-

$$\left. \begin{aligned}
 P &= 16 \\
 T &= 0.6 \\
 \phi &= 5 \times 10^{-2} \\
 N &= 450
 \end{aligned} \right\} \begin{aligned}
 k_p &= 0.966 \\
 k_d &= 0.96 \\
 \therefore k_w &= k_p \times k_d = 0.92336
 \end{aligned}$$

(i)

we know:-

$$f = \frac{PN}{120}$$

$$= \frac{16 \times 450}{120}$$

$$= 60 \text{ Hz}$$

(ii)

we know:-

$$E_{ph} = 4.44 k\omega \times \phi T f$$

$$= 4.44 \times 6.22796 \times 5 \times 10^{-2} \times 0.6 \times 60$$

$$= 1185.8332 \text{ V}$$

$$E_L = \sqrt{3} E_{ph}$$

$$= 2053.0243 \text{ V}$$

(iii)

we know:-

$$E_{max} = \sqrt{2} \times E_{ph} = \sqrt{2} \times 1185.8332 = 1677.022 \text{ V}$$

$$\omega = 2\pi f = 120\pi = 376.99 \approx 377 \text{ rad/s}$$

$$\therefore e_{aa'} = E_{max} \sin(\omega t) = 1677.022 \sin(377t) \text{ V}$$

$$e_{bb'} = E_{max} \sin(\omega t - 120^\circ) = 1677.022 \sin(377t - 120^\circ) \text{ V}$$

$$e_{cc'} = E_{max} \sin(\omega t + 120^\circ) = 1677.022 \sin(377t + 120^\circ) \text{ V}$$

Ans:-

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Stepper Motor!
Chin

Stepper Motor :-

⇒ Stepper motors are operate on discrete control pulses received and rotate in discrete (Angular rotation) Steps.

Step Angle (P)

⇒ The angle through which the motor shaft rotates each command pulse is called the step angle (P).

$$\beta = \frac{|N_s - N_r|}{N_s N_r} \times 360^\circ$$

$$\beta = \frac{360^\circ}{m N_r}$$

N_s = NO. of stator poles (teeth)

N_r = NO. of Rotor poles (teeth)

$$m = \frac{N_s}{N_s - N_r} = \text{Number of stator phases}$$

Resolution :-

⇒ Resolution is the number of steps need to complete one revolution of the rotor shaft.

$$\therefore \text{Resolution} = \frac{\text{No. of steps}}{\text{revolution}} = \frac{360^\circ}{\beta}$$

Mathematical Expression of Shaft speed (n)

$$n = \frac{\beta \times}{960} \text{ rps}$$

n = pulse frequency Resolution

β = stepping frequency or (pulse rate) in pps

#problem:- Q.1:-

⇒ A hybrid VR stepping motor has a 8 main poles which have been constructed to have 5 teeth each if rotor has 50 teeth. Calculate stepping angle.

⇒ In given:-

$$N_s = 8 \times 5 = 40 \text{ teeth}$$

$$N_r = 50 \text{ teeth}$$

We know:-

$$\begin{aligned} \beta &= \frac{|N_s - N_r|}{N_s \times N_r} \times 360^\circ \\ &= \frac{40 - 50}{40 \times 50} \times 360^\circ \\ &= 1.8^\circ \end{aligned}$$

Ans:-

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problem: ex. 2

⇒ A Stepper motor has a step angle of 2.5° . Determine.

- The resolution.
- The number of steps required for the shaft to make 25 revolutions.
- The shaft speed if the stepping frequency is 3600 rps

⇒ In given:—

$$\beta = 2.5^\circ$$

(i)

we know:—

$$\begin{aligned}\text{Resolutions} &= \frac{360^\circ}{\beta} \\ &= \frac{360^\circ}{2.5} \\ &= 144 \text{ steps/revolution}\end{aligned}$$

(ii)

In given:—

$$\text{Revolutions} = 25$$

we know:—

$$\begin{aligned}\text{No. of steps} &= \text{Revolutions} \times \text{Resolutions} \\ &= 25 \times 144 \\ &= 3600 \text{ steps.}\end{aligned}$$

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③

we know —

$$\text{Shaft speed } n = \frac{P.f}{360}$$

$$= \frac{2.5 \times 3600}{360}$$

$$= 25 \text{ rps}$$

#problem: 39.11:—

If $N_s = 8$, $N_R = 6$, $m = 4$ then calculate the steeping angle.

⇒ we know —

$$\beta = \frac{N_s - N_R}{N_s N_R} \times 360 = \frac{360}{m N_R}$$

$$= \frac{8 - 6}{8 \times 6} \times 360 = \frac{360}{4 \times 6}$$

$$= 15^\circ$$

Ans:—

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Transformer.
Ch: 21, 32

Transformer:-

⇒ It is a static device which transfers the electrical energy from one circuit to another circuit by raising or lowering the voltage without changing frequency.

Construction of Transformer:-

⇒ There are two basic parts of transformer.

(a) Windings core

(b) Magnetic Core.

Winding:-

⇒ In conventional transformers there have high voltage (HV) and low-voltage (LV) windings.

There are two types of winding

(i) Primary

(ii) Secondary.

Primary Windings:-

⇒ The winding at which source is connected is called primary winding.

Secondary windings:-

⇒ The winding at which load is connected is called Secondary winding.

Magnetic Cores:-

⇒ There are two basic type of magnetic cores.

① Core type

② Shell type.

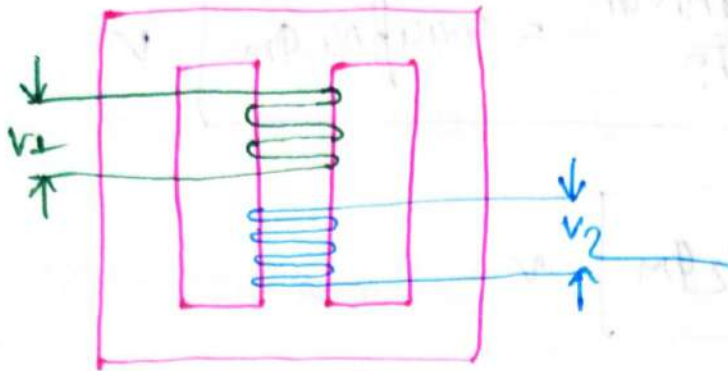
Core type:-

⇒ The windings surround a considerable part of the core.

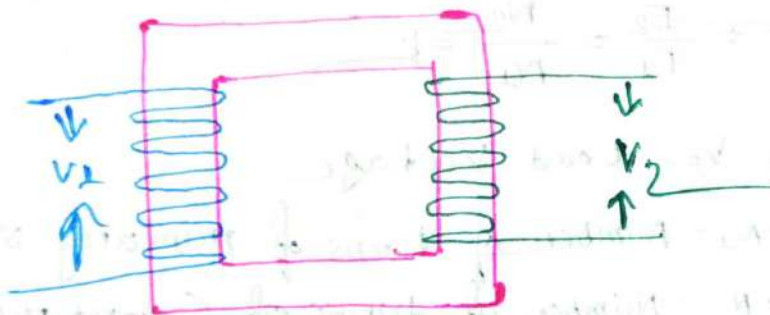
Shell type:-

⇒ The core surrounds the considerable portion of the windings.

Shell Type



Core type



EMF Equation of a Transformer

⇒ Let, the expression of developed flux in core due to the supply AC voltage:

$$\Phi(t) = \Phi_m \sin \omega t$$

From Faraday's Law

$$e_1(t) = -N_1 \frac{d\Phi(t)}{dt} = N_1 \times 2\pi f \times \Phi_m \sin(\omega t - 90^\circ) \text{ V}$$

$$e_2(t) = -N_2 \frac{d\Phi(t)}{dt} = N_2 \times 2\pi f \times \Phi_m \sin(\omega t - 90^\circ) \text{ V}$$

So, the RMS Value E_1 and E_2 are:-

$$E_1 = \frac{2\pi f N_1 \Phi_m}{\sqrt{2}} = 4.44 f N_1 \Phi_m \quad \checkmark$$

$$E_2 = 4.44 f N_2 \Phi_m \quad \checkmark$$

Voltage Transformation Ratio:-

$$\frac{V_2}{V_1} = \frac{E_2}{E_1} = \frac{N_2}{N_1} = k$$

V_2 = Load Voltage

$N_p = N_1$ = Number of turns of primary side

$N_s = N_2$ = Number of turns of secondary side.

k = transformation ratio / turns ratio.

Note:- If the losses of transformer are neglected, then

$$V_1 = E_1 \quad \text{or} \quad V_2 = E_2$$

$$\text{input VA} = \text{output VA}$$

So,

$$S_2 = S_1$$

$$V_2 I_2 = V_1 I_1$$

or,

$$\frac{I_2}{I_1} = \frac{V_1}{V_2} = \frac{E_1}{E_2} = \frac{N_1}{N_2} = \frac{1}{k}$$

I_2 = Load Current

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Problem:-

⇒ An ideal transformer has a turns ratio of 8:1 and the primary current 3A when it is supplied at 240V. Calculate the secondary voltage and current.

⇒ In given:-

$$K = 8:1 \text{ that means } \frac{N_p}{N_s} = \frac{8}{1}$$

$$I_p = 3A$$

$$E_p = 240V$$

We know:-

$$\frac{N_p}{N_s} = \frac{E_p}{E_s}$$

$$\therefore E_s = \frac{E_p N_s}{N_p}$$

$$= \frac{240 \times 1}{8}$$

$$= 30V$$

$$\frac{I_p}{I_s} = \frac{E_s}{E_p}$$

$$\therefore I_s = \frac{E_p I_p}{E_s}$$

$$= \frac{240 \times 3}{30}$$

$$= 24A$$

#problem:-

⇒ An ideal transformer, connected to a 240V mains, supplies a 12V, 150W lamp. Calculate the transformer turns ratio and the current taken from the supply.

⇒ In given:-

$$V_p = 240V$$

$$V_s = 12V$$

$$P_s = 150W = V_s I_s$$

$$\therefore I_s = \frac{150}{12} = 12.5A$$

We know:-

$$\frac{V_p}{V_s} = \frac{I_s}{I_p}$$

$$\therefore I_p = \frac{I_s V_s}{V_p} = \frac{12.5 \times 12}{240} = 0.625A$$

$$\frac{E_p}{E_s} = \frac{N_p}{N_s}$$

$$\Rightarrow \frac{N_p}{N_s} = \frac{240}{12}$$

$$\therefore N_p : N_s = 20 : 1$$

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#problem! 3:-

⇒ A ~~5kVA~~ Single-phase transformer has a turns ratio of 10:1 and is fed from a 2.5kV supply. Neglecting losses determine,

(a) the full load secondary current.

(b) the minimum load resistance which can be connected across the secondary winding to give full load kVA and

(c) primary current at full load kVA.

⇒ In given:-

$$S = 5k = 5 \times 10^3 \text{ VA}$$

$$\frac{N_p}{N_s} = \frac{10}{1}$$

$$V_p = E_p = 2.5k = 2.5 \times 10^3 \text{ V}$$

We know:-

$$\frac{N_p}{N_s} = \frac{E_p}{E_s}$$

$$\Rightarrow E_s = \frac{E_p N_s}{N_p} = \frac{2.5 \times 10^3 \times 1}{10} = 250 = V_2$$

$$S = V_2 I_2$$

$$I_2 = \frac{S}{V_2}$$

$$= \frac{5 \times 10^3}{250}$$

$$= 20 \text{ A}$$

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(b)

we know:-

Minimum value of load resistance

$$R_L = \frac{V_2}{I_2} = \frac{250}{20} = 12.5 \Omega$$

(c)

we know:-

$$S = V_1 I_1$$

$$\therefore I_1 = \frac{S}{V_1}$$

$$= \frac{5 \times 10^3}{2.5 \times 10^3}$$

$$= 2A$$

$$I_1 = I_p = 2A$$

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(b)

we know:-

Minimum value of load resistance

$$R_L = \frac{V_2}{I_2} = \frac{250}{20} = 12.5 \Omega$$

(c)

we know:-

$$S = V_1 I_1$$

$$\therefore I_1 = \frac{S}{V_1}$$

$$= \frac{5 \times 10^3}{2.5 \times 10^3}$$

$$= 2A$$

$$I_1 = I_p = 2A$$

Ans:-

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⇒ A 100kVA, 4000V/200V, 50Hz single-phase transformer has 100 secondary turns. Determine.

- The primary and secondary current.
- The number of primary turns.
- The maximum value of the flux

⇒ In given: _____

$$S = 100 \text{ kVA} = 100 \times 10^3 \text{ VA}$$

$$V_1 = 4000 \text{ V}$$

$$V_2 = 200 \text{ V}$$

$$f = 50 \text{ Hz}$$

$$N_2 = 100 \text{ turns.}$$

(a)

We know: _____

$$S = V_1 I_1$$

$$\therefore I_1 = \frac{S}{V_1} = \frac{100 \times 10^3}{4 \times 10^3} = 25 \text{ A}$$

$$S = V_2 I_2$$

$$I_2 = \frac{S}{V_2} = \frac{100 \times 10^3}{200} = 500 \text{ A}$$

(b)

We know: _____

$$\frac{V_1}{V_2} = \frac{N_1}{N_2}$$

$$N_1 = \frac{V_1 N_2}{V_2} = \frac{4000 \times 100}{200} = 2000 \text{ turns.}$$

(c)

We know:—For maximum value of flux

$$E_2 = 4.44 f \Phi_m N_2$$

$$\therefore \Phi_m = \frac{E_2}{4.44 f N_2}$$

$$= \frac{200}{4.44 \times 50 \times 100}$$

$$= 0.0009 \text{ mwb}$$

OR

$$E_1 = 4.44 f \Phi_m N_1$$

$$\therefore \Phi_m = \frac{E_1}{4.44 f N_1}$$

$$= \frac{4 \times 10^3}{4.44 \times 50 \times 2 \times 10^3}$$

$$= 0.0009 \text{ mwb}$$

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Classification of Transformer

Based on Voltage Level:-

⇒ there are two types:-

① Step-up Transformer

② Step-down Transformer

Step-up Transformer:-

⇒ If secondary side voltage (V_2) is greater than the primary side voltage (V_1) that means $K > 1$ then transformer is called step up transformer.

Step-down Transformer:-

⇒ If secondary side voltage (V_2) is smaller than the primary side voltage (V_1) that means $K < 1$ then transformer is called step ~~up~~ down transformer.

Regulation of Transformer :-

⇒ When the secondary of a transformer is loaded, the secondary terminal voltage, V_2 falls. As the power factor decreases, this voltage drop increases. This is called the regulation of the transformer and it is usually expressed as a percentage of the secondary no-load voltage, E_2 . For full load conditions:

$$\text{Regulation} = \left(\frac{E_2 - V_2}{E_2} \right) \times 100\%$$

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problem 1

3) A 400 KVA transformer has a primary winding resistance of 0.5Ω and a secondary winding resistance of 0.001Ω . The iron loss is 2.5 kW and the primary and secondary voltage are 5 kV , 320 V . If the power factor of the load is 0.85 determine.

(i) on full load efficiency of the transformer.

(ii) on half load efficiency of the transformer.

⇒ In given:

$$S = 400 \times 10^3 \text{ VA}$$

$$R_1 = 0.5 \Omega$$

$$R_2 = 0.001 \Omega$$

$$\text{iron loss} = 2.5 \times 10^3 \text{ W}$$

$$V_1 = 5 \times 10^3$$

$$V_2 = 320$$

$$\text{p.f.} = 0.85$$

(a)

we know:

$$S = V_1 I_1 = V_2 I_2$$

$$\therefore I_1 = \frac{S}{V_1} = \frac{400 \times 10^3}{5 \times 10^3} = 80 \text{ A}$$

$$I_2 = \frac{S}{V_2} = \frac{400 \times 10^3}{320} = 1250 \text{ A}$$

$$\text{Total Copper loss} = I_1^2 R_1 + I_2^2 R_2$$

$$= (80)^2 \times 0.5 + (1250)^2 \times 0.001$$

$$= 1562.5 + 1250$$

$$= 4762.5 \text{ W}$$

on full load, total loss = copper loss + iron loss

$$= 4762.5 + 2.5 \times 10^3 \text{ W}$$

$$= 7262.5 \text{ W}$$

$$\begin{aligned} \text{Total output power on full load} &= S \cos \theta_2 \\ &= 400 \times 10^3 \times 0.95 \\ &= 340 \text{ kW} \end{aligned}$$

$\therefore$ Input power = output + loss

$$= 340 \text{ kW} + 7.2625 \text{ kW}$$

$$= 347.2625 \text{ kW}$$

we know —

$$\text{Efficiency } \eta = \left(1 - \frac{\text{Loss}}{\text{input}} \right) \times 100\%$$

$$= \left(1 - \frac{7.2625 \text{ kW}}{347.2625 \text{ kW}} \right) \times 100\%$$

$$= 98\%$$

Ans.

(6)

$$\begin{aligned}\text{Total Copper loss on half load} &= \text{full loss} \times \left(\frac{1}{2}\right)^2 \\ &= 4762.5 \times \left(\frac{1}{4}\right) \\ &= 1190.625 \text{ W}\end{aligned}$$

So, total loss on half load:—

$$\begin{aligned}&= 1190.625 + 2.5 \times 10^3 \\ &= 3.691 \text{ kW}\end{aligned}$$

$$\begin{aligned}\text{Output power on half load} &= \text{full load} \times \frac{1}{2} \\ &= \frac{340}{2} \text{ k} \\ &= 170 \text{ k}\end{aligned}$$

∴ Input power = output + loss

$$\begin{aligned}&= 170 \text{ k} + 3.691 \text{ k} \\ &= 173.691 \text{ kW}\end{aligned}$$

we know,

$$\begin{aligned}\eta &= \left(1 - \frac{\text{loss}}{\text{input}}\right) \times 100\% \\ &= \left(1 - \frac{3.691}{173.691}\right) \times 100\% \\ &= 97.71\%\end{aligned}$$

Problem 04:-

⇒ A single phase transformer has 800 turns in the primary and 1500 turns in the secondary. The cross sectional area of the core is 84 cm^2 . If the primary winding is connected to a 50 Hz supply at 550 V. Calculate

(a) Peak flux density

(b) Induced voltage in the secondary side.

⇒ In given:-

$$N_p = 800 \text{ T}$$

$$N_s = 1500 \text{ T}$$

$$A = 84 \times 10^{-4} \text{ m}^2$$

$$f = 50 \text{ Hz}$$

$$V_p = 550$$

we know:-

maximum flux:-

$$\Phi_{\text{max}} = \frac{E_s}{4.44 f N_s}$$

$$= \frac{550}{4.44 \times 50 \times 800}$$

$$= 3.0068 \times 10^{-3} \text{ wb}$$

Flux Density:-

$$B = \frac{\Phi}{A} = \frac{3.0068 \times 10^{-3}}{84 \times 10^{-4}}$$

$$= 0.3686 \text{ T}$$

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(b)

⇒ We know:-

$$\frac{V_p}{V_s} = \frac{N_p}{N_s}$$

$$\Rightarrow V_s = \frac{V_p N_s}{N_p}$$

$$= \frac{550 \times 1500}{800}$$

$$= 1031.25 \text{ V}$$

Ans:-